



EXCELLENCE IN
DECEASED DONATION
COURSE

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Prescribed reading

1. Organ and Tissue Donation Policy Framework
2. SAMJ – South African Guidelines on Death Determination
3. Brain Death Checklist
4. Advice Sheets for Brain Death Testing
5. Circulatory Death Checklist
6. Donation after Circulatory Death – DCD
7. Care and Management of a Brain Dead Patient – DBD
8. Donor Management Checklist
9. Best Practice for Offering Organ and Tissue Donation – Donate Life
10. Family Conversation Planning Template
11. Family Conversation Debriefing Template
12. National Strategic Roadmap – Deceased Donor Definitions and Pathways

Course Training Tools

1. Organ Donation from Death to Life – Online Training Platform
2. Constructive Simulation Debriefing Sheet

Supporting texts

1. SAMJ – Medicine and the Law – Human tissue and organ transplant provisions: Chapter 8 of the National Health Act and its Regulations
2. SAJBL – Organ donation after circulatory death – legal in South Africa and in alignment with Chapter 8 of the National Health Act and Regulations relating to organ and tissue donation
3. HPCSA – Informed Consent – Booklet 4
4. HPCSA – Guidelines on Palliative Care – Booklet 17

Checklists

1. Brain Death
2. Circulatory Death
3. Advice Sheets on Brain Death Testing
4. Donor Management
5. Family Conversation Planning
6. Family Conversation Debriefing

Online resources

- a. Death Determination Checklists Download (<https://criticalcare.org.za/wp-content/uploads/2021/02/Death-Determination-Checklists-South-African-Guidelines-Brain-Death-and-Circulatory-Death.pdf>)
- b. Organ and Tissue Donation in South Africa – Creating a National Strategic Roadmap (<https://sats.org.za/wp-content/uploads/2024/09/Organ-and-Tissue-Donation-in-South-Africa-Strategic-Roadmap.pdf>)
- c. Best Practice Referral Download



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Circular H84/2025

ORGAN AND TISSUE DONATION POLICY IN THE WESTERN CAPE

This document outlines a comprehensive framework for the Western Cape Department of Health and Wellness to support healthcare teams in assisting patients and families in the best possible way at the end of life in a dignified manner and ensuring best practice approaches for consent to donation in the complex end-of-life space.

The policy is underpinned by several key principles that are both timely and essential in supporting the dignity of individuals and their families during the process of end-of-life care and donation.

The following highlights are pivotal in the evolution of this policy:

- **Compassion for Families:** Recognizing the emotional and psychological weight that families bear during this time, the policy emphasizes compassionate care for those who may have to make decisions regarding organ donation. This is aimed at ensuring that families are fully supported, informed, and guided through the process.
- **Consistent End-of-Life Care and Counselling:** The policy underlines the need for consistent and well-offered end-of-life care and counselling, which is essential for providing dignity in death and reducing the moral distress experienced by healthcare workers.

- **Dignity in Dying:** This policy seeks to ensure that individuals and their families who wish to, or decline to donate organs and tissues are treated with the utmost respect and dignity throughout the process, thereby facilitating a compassionate and respectful end-of-life experience.
- **Equity and access:** The policy sets forth guidelines that ensure consistent practices across the Western Cape, reducing variability and ensuring that all individuals receive the same high level of care, regardless of their geographic location or circumstances.
- **Best Practice:** Adopting internationally recognized best practices, the policy promotes the use of evidence-based approaches and the integration of donation as a normal part of end of life care. This approach will help ensure that we offer donation in the best way possible for all our patients and families.
- **Solace Through Giving:** One of the most profound aspects of the policy is its recognition of the solace that comes from being able to support organ and tissue donation after death. This act of giving offers comfort to grieving families, knowing that their loved one's legacy can save lives and provide hope for those in need of transplants. It is something all our healthcare teams can influence and can take pride in supporting well.

This policy marks a crucial step in strengthening the Western Cape's commitment to improving and leading on organ and tissue donation practices while ensuring that all individuals involved are treated with the dignity, respect, and compassion they deserve.

Staff are encouraged to engage with the document, explore linked training programs and become champions within your own roles to actively support donation. This policy must be widely disseminated and implemented with immediate effect.

The newly formed Provincial Organ and Tissue Donation Committee will be working on building from this policy and working with all staff. Should you have any queries, kindly contact Dr David Thomson at david.thomson@uct.ac.za



DR SAADIQ KARIEM

DDG: CHIEF OF OPERATIONS

DATE: 19/06/2025



Western Cape
Government

Organ and tissue donation policy in Western Cape

*Ensuring standards of care in a
dignified death and improved
donation support*

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1. Definitions

DBD – Donation after brain death

DCD – Donation after circulatory death

TC – Transplant coordinator (who is a SNOD – Specialist nurse in organ donation)

2. Executive summary

The aim of this practical guideline is to ensure that within the Western Cape every patient is given a dignified death which includes the option to be an organ or tissue donor at the end-of-life and that this decision is honoured wherever possible.

Organ donation refers to the donation of a solid organ or organs including, though not limited to, heart, lungs, liver, kidneys, pancreas and small bowel. Tissue donation includes the donation of corneas, skin, bone, tendons, cartilage and heart valves.

The consideration of donation after death should be considered a normal part of end-of-life care in the Department of Health and Wellness of the Western Cape. Best practice is that when a patient is close to death and their views cannot be determined, medical teams should be prepared to facilitate exploration with those close to the patient (their surrogate decision makers) whether the patient would have wished to support organ and or tissue donation by contacting the transplant coordinator who is best placed to clarify the potential for donation and to advise and help fully explore donation with the next of kin.

Healthcare workers should make every effort to support exploring the views of the potential donor and then to support their decision being fulfilled.

Although organ recovery occurs after death there are steps that health professionals will need to take before the death of the patient, if donation is to take place. Key to this is the timely referral and involvement of the transplant coordinators (specialist nurses in organ donation), whose expertise will guide and support families and health care professionals through this challenging and emotional process.

The successful donation of an organ or tissue requires collaboration across the hospital system and thus necessitates that all infrastructure and supports are in place to support a dignified death with excellent family communication. A key priority is to ensure that all parts of the healthcare ecosystem are knowledgeable about, and support donation and it is the responsibility of the hospital managers and department, division and unit heads to champion and support donation and transplantation within their areas and staffing complements.

Patients who are eligible to donate their organs are usually those who pass away whilst receiving, or immediately following the withdrawal of, mechanical ventilation. Most organ donors will therefore be cared for in the emergency department, trauma unit or ICU with a small number from other areas such as theatres or the general wards.

Tissue donation is able to be considered for patients who die without a mechanical ventilator in their end-of-life care pathway. It is possible to recover tissues for some time after death

(cornea within 12 - 24 hours, heart valves within 48 hours, skin within 48 hours and bone and tendons within 5 days after death) and tissue donation allows a way to allow more people to support donation after death. Tissue donation can occur in hospital or at the mortuary or forensic services and is supported by the Vitanova Tissue Bank a registered non-profit in the Western Cape.

3. Scope of policy

This guideline applies to all employees of Western Cape Department of Health and Wellness hospitals in all locations. Specific recipients who should:

Be <u>aware</u> of the document and where to access it	Head of Health, CEO, COO, Medical Managers and Operational Managers
<u>Understand</u> the document	Head of Health, CEO, COO, Medical Managers and Operational Managers
Have a <u>good working knowledge</u> of the document	Staff working in all clinical areas, especially critical care units, emergency departments and operating theatres.

As a provincial health department, we should pride ourselves on leading with best practice in end-of-life care practices which includes organ and tissue donation.

4. Problem statement

There is marked variability in offering patients and families the option to donate organs and tissues at end-of-life. The benefits of supporting donation are multiple and should be pursued whenever possible. End-of-life care and counselling needs to be offered consistently and well. Providing dignity in death lessens the moral distress for healthcare workers providing care to dying patients. A framework to support clinicians to best offer this support is needed. The risk is that we do not do this well or have a supporting framework and educational program we may increase the moral distress of our clinicians by not allowing our health system to offer the solace of donation to patients and grieving families.

5. Identification of potential organ donors

To support the identification of potential organ donors, clinical triggers should be used to prompt discussion with a transplant coordinator (specialist nurse in organ donation). While recognising that clinical situations vary, the following triggers should be considered for all (paediatric and adult) mechanically ventilated patients where:

- A decision has been made to perform brain death tests.
- There is a catastrophic brain injury with GCS 3-4 (which cannot be explained by sedation) after completion of a period of observation to allow prognostication.

- The intention is to discuss with the family the palliative withdrawal of life-sustaining treatment which will, or is expected to, result in circulatory death.

The earliest possible discussion with the transplant coordinator, following meeting one of these triggers, is recommended. This will enable the transplant coordinator to screen the patient for donation, identifying when organ donation is not feasible and when appropriate, mobilise to support the team thus minimising any delays to the patient's family and the donation process. The initial discussion with the transplant coordinator should occur prior to raising the subject of end-of-life care with the patient's family/nearest relative. This is to prevent premature discussions regarding donation, or consideration of donation where it is not feasible (some referrals may be screened out as unsuitable for organ donation), allow a check of the national priority listing for patients who would accept marginal organs and to check if the patient has expressed intent to donate by signing up as a donor with the Organ Donor Foundation.

Most potential organ donors are cared for within the ED, trauma unit or ICU although some are cared for in other clinical areas, e.g. the general wards or theatres. The above principles are applicable across all clinical areas.

While assessing if a patient is a potential donor, it is appropriate to resuscitate and stabilise the patient within the current setting. Life-sustaining treatments should not be withdrawn or limited until the patient's (or their surrogate decision maker's) decision around organ donation has been clarified with them directly by (or in consultation with) a transplant coordinator. The assessment of donation potential for organs or tissues should be documented in the patient's notes in accordance with best practice which is after a referral and discussion with a transplant coordinator. Appropriate support will be facilitated by a transplant coordinator or a donation support physician.

Appendix 1 – Donor identification and referral pathway

Appendix 2 – Deceased donor organ donation pathway

6. Approaching the family

Approaching families regarding donation can be challenging and whilst requiring tailoring to individual circumstances, best practice includes:

- Early discussion with the **transplant coordinator** to clarify the potential for organ and tissue donation.
- The **treating clinical team** to communicate the prognosis clearly and honestly to allow the family time to come to terms with the death or imminent death of their loved one.
- Decoupling of prognosis discussions (separate) from any request for consent to donate
- Confirming the family understands and are accepting of their loss prior to raising the potential of organ/tissue donation. **(collaborative)**
- Checking of the organ donor register by the **transplant coordinator**

- The planning of the approach for consent, prior to meeting with the family, with available / needed team members (e.g. social worker, psychologist, palliative care, specialist nurse in organ donation / transplant coordinator, religious leader etc.) (**collaborative**)
- Making a collaborative, multi-disciplinary approach with the treating clinical team and the transplant coordinator as a team and others as deemed appropriate, e.g. faith representatives. (**collaborative**)

Palliative care is a key support structure in all end-of-life care pathways in the Western Cape. They are able to assist in advanced care planning, counselling, supporting the dying process, pain and symptom control, and post death bereavement support. Integration of palliative care is done on a case-by-case basis and as a support to the treating clinical team. *(Refer to the Provincial Palliative Care Policy)*

Appendix 3 - NHSBT "Approaching the families of potential organ donors: - Best practice guidance"

7. Obtaining forensic pathology authorisation

After family consent the transplant coordinator will obtain authorisation to proceed with donation from Forensic Pathology Services in cases which meet the criteria for a forensic postmortem.

Forensic Pathology Services aim to support organ and tissue donation wherever possible and in circumstances where the potential donor's death is required to have a forensic post-mortem will inform the transplant coordinator of specific requests and limitations, they wish to apply related to organ and tissue recovery.

Legally all patients that meet the criteria for a forensic postmortem must be discussed with the forensic pathologist on call and authorisation to proceed with donation must be obtained prior to the organ recovery operation. This is done by the transplant coordinator after family consent.

Any discussions around the necessity for a forensic postmortem should involve the forensic pathologist on call whenever it is unclear whether a forensic postmortem is needed.

The consent required prior to organ and tissue donation is very specific and is done by or coordinated by the transplant coordinator. Therefore, the transplant coordinator must be fully informed as to the cause of death and the need for a forensic postmortem.

The on-call forensic pathologist may have limited experience of organ or tissue donation. Where donation is declined by forensic pathology the transplant coordinator will engage to ensure the on-call forensic pathologist has discussed with senior cover or colleagues. This may be particularly relevant in sensitive situations including paediatric cases. Forensic support and input can and should assist in giving guidance in medico-legally complex scenarios. The

recovering surgical teams provide a report to forensics and can additionally supply biopsy material if needed.

8. Obtaining medical manager authorisation

The head of the medical institution where the organ recovery is taking place must give authorization for the organ recovery in alignment with the principles of the National Health Act 61 of 2003.

The institution must maintain a record of donation activity and the medical manager will check that family consent has been obtained, and that forensic authorisation is in place in cases of an unnatural death. An estimated theatre time is also supplied.

The on-call medical manager may have limited experience of organ or tissue donation. Where donation is declined the transplant coordinator will pursue clarification to ensure the on-call medical manager has discussed with more senior cover where available and that it is a fully informed decision that will deprive a donor the right to donate has been made with all avenues to support donation fully explored. Inclusion of missed donation opportunities should be included in hospital morbidity and mortality meetings to allow review of donor statistics, constructive debriefing and quality improvement initiatives.

9. Care of the organ donor

Most organ donors need intensive support to maintain the option of successful transplant outcomes. This requires the treating team to actively manage the potential donor and consented donor. It is only in cases of a consented brain-dead donor that the transplant team can legally take over donor management in accordance with the National Health Act 61 of 2003. Donor care is preferentially delivered in the hospital where the donor is identified, given the risk of donor loss in transfer, proximity to the family and primary care team. Support for these complex cases will always be provided, and additional critical care expertise (donation support physician) can be sourced to assist in all consented donors within the healthcare ecosystem.

10. Organ and tissue donation after brain death (DBD)

All patients in whom brain death is suspected should have brain death testing performed. Brain death testing should be completed in keeping with the guidelines from the Critical Care Society of Southern Africa published in the South African Medical Journal and South African Journal of Critical Care, the South African Guidelines on the Determination of Death.

[South African guidelines on the
1:37\(1\):10.7196/SAJCC.2021v37i1b
PMCID: PMC10193841](#)

If the family agree to donation, then supportive care should be continued to optimise organ function with the implementation of the Donor Optimisation Extended Care Bundle. Decisions around cardiopulmonary resuscitation should be made clear.

Appendix 4 – South African Determination of Death Checklists (SAMJ)

Appendix 5 – Donor optimisation extended care bundle.

11. Organ and tissue donation after circulatory death (DCD)

After an independent decision by the treating team to palliate a patient on mechanical ventilatory support with compassionate extubation and in alignment with principles of palliative care, it is appropriate to consider organ donation in cases where death is felt likely to be imminent.

If the family consents to donation, then supportive care should be continued whilst the preparations are made for the organ retrieval. Withdrawal of supportive care should occur in accordance with unit practice and with two doctors immediately available to certify death. Death following cardiorespiratory arrest should be confirmed in accordance with the guidance from the Critical Care Society of Southern Africa published in the South African Medical Journal and South African Journal of Critical Care, the South African Guidelines on the Determination of Death. Only after the declaration of circulatory death by two doctors independent of the transplant team (and one with more than 5 years' experience) the transplant team can proceed with organ recovery.

If lung donation is being considered and the patient has been extubated as part of their withdrawal of non-beneficial treatment, then the trachea should be re-intubated following confirmation of death. This may be done by either the certifying doctor or the retrieval team. Lungs should not be re-inflated until 10 minutes after confirmation of death. At this point one recruitment breath may be given and the lungs held open with CPAP. No cyclical ventilation should commence until cerebral circulation is isolated. This is in accordance with best ethical practice and the transplant coordinator and thoracic retrieval team will advise as necessary. Given donation after circulatory death is relatively unfamiliar at present, a focus on training, advocacy and debriefing is facilitated by the transplant coordinator for every case.

Appendix 6 – Donation after circulatory death pathway.

12. Tissue only donation

Patients who die in the community, or a hospital (without mechanical ventilation in their end-of-life care pathway), or in a hospice can and should also be offered the option to support donation. This is supported through non-profits such as the Vitanova Tissue Bank and the Centre for Tissue Engineering who work closely with funeral homes, forensic pathology services and hospitals.

In these cases, tissue donation which can include the donation of skin, bone, tendons, cartilage, heart valves and corneas is possible. Unlike organ donation, tissue donation does not need a predicted or anticipated death as there is no need to preserve metabolic function of the retrieved tissue up until recovery. Tissue donation can take place up to 12 - 72 hours after death and can therefore be considered for each and every death.

There is a great clinical demand for tissue donation; heart valves and skin can be lifesaving while other tissues tend to be greatly life-enhancing. At present, there is a massive ongoing unmet clinical demand for these tissues in the Western Cape and South Africa.

Families of all deceased patients in Western Cape should be offered the option of tissue donation as a usual part of end-of-life care planning and support after death.

A tissue donation leaflet should be given to all families alongside the bereavement information on how to register a death. Where appropriate the leaflet should initiate discussion about the option of tissue donation. Where families are in support of tissue donation the on-call tissue donation coordinator (TDC) can be contacted 24/7.

There is a close working relationship between the transplant coordinators and the tissue donation coordinators which means that when a patient is proceeding to become an organ donor, tissue donation is also considered and progressed when able. However, when tissue only donation is being considered the tissue donation service will be engaged directly.

As with organ donation if there is an indication for referral to the forensic pathologist this authorization is secured before tissue donation can proceed.

13. Theatres

a) Operating theatre responsibilities

The donor hospital will provide a fully equipped operating theatre for the retrieval procedure, including appropriate anaesthetic equipment and medications to support the donor.

The donor hospital will provide trained theatre staff who are familiar with the theatre facilities including access to the equipment, instruments and medications required by the retrieval team. These individuals will remain in theatre during the retrieval procedure to provide assistance to the transplant coordinator and retrieval team.

The donor hospital is responsible for the safe transfer of the donor to the operating theatre.

The donor hospital will provide an anaesthetist and anaesthetic floor nurse to provide care for the donor during the retrieval operation where the donation is following brain death (DBD).

The donor hospital will provide a scrub sister and floor nurse for the organ recovery operation in cases of donation following brain death or circulatory death determination.

It is the responsibility of the theatre staff to ensure that local theatre policies are adhered to, and appropriate legal documentation is completed for theatre records.

The transplant coordinator will be present throughout the retrieval operation to help ensure smooth running of the retrieval process and to support the theatre staff.

Where the family has requested to view the deceased following the donation, this should be arranged jointly by the transplant coordinator, theatre staff and/or critical care staff to arrange an appropriate area for the family to spend time with their loved one. Following this the deceased should be transferred to the mortuary, as per hospital policy.

b) Theatre booking and priority

Theatre should be made aware that there is the possibility of an organ donor at the earliest opportunity.

Solid organ retrieval and transplant implantation procedures should be considered as emergency cases. Organ donation should take priority over elective work where there is a conflict. If there is any doubt, the final decision should be made by the medical manager of the institution after discussion with the transplant coordinator and surgical teams given that recipient operations are being concurrently scheduled.

For organ retrieval, it is the responsibility of the surgical recovery team to book the case in theatre and speak to the on-call anaesthetist and to liaise regarding the timing of the retrieval.

c) Donation following circulatory death (DCD)

DCD occurs following the withdrawal of non-beneficial treatment and confirmation of death by loss of cardiac and respiratory function. Ischaemic injury to the organs occurs very rapidly and therefore the retrieval process must begin as soon after confirmation of death as possible. The patient may have treatment withdrawn in theatre or the ICU if the family wishes to be present, in which case the patient is required to be rapidly transferred to theatre after death. The accurate timing of the organ retrieval is difficult to predict in the DCD setting as the palliative care process happens independently and no action is taken to hasten the dying process.

An anaesthetist may be required for the DCD organ retrieval if lung retrieval is considered. The most appropriate fully equipped and staffed theatre should be identified and coordinated to be available for the retrieval process at the time point of compassionate extubation. Theatre teams will stand down and DCD organ recovery will be deemed not possible if after a prespecified period of time (usually 1 hour from the point of functional warm ischaemic time) it is felt the donor will not arrest. Palliative care will then continue in a pre-arranged location, by the treating team, outside of theatre and the family informed that organ donation was not possible (which was already discussed as part of informed consent). The referring treating team continues the palliation and comfort care in this setting as the donor would have only been handed over to the transplant team for organ recovery after a declaration of circulatory death.

d) Donation following brain death (DBD)

DBD occurs following confirmation of death by neurological criteria. The heart will still be beating, and the patient will be receiving mechanical ventilation. It will usually be possible to provide an approximate time for organ retrieval and an anaesthetist is required for DBD organ retrieval.

DBD organ retrieval usually takes place in the emergency theatre though the most appropriate fully equipped and staffed theatre should be identified and made available for the retrieval process which can take a few hours.

If conflict occurs with other emergency cases, appropriate prioritisation should be made by the Consultant Anaesthetist and Consultant Surgeons involved and the opening of another theatre should be considered.

14. Donation from the Emergency Department and Trauma Unit

Patients dying in the Emergency Department and Trauma Unit should be given the same opportunity to donate organs and tissues as those in Critical Care units. The Emergency Department (ED) and Trauma units should maintain an awareness of organ and tissue donation as a possibility in appropriate ED and trauma unit patients who have been assessed as having a non-survivable diagnosis. It is recommended that organ and tissue donation be viewed as an integral part of end-of-life care in the ED and trauma unit. Suitable individuals and their families should have the opportunity to donate organs and tissue if this is their decision. Exploring this decision must be facilitated by the treating team by including and supporting the transplant coordinator whenever possible.

Mechanically ventilated patients deemed to have a devastating brain injury i.e. from traumatic head injury or haemorrhagic or ischaemic stroke may go on to become organ donors. Such patients should be discussed with neurosurgery in the first instance but if "not for neurosurgical intervention," they should be referred to the transplant coordinator when a decision has been made to withdraw life sustaining therapy in accordance with palliative care principles. Organ donation should be considered prior to any withdrawal of treatment.

Patients who die in the ED or trauma unit and who are not suitable for organ donation may still be considered for tissue donation (see tissue donation, section 7). The criteria for life enhancing tissue donation differs from that of organ donation. (Appendix 7). The on-call Tissue Donor Coordinator (TDC) is contacted by phone to determine the potential donor's suitability and if they had registered their decision with the Organ Donor Foundation. The approach to the family is then made either by the Tissue Donation Coordinator, local transplant coordinator or by the ED or trauma unit medical staff. Specific bloods are required and blood tubes for this purpose are available in all ED's and supported through donation cost centres. It is important that the deceased patient is transported to the mortuary timeously.

15. Critical care capacity

Care of the consented brain-dead organ donor will normally be provided within the ICU if the hospital has an ICU.

If there is an intubated patient outside of ICU who is being considered for organ donation after a declaration of brain death, then the following should occur in order:

- Care should continue in the current location by the treating team, in liaison with the ICU team and the transplant coordinator whilst ICU resources are explored.
- ICU staffing should be explored to enable the patient to be cared for within the hospital's ICU.
- If an ICU bed is not available, the transplant coordinator will continue management in a dedicated area of the hospital with suitable monitoring and ICU team support.
- Every effort will be made to move the brain-dead consented donor out of the ED or trauma unit in cases where needed operationally and when organ retrieval is not imminent.

Consented potential donors after circulatory death are not transferred to ICU as the end-of-life care and palliation is typically overseen by that treating team and not handed over to another team.

16. Training

Regular training for medical and nursing staff will be performed in the following areas by the resident transplant coordinators or supporting medical staff:

- Critical Care
- Anaesthetics
- Emergency Department
- Trauma Unit
- Theatres
- Paediatric facilities
- Palliative Care
- Other

17. Quality improvement, reporting, and policy oversight

The transplant coordinators will engage with referring staff after both successful and unsuccessful organ donor referrals to offer engagement and education around the particular case and create feedback loops to allow quality improvement and constructive discussion to improve processes that are complicated and unfamiliar to most. This will also offer an opportunity for staff to debrief and have any concerns answered.

All missed potential solid organ donors are audited and will be discussed by transplant coordinators and supporting medical staff at unit morbidity and mortality meetings. If issues are identified feedback and further training will be given to those units.

A hospital Organ and Tissue Donation committee will provide oversight of the local policy and an additional forum for engagement with multidisciplinary input from all role players on a regular basis.

18. Equality and diversity

It is important that no community is excluded from being offered the option to donate. Long term outcomes are improved by having a large pool from which to allocate organs.

Donation is possible from all communities and cultures in South Africa. Equity of access to donation support pathways is a crucial component of healthcare delivery across the entire healthcare ecosystem. It is important to create an environment where open discussion can be held to answer all queries related to donation as it is a source of solace to families to donate. It is shown that having discussions about donation as part of end-of-life care assists families independent of their ultimate decision regarding donation. There is an ethical need to offer each family the ability to make a fully informed decision about donation and not to have conversations ended prematurely, prior to adequate information being provided.

For this reason, it is important that the specialist requestor, in our case a transplant coordinator, is brought in to handle these discussions as they are in a position to spend as much time as is needed by the family to come to an informed decision.

They should be introduced as part of the team who help in end-of-life care situations by the treating team. There must be a planning meeting prior to this introduction to fully discuss each donor and family situation.

The transplant coordinator will always start discussions by assessing the families understanding of the situation and for this reason it is important that the treating team is present for the first part of the discussion in order to answer any misunderstandings that may have arisen out of prior communication. The consent rate is a reflection of both the transplant coordinator and the treating team's engagement around end-of-life communications.

It is important for the treating team to forewarn the family about the poor prognosis before communication of a brain death diagnosis or a decision to palliate and compassionately extubate the patient. Making the transplant coordinator aware of prior communications and including them in the scheduling of subsequent discussions is key to supporting the family appropriately and giving them time to come to terms with the situation.

The transplant coordinators are committed to promoting awareness amongst all communities directly regarding the need for organ donor registration through outreach activities and are also able to help increase clinicians' understanding of religious and cultural needs with families from all communities at the end-of-life.

In accordance with equality and diversity requirements it is essential that potential donors from all ethnic backgrounds are referred to the transplant coordinator. Paediatric donors (<18 years) are preferentially allocated to paediatric recipients (<18 years). It should not be assumed that because a patient or their family come from a specific ethnic, cultural or spiritual background that referral for donation should not be made. The responsible Consultant and transplant coordinator may explore including a faith representative or hospital chaplain to support the families during these discussions.

19. Media

The hospital and provincial public relations officer/s will support organ and tissue donation by promoting awareness of the donation process. National and local campaigns to encourage people to sign up to the organ donation register through all available media channels.

There is a strong commitment to work proactively with both broadcast and print media to support organ donation and also to raise awareness of any issues that are raised relating to hospital processes.

The donor family's privacy and that of the recipients is always protected and paramount. The transplant coordinator coordinates anonymous "thank you" letters to the donor family and offers bereavement support where appropriate.

We do not actively publicise donor details although the family may wish to do so themselves. A period of time should pass prior to public relations engagements to ensure publicity is remote to an acutely grieving family and possible adverse transplant outcomes.

20. Implementation and support

The province has constituted a Provincial Organ and Tissue Donation Committee to oversee the Western Cape Organ and Tissue Donation Policy implementation, monitoring and review. The Committee will provide a consultative forum to address quality, safety, efficiency, and other governance issues arising within organ and tissue donation services in the Western Cape. The Committee will oversee and monitor provincial activities relating to organ and tissue donation. The Committee will constitute working groups to support implementation and system improvements from this policy.

The Terms of Reference and membership of the committee has been finalised and can be provided upon request.

The Committee will provide a report to the Provincial Health Department leadership annually. The report will include a summary of the province's donation activity from the previous year along with an executive report detailing system improvements and any action areas for improvement or learning.

21. Media

The hospital and provincial public relations officer/s will support organ and tissue donation by promoting awareness of the donation process. National and local campaigns to encourage people to sign up to the organ donation register through all available media channels.

There is a strong commitment to work proactively with both broadcast and print media to support organ donation and also to raise awareness of any issues that are raised relating to hospital processes.

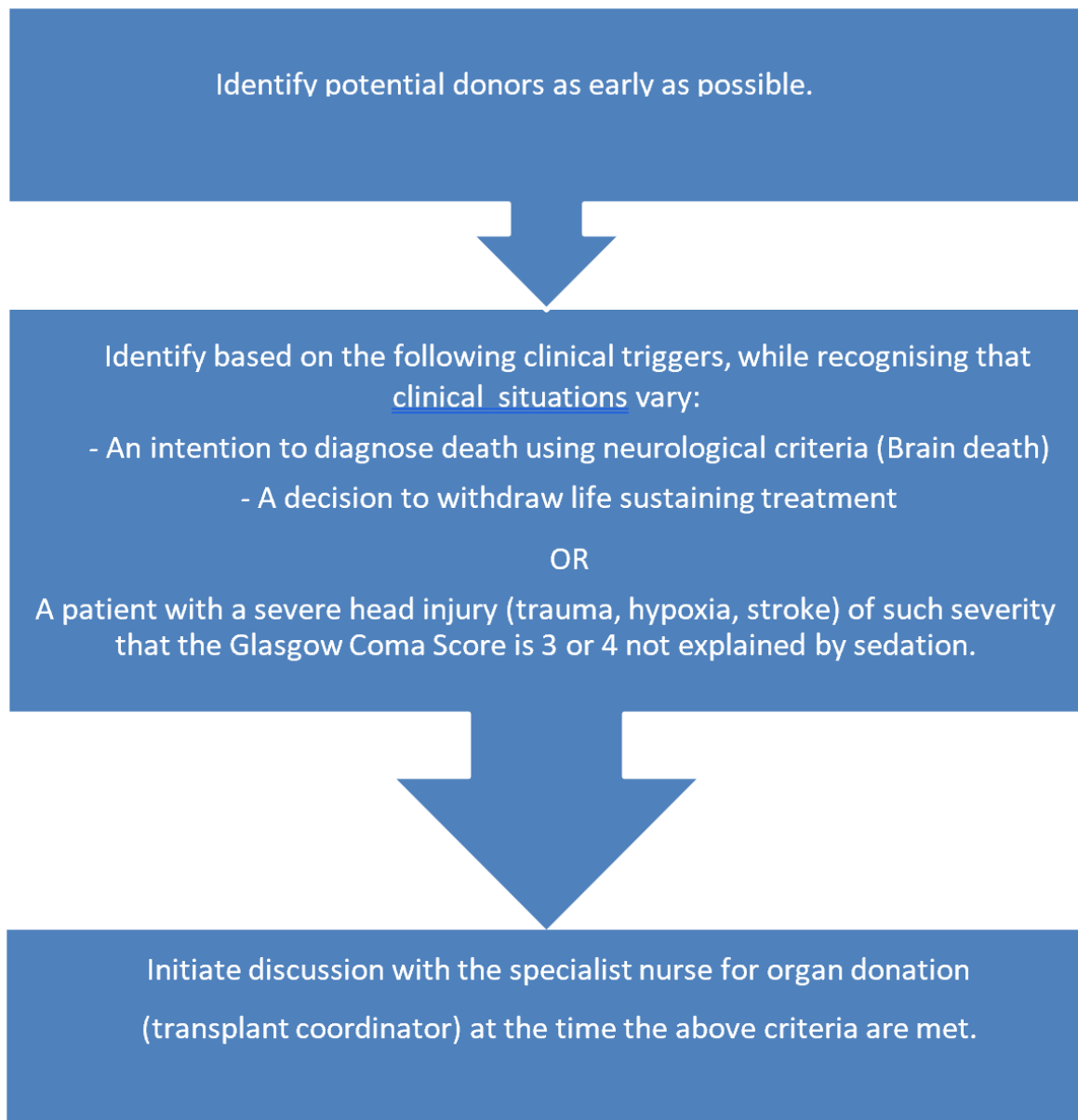
The donor family's privacy and that of the recipients is always protected and paramount. The transplant coordinator coordinates anonymous "thank you" letters to the donor family and offers bereavement support where appropriate.

We do not actively publicise donor details although the family may wish to do so themselves. A period of time should pass prior to public relations engagements to ensure publicity is remote to an acutely grieving family and possible adverse transplant outcomes.

22. References

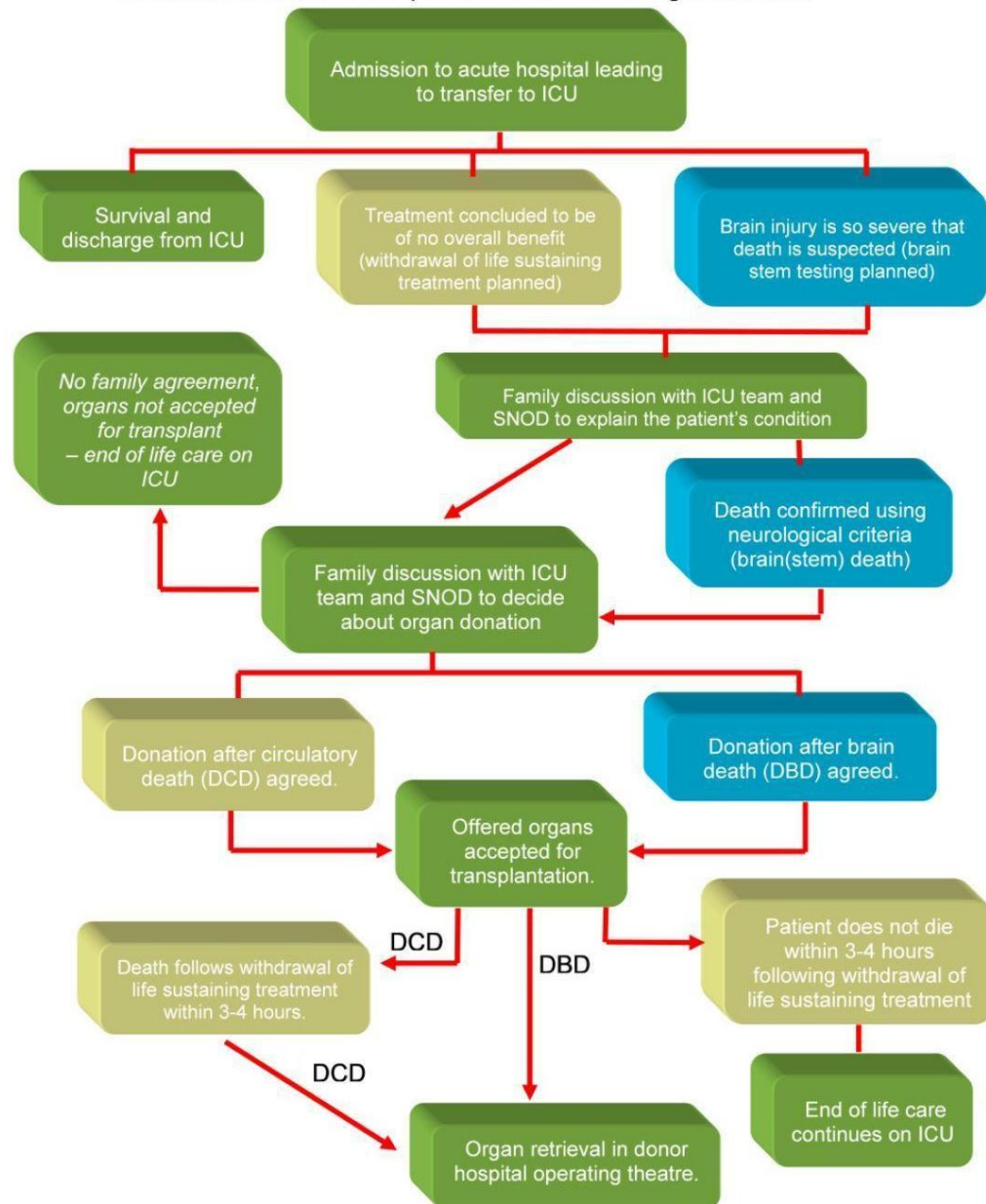
1. https://www.gov.za/sites/default/files/gcis_document/201409/a61-03.pdf
2. http://odt.nhs.uk/pdf/family_approach_best_practice_guide.pdf
3. http://www.odt.nhs.uk/pdf/dbd_care_bundle.pdf
4. Thomson, D., Joubert, I., & de Vasconcellos, K. South African guidelines on the determination of death. South Afr J Crit Care 2021; 37 (1b): 41-54.

23. Appendix 1: Donor identification and referral pathway



24. Appendix 2: Deceased donor pathway in an ideal setting with access to ICU

Schematic overview of the process of deceased organ donation.

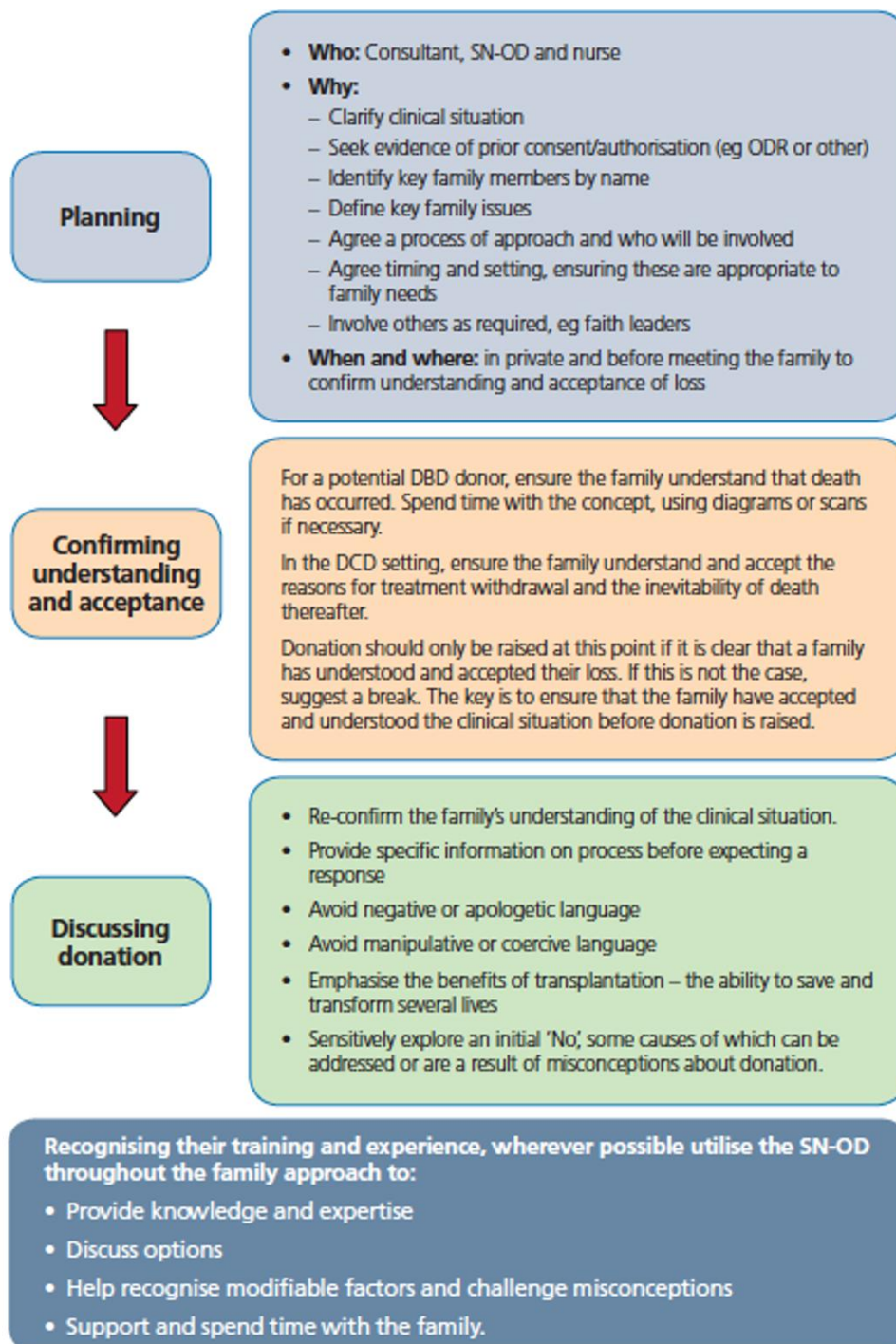


TC – Transplant coordinator = SNOD – specialist nurse in organ donation

DCD – Donor after circulatory death

DBD – Donor after brain death

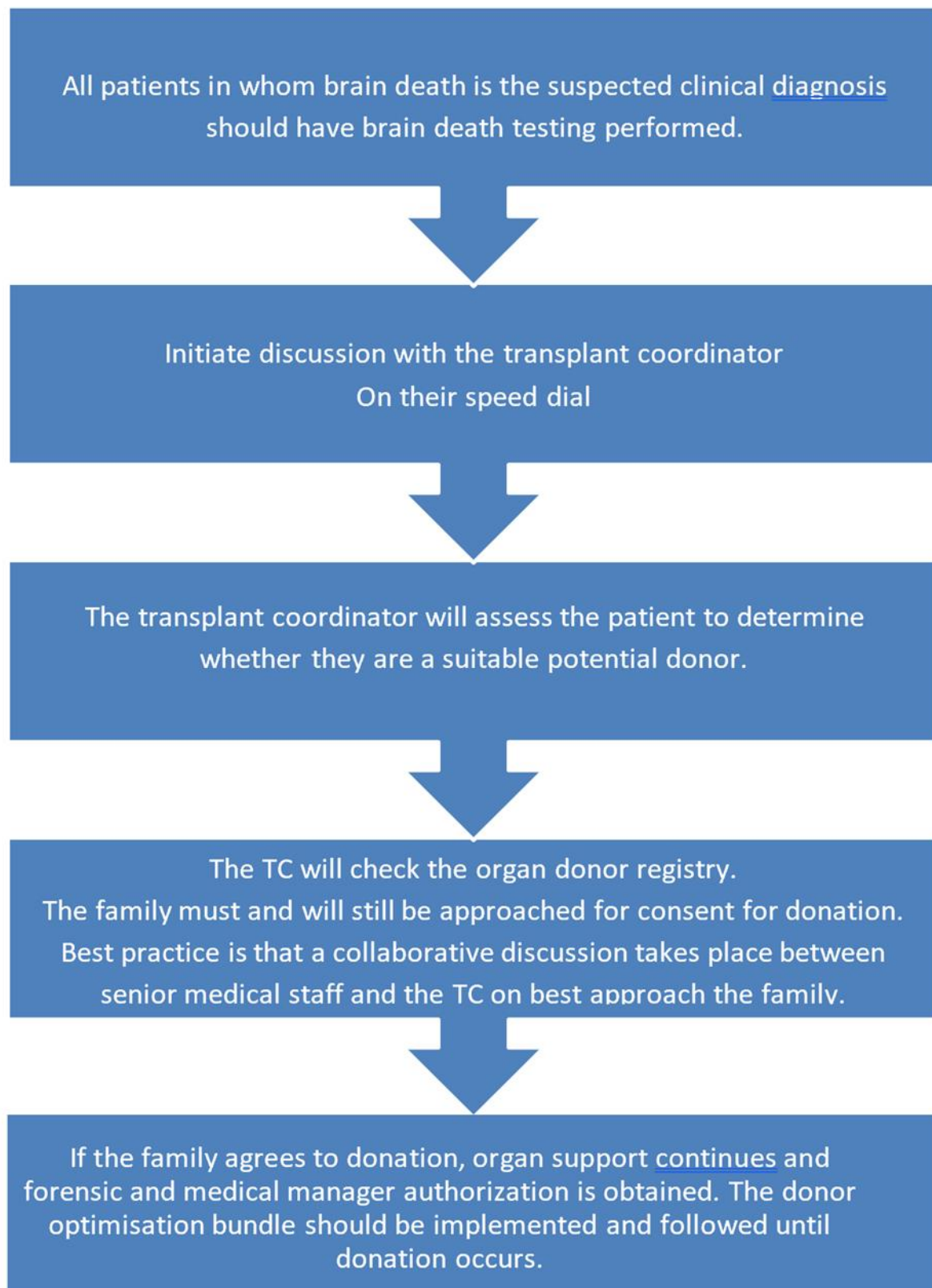
25. Appendix 3: Approaching the family of a potential donor



TC – Transplant coordinator = SNOD – Specialist nurse in organ donation

Adapted from Approaching the families of potential organ donors: Best practice guidance (2013). NHSBT.

27. Appendix 5: Donation after Brainstem Death DBD/DNC pathway



28. Appendix 6: Donor optimisation extended care bundle

GENERAL CARE AND SUPPORT	
☐	Instrumentation: endotracheal tube, nasogastric tube, urinary catheter, multilumen central line, arterial line, large peripheral intravenous line
☐	Monitoring: ECG, pulse oximetry, intra-arterial pressure, core temperature, urine output (hourly)
☐	Routine investigations: CXR, ECG, cardiac echo, blood group, coagulation
☐	Regular blood testing: FBC, ABG, U&Es, LFTs q6h and as necessary blood gases
☐	Review all medications; cease unnecessary orders, continue antibiotics and thromboprophylaxis
☐	Maintain core temperature 36–37°C
☐	Eye care, mouth care and all other nursing care
RESPIRATORY SUPPORT	
☐	Aim for S_pO_2 92–97% by FiO_2 (less than 0.6) and PEEP adjustment (minimum PEEP 5cm H_2O)
☐	Aim for P_aCO_2 35–45 mmHg by ventilator adjustment (TV 6–8mL/kg ideal body weight, plateau pressure <30 mmHg)
☐	Regular turns from side to side; elevate head of bed 30° - 45°
☐	Regular tracheal suctioning, chest physio, lung recruitment after apnoea test, bronchoscopy and further recruitment required
CARDIOVASCULAR SUPPORT	
☐	Ensure patient is neither hypovolemic nor fluid overloaded
☐	Aim for MAP 60–80 mmHg, good peripheral perfusion, normal (or decreasing) lactate
☐	Aim for urine output ~1 mL/kg/hr (range 0.5–3 mL/kg/hr)
☐	Commence vasopressin (infusion of 0.01 to 0.1 units/minute) or noradrenalin if needed
☐	Fluid resuscitation with boluses of 500ml of Ringer's lactate or Plasmalyte should be guided by positive response to passive leg raise or strokevolume variation or pulse pressure variation
FLUIDS AND METABOLIC MANAGEMENT	
☐	Give maintenance fluid as low sodium fluids e.g. glucose 5% 1mL/kg/hr, continue NG feeding or TPN, insulin infusion to keep blood glucose between 6 - 10 mmol/L
☐	Give methylprednisolone 15mg/kg stat
☐	If polyuric >300 mL/hr assume diabetes insipidus and immediately give desmopressin (DDAVP) as a bolus 1 to 4 micrograms stat followed by 1 - 2 micrograms intravenously 6 hrly
☐	If Na >150 mmol/L give additional free water
☐	Aim for Hb>70g/L

Adapted from NHSBT Donor optimisation extended care bundle taken from http://www.odt.nhs.uk/pdf/dbd_care_bundle.pdf

29. Appendix 7: Donation after Circulatory Death (DCD) pathway



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South African guidelines on the determination of death

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Death is a medical occurrence that has social, legal, religious and cultural consequences requiring common clinical standards for its diagnosis and legal regulation. This document compiled by the Critical Care Society of Southern Africa outlines the core standards for determination of death in the hospital context. It aligns with the latest evidence-based research and international guidelines and is applicable to the South African context and legal system. The aim is to provide clear medical standards for healthcare providers to follow in the determination of death, thereby promoting safe practices and high-quality care through the use of uniform standards. Adherence to such guidelines will provide assurance to medical staff, patients, their families and the South African public that the determination of death is always undertaken with diligence, integrity, respect and compassion, and is in accordance with accepted medical standards and latest scientific evidence. The consensus guidelines were compiled using the AGREE II checklist with an 18-member expert panel participating in a three-round modified Delphi process. Checklists and advice sheets were created to assist with application of these guidelines in the clinical environment (<https://criticalcare.org.za/resource/death-determination-checklists/>).

Key points

- Brain death and circulatory death are the accepted terms for defining death in the hospital context.
- Death determination is a clinical diagnosis which can be made with complete certainty provided that all preconditions are met.
- The determination of death in children is held to the same standard as in adults but cannot be diagnosed in children <36 weeks' corrected gestation.
- Brain-death testing while on extra-corporeal membrane oxygenation is outlined.
- Recommendations are given on handling family requests for accommodation and on consideration of the potential for organ donation.
- The use of a checklist combined with a rigorous testing process, comprehensive documentation and adequate counselling of the family are core tenets of death determination. This is a standard of practice to which all clinicians should adhere in end-of-life care.

S Afr Med J 2021;111(4b):367-380. <https://doi.org/10.7196/SAMJ.2021.v111i4b.15200> | *South Afr J Crit Care* 2021;37(1b):41-54. <https://doi.org/10.7196/SAJCC.2021v37i1b.466>

Death is a medical occurrence that has social, legal, religious and cultural consequences requiring common clinical standards for its diagnosis and legal regulation.^[1] There is no documented case of a person who fulfils the preconditions and criteria for brain death ever subsequently developing any return of brain function.^[2,3]

Clear medical standards for death certification augment the quality and rigor of death determination.^[4-6] Currently there are no clinical guidelines on death determination in South Africa (SA), with clinicians using available international guidelines, which vary markedly and are not always applicable to the SA context.^[7-10] The World Federation

of Societies of Intensive and Critical Care Medicine (WFSICCM) recently published a document, based on current literature, aiming to standardise terminology and establish minimum testing standards across the world.^[5] In order to be applicable over a range of legal jurisdictions, some sections of this document are necessarily broad, making application to the local context challenging. There is therefore a need to align the WFSICCM document with the SA context and legal system, and to provide clear guidance to SA practitioners and the public.

The aims of this document are to provide guidelines for healthcare providers for the testing process, and to answer key questions that commonly arise when medical professionals are called upon to determine death in SA. The objectives are to:

- provide clear medical standards for healthcare providers in the determination of death, in order to promote safe practice and avoid diagnostic errors in the determination of death
- provide assurance to patients, their families and the SA public that determination of death is undertaken with diligence, integrity, respect and compassion and in accordance with accepted medical standards and societal expectations
- create a checklist for brain death and circulatory death determination.

Methods

An expert panel was constituted and reviewed current evidence and compiled consensus-based recommendations for the minimum standards required for death determination, applicable in the SA context, using a modified Delphi process. All statements in this guideline are considered strong recommendations. In cases where evidence is less strong, the term 'suggested' is used to indicate a lower level of evidence. Statements are intentionally similar to international documents in the interests of commonality of approach. Recommendations are aligned with the best interests of patients and their families.

Panel recruitment

Expert panel members were recruited through the Critical Care Society of Southern Africa (CCSSA) as experts in the field of death determination and end-of-life care representing a broad range of disciplines within critical care (neurosurgery, paediatrics, obstetrics and gynaecology, surgery, anaesthesiology, nursing, ethics). Informed consent was given by the expert panel with acknowledgement that they would be identifiable in the publication (Appendix 1 - 3: <http://samj.org.za/public/sup/15200-1.pdf>). The study methodology was approved by the University of Cape Town Human Research Ethics Committee (HREC 476/2019).

Delphi process

A literature search of Pubmed, Web of Science, SCOPUS and the grey literature was conducted by the steering committee of the expert panel and a professional librarian to identify national and professional society guidelines in death determination, and their supporting evidence, for review (<http://samj.org.za/public/sup/15200-2.pdf>). A modified Delphi approach was used with adherence to the Appraisal of Guidelines for Research and Evaluation (AGREE II) instrument to compile the guidelines.^[11]

Key questions related to current principles and practice regarding death certification were drafted by the meeting steering committee. During three rounds of a modified Delphi process^[12] using SurveyMonkey (SVMK Inc., USA), a web-based application, the expert panel progressively modified, deleted or added questions and components. Participants were asked to rate agreement with each component between 1 and 9 on a Likert scale, with 1 - 3 being 'not important', 3 - 6 being 'important but not critical' and 7 - 9 being

'critically important', or state if they were unable to comment. This scale is recommended by the Grading of Recommendations Assessment, Development and Evaluation working group.^[11,13] Participants were invited to suggest additional questions for consideration for the round table in each round using free-text responses. Although five to 10 experts are considered adequate for content validation, we used 18 to ensure a broadly representative group.^[14]

Precise terminology was reviewed and finalised in order to improve clarity in death determination discussions and debate. We defined consensus for the Delphi *a priori* based on guidance in *The COMET Handbook*.^[15] For inclusion in the consensus outcome statement (COS), outcomes required at least 70% of participants in each stakeholder group to score the outcome as critically important and <15% to score the outcome as not important. Outcomes excluded from the COS required at least 70% of participants in each stakeholder group to score the outcome as not important and <15% to score the outcome as 'critical'. If outcomes did not meet either criterion, they were presented at the round-table for discussion. The expert panel members summarised the responses and the available evidence and formulated draft recommendations that were presented and discussed at a face-to-face round table meeting prior to the CCSSA national conference in Cape Town, SA, in 2019. The results were presented at a plenary meeting of the conference.

All questions and feedback from this meeting were then reviewed with the steering committee. The combined consensus statement and proposed guideline were circulated to all society members for comment over three months. All expert panel members approved the final document for publication. The guidelines were then submitted for external review and endorsement by other South African professional medical societies (Table 1).

Updating of the guidelines

A review period of five years was set after publication of this document, the review to be undertaken by the guideline development group of the CCSSA, unless an earlier revision is mandated by emerging high-quality medical evidence or legislative changes.

South African guidelines on the determination of death

1. General statements

Death is the clinical point of irreversible loss of the capacity for consciousness and the irreversible loss of the capacity to breathe.^[16]

Death is determined by either neurological or circulatory criteria and must be made in accordance with accepted medical standards.^[1]

Table 1. List of endorsing societies

Trauma Society of Southern Africa
The Association of Surgeons of South Africa
Radiology Society of South Africa
Southern African Transplantation Society
Islamic Medical Association
South African Medical Association
Resuscitation Council of South Africa
South African Thoracic Society
Colleges of Medicine of South Africa - Committee of Critical Care
Neurological Association of South Africa
Society of Neurosurgeons of South Africa
South African Paediatric Association (Affirmation of Value)

A correctly performed clinical examination can determine the point of death with complete certainty. It is not necessary to wait for pathognomonic signs of death (hypostasis, rigor mortis, decay) to be present in the healthcare setting.^[16,17]

In cases where there are confounders affecting the clinical examination, testing should be deferred until such confounders are resolved or ancillary testing confirms the diagnosis of death.^[17]

2. Definitions

Ancillary test: an additional test that can assist with the clinical diagnosis of brain death.^[18]

Accommodation: a period of somatic support to allow for family to process the diagnosis of death.

Brain death/death by neurological criteria: the preferred term when death is determined on neurological grounds. (This is in preference to the term brain-stem death and is in alignment with the latest guidelines.)^[5]

Circulatory death/death by circulatory criteria: the preferred term when death is determined on circulatory grounds. (This is in preference to terms such as cardiac or cardiorespiratory death and is in alignment with the latest guidelines.)^[1]

Coma: the absence of wakefulness, awareness and the capacity for sensory perception and responsiveness to the external environment.^[1]

Confounder: a situation during which a diagnostic test may be unreliable. Repeat testing when the confounder is no longer present, or ancillary testing, is required to diagnose death in these settings.

Somatic support: management to support the body and organs, excluding the brain, after brain death has been confirmed.

3. Brain death

3.1 Preconditions for brain death testing

There must be an established aetiology compatible with complete and irreversible loss of all brain function prior to commencement of the determination of brain death.

Preconditions for brain-death testing to be valid are:

- a minimum temperature of 36°C^[2,5]
- a systolic blood pressure (BP) ≥100 mmHg or a mean arterial pressure (MAP) ≥60 mmHg (or age-appropriate haemodynamic targets in the paediatric population)
- exclusion of the influence of central nervous system (CNS) depressing drugs. This is done by at least one of the following:
 - allowing five elimination half-lives of the drug to pass before making an evaluation of brain death (taking into consideration the dose and the elimination half-life, which may be influenced by age, organ dysfunction or prior hypothermia)^[5]
 - administering an appropriate drug antagonist
 - measuring drug levels^[5]
 - performing ancillary testing in addition to a complete clinical examination if there is concern about prolonged or unknown drug elimination
- intact neuromuscular function
 - If doubt exists regarding the effects of pharmacological paralysis, a train-of-four nerve stimulator can be used. A normal response to stimulation should be observed.^[3]
- correction of severe metabolic, acid-base and endocrine derangements that could affect the examination
 - If these derangements cannot be corrected and are judged to be potentially contributing to the loss of brain function, ancillary testing must be used after a clinical examination of brain death has been completed.

3.1.1 Additional remarks: Preconditioning^[5]

Neuroimaging is recommended when available to assist with establishing an aetiology and confirming brain injury.

Neuroimaging is not required for the diagnosis of brain death where an obvious cause is known.

Drug levels should not exceed the therapeutic range and, if within a therapeutic range, not be associated with an altered level of consciousness.

Barbiturates have long and variable elimination half-lives, so blood levels should be measured and documented to be below that of clinically significant effects (<10 mg/L) or alternatively ancillary testing should be used to diagnose brain death.

If there is concern about severe alcohol intoxication being the primary cause of coma, the blood alcohol level should be shown to be <80 mg/dL.

Consensus was not achieved on a standardised observation period prior to brain-death testing. The appropriate observation period was felt to be specific to each individual case. However:

- in cases of anoxic brain injury, this observation period should be 24 hours
- in cases where therapeutic hypothermia was used, there should be a 24-hour period of normothermia prior to an examination for brain death
- in paediatrics, a cautious approach is advised, given the frequent presence of confounders in this group.

It must be possible to adequately examine the brain-stem reflexes. It must be possible to examine at least one eye and one ear and safely perform apnoea testing.^[3,5]

Apnoea testing may be precluded by severe hypoxic respiratory failure or a high cervical cord injury, in which case ancillary testing can be used after confirming that brain-stem reflexes are absent.^[3,5]

While trying to provide broad guidance on the magnitude of metabolic and endocrine disorders which are likely to influence the testing of brain-stem reflexes, it is essential to bear in mind that the most important factor is the establishment of an unequivocal cause for the individual's coma.^[17]

It is recognised that circulatory, metabolic and endocrine disturbances (e.g. hypotension, hypernatraemia, diabetes insipidus) are likely accompaniments of death as a result of cessation of brain-stem function. It is important to emphasise that these may be the effect rather than the cause of cessation of brain function and do not preclude the diagnosis of death by neurological testing.^[17]

It is recognised that:^[17]

- Sodium can cause unresponsiveness at levels <115 or >160 mmol/L.
- Serum potassium levels can cause flaccid paralysis below 1 mmol/L. Therefore, we recommend a level >3 mmol/L be confirmed.
- Profound elevation or lowering of phosphate or magnesium can be associated with severe neuromuscular weakness at levels <0.5 or >3.0 mmol/L and may need to be corrected.
- Hyperglycaemia in diabetic ketoacidosis or hyperosmolar non-ketotic coma may cause a state of unresponsiveness which mimics irreversible cessation of brain-stem function, but this state is extremely unlikely with blood glucose levels <20 mmol/L.
- Severe hypoglycaemia is associated with coma or stupor and testing of brain-stem reflexes should not be undertaken if the glucose level is <4 mmol/L. As blood glucose concentrations change rapidly in critically ill patients, a blood glucose measurement should always be made immediately prior to the testing of brain-stem reflexes.

The following endocrine conditions are extremely rare and unlikely to co-exist in the presence of known primary pathologies. If there is any clinical reason to expect these disturbances, then it is obligatory to ensure appropriate hormonal assays are undertaken.

- Myxoedema may cause a deep unresponsive coma.
- An Addisonian crisis may be associated with severe neuromuscular weakness, causing an acute ascending paralysis or encephalopathy proceeding to coma.

3.2 Braindeath testing^[3,5,17]

The testing process comprises assessment for:

1. coma
2. absence of brain-stem reflexes
3. inability to breathe.

A detailed explanation of the testing process is outlined (Table 2).

Table 2. Clinical testing for brain death – coma, brain stem reflexes, apnoea test

Clinical testing for	Test and interpretation	Cautionary remarks
1. Coma	Test: Assess both centrally and peripherally. Apply deep pressure to all of the following: • the condyles at the level of the temporomandibular joints • the supra-orbital notch bilaterally • the sternal notch • all four extremities, via deep nail bed pressure. Interpretation: Noxious stimuli should not produce grimacing, facial muscle movement or a motor response of the limbs other than spinally mediated reflexes. Any non-spinal reflex response is incompatible with a brain death diagnosis.	A multilevel assessment of motor function is important to exclude focal lesions. The clinical differentiation of spinal responses from brain-mediated motor responses requires expertise. Consultation with an experienced practitioner is recommended if the origin of a response is unclear. Alternatively, if interpretation is unclear, ancillary testing is recommended.
2. Brain-stem reflexes	Testing of the brain-stem reflexes comprises examination of the cranial nerves: pupils, ocular movements, facial sensation and movement, pharyngeal and tracheal response. These are tested sequentially and bilaterally when possible. All brain-stem reflexes must be absent to determine brain death.	
2.1. Pupillary light reflex – cranial nerves II and III	Test: Shine a bright light into the eye and look for a pupillary constrictor response. Interpretation: There should be no pupillary response.	The pupils should not be pinpoint. Pupils do not have to be fully dilated. Anticholinergic drugs such as atropine can cause pupillary dilatation. Cataract or iris surgery is not a contraindication to clinical testing.
2.2. Corneal reflex – cranial nerves V and VII	Test: Touch the corneas with soft cotton wool or gauze and examine the eyes for blinking or a withdrawal response. Interpretation: No blinking or withdrawal response.	Touching the sclera is not sufficient. Remove contact lenses. Examine the cornea gently as it is easily damaged.
2.3. Response to pain in the trigeminal distribution – cranial nerves V and VII	Test: Apply pain over the trigeminal distribution with deep pressure over the supra-orbital nerve bilaterally and to the condyles at the level of the temporomandibular joints. Interpretation: No grimacing, facial muscle movement or motor response of the limbs other than spinally mediated reflexes.	In patients with spinal cord injuries, peripheral sensation and motor function can be lost and it is essential to adequately assess response to stimuli through brainstem-mediated sensation and motor response.
2.4. Vestibulo-ocular reflex – cranial nerves III, IV, VI and VIII	Test (cold caloric): Inspect the external auditory canal with an otoscope to confirm that the eardrum is visible. If the eardrum is not visible, the canal must be cleared before testing can occur. Elevate the head to 30° to align the semi-circular canal and generate a maximal response. Flush 50 mL of ice-cold water into the ear canal using a syringe. Hold eyelids open and observe for eye movement for a minimum of 60 seconds. Interpretation: No eye movement in response to the cold water; the eyes remain in the midline within the socket.	Presence of a ruptured eardrum does not invalidate the test. Fractures to base of skull or petrous temporal bone may obliterate the response on the side of the fracture. Testing should not proceed on that side if there is cerebrospinal fluid (CSF), blood or brain tissue in the external auditory canal.
Vestibulo-ocular reflex – cranial nerves III, IV, VI and VIII	Not required/recommended: Testing for the oculo-cephalic (head turning/doll's eye) reflex examines the same reflex pathways as cold caloric testing but is a sub-maximal stimulus and is not recommended. It may also aggravate a pre-existing cervical spinal injury.	

...continued

Table 2. (continued) Clinical testing for brain death – coma, brain stem reflexes, apnoea test

Clinical testing for	Test and interpretation	Cautionary remarks
2.5. Gag reflex – cranial nerves IX and X	Test: Stimulate the posterior pharyngeal wall, on both sides, with a tongue depressor or Yankauer suction. Interpretation: No gag response seen.	
2.6. Cough reflex – cranial nerve X	Test: Stimulate the tracheobronchial wall with a soft suction catheter. Interpretation: No cough response seen.	The efferent limbs for this reflex are the phrenic nerve and the innervation of the thoracic and abdominal musculature. Therefore, it cannot be assessed in patients with a complete high cervical injury.^[21]
3. Apnoea test	Only proceed with the apnoea test if all above reflexes are absent. Apnoeic oxygenation is used to demonstrate lack of ventilatory drive. This involves the supply of 100% oxygen to the trachea, without providing ventilatory assistance. Through mass-movement, oxygen reaches the alveoli, allowing for transfer to the blood. In the absence of ventilation, $p_a\text{CO}_2$ rises and stimulates the brain-stem respiratory centres, causing spontaneous breathing. As the $p_a\text{CO}_2$ rises, the ventilatory centre is maximally stimulated by a $p_a\text{CO}_2 > 60$ mmHg (8 kPa) and pH < 7.30. An attempt at breathing is defined as any respiratory muscle activity that results in abdominal or chest excursions or activity of accessory respiratory muscles. Test: Pre-oxygenate the patient with 100% oxygen for 10 minutes to allow for elimination of nitrogen and to prevent hypoxaemia during the test. Perform a baseline blood gas measurement. Disconnect the patient from the mechanical ventilator. Supply continuous oxygen via a T-piece (preferred) or through a catheter inserted through the endotracheal tube and placed above the carina. Observe continuously for any spontaneous breathing. At the end of the period without mechanical ventilation, apnoea must persist in the presence of an adequate stimulus to spontaneous ventilation, i.e. an arterial $p_a\text{CO}_2 > 60$ mmHg (8 kPa) and an arterial pH < 7.30. Take an arterial blood gas to document the rise in $p_a\text{CO}_2$ and change in pH. At the end of the test, reconnect the patient to the mechanical ventilator. Interpretation: No breathing effort is seen at any point. Apnoea testing should be aborted if: • spontaneous respirations are witnessed during apnoea testing • systolic blood pressure becomes < 100 mmHg or mean arterial pressure becomes < 60 mmHg despite titration of inotropes/vasopressors (age-appropriate targets for paediatrics) • there is sustained desaturation below 85% • an unstable arrhythmia occurs. It is suggested that prior to aborting the apnoea test because of cardiorespiratory instability, an arterial blood gas (ABG) be sent for testing. If the $p_a\text{CO}_2$ target is met, the apnoea test can be considered positive (consistent with brain death).	
	Throughout the procedure, monitor the patient's SpO_2. An option to minimise the time required for the $p_a\text{CO}_2$ to rise to the desired level, is to adjust the minute ventilation to mild hypercarbia ($p_a\text{CO}_2 \sim 45$ mmHg (6 kPa)) before disconnecting the patient from the ventilator. If using a catheter, supply oxygen at 4 - 6 L/min. The diameter of the catheter must be $< 70\%$ of the diameter of the endotracheal tube. When using the tracheal insufflation method, care should be taken to avoid high oxygen flows and wedging of the catheter insufflating oxygen as high intrapulmonary airway pressure may cause barotrauma. The period of observation to achieve an adequate threshold of stimulus of the respiratory centre is variable. Usually $p_a\text{CO}_2$ rises by ~ 3 mm Hg (0.4 kPa) for every minute of apnoea. If starting from normocapnia, the $p_a\text{CO}_2$ is likely to be > 60 mmHg (8 kPa) after 10 minutes. If this is not the case at 10 minutes and the patient is stable, wait a further 5 minutes and repeat the arterial blood gas. In patients with pre-existing hypercapnia, it is recommended to wait for a $p_a\text{CO}_2$ rise > 20 mmHg (2.7 kPa) above the chronic level, with a pH < 7.30. Failure of the $p_a\text{CO}_2$ to rise is most likely due to an inappropriately high oxygen flow rate via a tracheal catheter. If hypoxia occurs, 1 - 2 positive pressure breaths can be given, and apnoea testing continued. Adequate pre-oxygenation and recruitment usually avoids this problem.	

Two doctors are required to confirm the diagnosis of brain death. One of the doctors must have more than five years' experience (since qualification as a medical practitioner). Neither doctor must be involved with a transplantation team.^[19]

Two doctors should perform the brain death testing together. If testing is done separately, there is no need for a delay between the tests; however, it is strongly advised that a single apnoea test should be done by the two doctors together.^[5,20]

3.2.1 Additional remarks: Apnoea testing

We do not recommend using a spontaneous breathing mode and remaining connected to the ventilator during apnoea testing. Performing testing in such a manner typically requires turning off the ventilator's default safety features to prevent back-up apnoea ventilation. If a patient remains connected to a mechanical ventilator, auto-triggering can give a false impression that a person is breathing spontaneously. This occurs when the ventilator is set on a spontaneous breathing mode (pressure support ventilation) and either extrinsic or intrinsic factors generate sufficient change in airflow or negative pressure that exceeds the trigger threshold, leading to a mechanically delivered breath. Extrinsic causes for auto-triggering include excessive condensation in ventilator tubing, endotracheal tube leak, chest tubes, and random artefacts or noise in the ventilator circuit, while intrinsic causes include cardiogenic oscillation, especially in a hyperdynamic cardiovascular state.^[5]

3.2.2 Additional remarks: Clinical observations

Observations compatible with a diagnosis of brain death are:^[3]

- spinal reflexes (which can be spontaneous or elicited by stimulation)^[22,23]
 - extension-pronation movements of the upper limbs
 - nonspecific flexion of the lower limbs
 - presence of deep tendon reflexes
 - plantar responses, either flexor or extensor
 - respiratory-like movements (shoulder elevation and adduction, back arching or intercostal expansion) without significant tidal volume
 - undulating toe reflex (plantar flexion of great toe, followed by brief plantar flexion sequentially of second to fifth toes)
 - Lazarus sign (bilateral arm flexion, shoulder adduction, hand raising to above the chest, and may include flexion of trunk, hips and knees)
 - head turning
- sweating, blushing, tachycardia
- normal blood pressure without the need for pharmacological support
- absence of diabetes insipidus (DI) (preserved osmolar control mechanism).

Observations incompatible with brain death are:

- decerebrate or decorticate posturing
- true extensor or flexor motor responses to painful stimuli
- witnessed seizures.

3.3 Ancillary testing

The clinical exam is considered the most sophisticated way of testing neurological function in that it deliberately delivers a stimulus to provoke central processing followed by an efferent response. All ancillary tests infer the integrity of this stimulus-integration-response arc, but do not observe it directly.^[5]

It is recommended that the clinical exam be completed to the fullest extent possible prior to conducting an ancillary test.^[5]

Making an appropriate choice of an ancillary test depends on the clinician and the clinical circumstances. The most effective method to address a confounder must be identified using an educated appraisal of the strengths and weaknesses of those tests that are readily available.^[5]

It is recommended that ancillary testing is required in the following circumstances:^[5]

- inability to complete a clinical neurological determination of death, including the apnoea test
- confounding conditions which would invalidate clinical testing that cannot be resolved
- uncertainty regarding interpretation of spinal-mediated motor reflexes
- pre-existing severe neuromuscular disorder, such as amyotrophic lateral sclerosis or a pre-existing severe sensory neuropathy.^[24,25]

Ancillary testing which establishes the absence of intracranial blood flow is the preferred method of testing where available.^[5]

- radionuclide studies
- conventional four-vessel cerebral angiography
- transcranial Doppler.

It is suggested that electrophysiological testing with an electroencephalogram (EEG) not be utilised routinely as an ancillary test, given its low specificity.^[5,26] EEG testing may, however, be preferred where craniovascular impedance has been affected by an open skull fracture, decompressive craniectomy, or an open fontanelle/suture in infants.^[5,27]

It is suggested that computed tomography angiography (CTA) and magnetic resonance angiography (MRA) which may be used in the investigation of these patients not be used in isolation to support a diagnosis of cerebro-circulatory arrest at present, pending further research into the sensitivity and specificity of these modalities.^[5,28-35]

3.3.1 Technical aspects of ancillary testing

Radionuclide/brain scintigraphy studies

To confirm brain death with radionuclide/scintigraphic techniques, the study must demonstrate absence of intracranial isotope.^[5,36-38]

- Diffusible radiopharmaceuticals (brain-specific tracers) should be used preferentially.
- Single photon emission computed tomography (SPECT) is preferred over planar imaging.
- If SPECT is not available, perfusion scintigraphy with anterior and lateral planar imaging should be used with appropriate time intervals to demonstrate static filling of the posterior fossa.

Cerebral angiography

If used as an ancillary test to confirm brain death, four-vessel cerebral angiography must demonstrate absent filling at the points where the internal carotid and vertebral arteries enter the skull base, with demonstration of patent external carotid circulation.^[39]

Transcranial doppler

Transcranial doppler testing is operator training dependent, needs specialised probes and requires that:^[40,41]

- two examinations be performed ≥ 30 minutes apart
- the examinations be performed bilaterally, anteriorly and posteriorly to include both internal carotid arteries as well as the vertebrobasilar circulation
- the exams illustrate biphasic oscillating flow and systolic spikes with reversal of flow in diastole in order to make a declaration of brain death.

Transcranial doppler testing is not validated in paediatrics^[42] and is not widely available in SA.

Electroencephalography (EEG)

If EEG testing is used, it should be used in conjunction with somatosensory and brainstem auditory evoked potentials.^[43]

EEG should be interpreted as demonstrating absence of brain activity when a 30-minute study is isoelectric above 2 microvolts with a sensitivity of 2 microvolts/mm and filter of 0.1 or 0.3 seconds and 70 Hz.^[5,44]

3.4 Paediatric considerations in brain death

A cautious approach with a low threshold for serial examinations is recommended in this population group.^[5,45]

Brain death cannot be diagnosed in neonates <36 weeks' corrected gestation.^[3,4]

Clinical assessment of brain death in neonates and the paediatric population is the same as in adults, with age-appropriate haemodynamic targets,^[46,47] except that in neonates (<4 weeks) the sucking and rooting reflex should be absent in addition to the other brain-stem reflexes.^[5]

Ancillary testing (as with adults) is not routinely required in this population group when clinical testing confirms brain death.^[45]

A T-piece should be used for oxygenation during the apnoea test in paediatric patients in preference to the tracheal insufflation method, owing to the risk of barotrauma with this method.^[5]

In patients with chronic hypoxaemia owing to cyanotic heart disease, apnoea testing should not be performed and instead an ancillary study be conducted to assist with the determination of brain death.^[5]

3.5 ECMO and brain death^[48-50]

The same fundamental principles of brain-death testing apply to patients on extracorporeal membrane oxygenation (ECMO).

The apnoea test should be performed in patients on veno-arterial (VA) or veno-venous (VV) ECMO unless unable to be completed owing to haemodynamic instability.

In VA ECMO, the flow rate may be adjusted to maintain a MAP ≥ 60 mmHg during testing.

Preoxygenation prior to apnoea testing should be done by increasing the inspired oxygen via the mechanical ventilator to 100% and to the membrane lung of the ECMO machine to 100% for 10 minutes. The sweep gas flow should be titrated to <1 L/min while maintaining oxygenation in order to allow a rise in carbon dioxide (CO_2). Apnoea must persist in the presence of an adequate stimulus to spontaneous ventilation, i.e. an arterial $\text{p}_a\text{CO}_2 > 60$ mmHg (8 kPa) or 20 mmHg above the patient's baseline and an arterial pH <7.30. Serial blood gases may be required as achieving the rise in CO_2 may take longer to achieve than in a person without ECMO support.

3.6 Pregnancy and brain death^[51-54]

If a decision is made to offer somatic support to a brain-dead pregnant patient, it is recommended that a multidisciplinary team of intensivists, obstetricians, social workers, psychologists and neonatologists be involved.

The consent to continue providing somatic support to a brain-dead pregnant patient should be made in keeping with the HPCSA professional rules on consent.^[55] Due consideration to duration of support required, the high-risk nature of the delivery and an assessment of the home and legal circumstances needs to be conducted by a multidisciplinary team.

The fetus should be routinely monitored with at least daily heart rate checks, given that fetal health may affect decision-making.

Antenatal steroids should be administered to facilitate fetal lung maturation with preparations for delivery made between 26 and 33 weeks when fetal lung maturity is reached.

3.7 Family accommodation in brain death

While it is reasonable to provide accommodation for a finite period of time, assuming that the specific timeframe for doing so is brief, that resources allow, and that a family is informed of the timeframe in advance, accommodation ordinarily should not be provided for a period greater than 24 hours.

It is ethically and legally appropriate for the treating team to end somatic support for a body when the family has been adequately counselled on the diagnosis of death and the option of organ donation explored.

An additional clinician in the hospital can provide the family with a second opinion regarding determination of brain death if it is felt that this may assist the family in accepting the person's death.

Even in the setting of requests for accommodation, support should be discontinued if a person has been declared brain-dead and the hospital bed is required for a living patient where no other bed is available.^[5]

4. Circulatory death

4.1 Preconditions for circulatory death testing

To declare death on circulatory grounds, one of the following criteria must be met:

- it is inappropriate to attempt cardiopulmonary resuscitation
- attempts at cardiopulmonary resuscitation have failed
- treatment aimed at sustaining life has been withdrawn.

Treatment may be withdrawn because:^[56]

- it has been assessed to be of no further benefit to the patient (non-beneficial/futile) and is not in his or her best interest to continue
- it is in respect of the patient's wishes via an advanced directive to refuse life-sustaining treatment
- it is in respect of the patient's wishes as expressed by their legal surrogate decision-maker.

4.2 Circulatory death testing

The patient should be observed by the person responsible for confirming death for a minimum period of five minutes to establish that irreversible circulatory arrest has occurred.^[57]

The absence of mechanical cardiac function should be confirmed using a combination of the following:

- absence of a central pulse on palpation
- absence of heart sounds on auscultation.

In the hospital setting, clinical assessment of absent mechanical cardiac function can be supplemented by one or more of the following:

- absence of pulsatile flow using direct intra-arterial pressure monitoring
- absence of contractile activity using echocardiography.

Once mechanical cardiac function is confirmed as absent by clinical assessment, intra-arterial pressure monitoring or echocardiography, the 5-minute waiting period can begin.

Any spontaneous return of circulatory or respiratory activity during the five-minute observation period should prompt a reset of the observation period from this point.^[58]

Spontaneous return of circulatory or respiratory activity is not an indication to begin resuscitation efforts in a context where this has been determined to be inappropriate.

After the 5-minute period of continued circulatory arrest has passed, the absence of pupillary responses to light and of any motor response to supra-orbital pressure should be confirmed.^[59,61] The time of death is recorded as the time at which these criteria are fulfilled.^[11]

It is inappropriate to initiate any intervention that has the potential to restore cerebral perfusion after death has been confirmed.^[3,62] In cases where organ donation after circulatory death takes place, a second doctor is required to certify the death.^[19]

5. Organ donation

End-of-life care should, as standard of care, explore the patient's wishes regarding organ and tissue donation.^[63]

The recommended time for a clinical assessment of organ donation potential (with the transplant co-ordinator) is when the treating team makes a decision to perform brain-death testing or to initiate discussions with the family to withdraw life-sustaining treatment. This allows clarification of the potential for organ and tissue donation prior to end-of-life discussions, and an informed approach for consent from the family.^[64]

Somatic support of a brain-dead patient is appropriate to allow the possibility of organ and tissue donation to be fully explored with the family.^[3]

6. Training and documentation

All doctors should be trained in the determination of death. It is recommended that a standardised checklist be used for death determination and its documentation.^[65,66]

Consultation with a medical practitioner experienced in the diagnosis of brain-death is advised in situations where the diagnosis and testing are uncertain.

All doctors training in disciplines that manage patients with severe brain injuries should receive detailed training in the preconditions for brain-death testing, clinical testing procedures, indications for and utility of ancillary testing, somatic support of the brain-dead patient, and techniques for effective counselling of families.^[5,65,67]

All phases of the brain death determination should be documented in the medical record, including:^[5,68]

- aetiology of the coma
- absence of confounders
- full details of the clinical testing performed, including apnoea testing and laboratory values
- neuroimaging results if done
- the reason for ancillary testing if performed and the findings
- time and date of death
- identity of the practitioners performing the evaluation, with their HPCSA numbers.

In cases where brain death can be determined with a neurological exam and ancillary testing is not needed, the time of death must be documented as the time the arterial $p_a\text{CO}_2$ reaches the target, with no spontaneous respirations seen, during the apnoea test confirmed by two doctors.^[3,5]

In cases where the doctors were not able to complete the apnoea test together, the time of death will be upon completion of the second apnoea test.^[3,5]

In cases where ancillary testing is performed, the time of death must be documented as the time that the ancillary test results are formally interpreted and documented.^[3,5]

Determination of death on the basis of circulatory criteria should be documented in the medical record, including:

- in cases where resuscitation was attempted, a record of the resuscitation attempts and the time resuscitation was stopped

- in cases where resuscitation was not attempted, the rationale
- confirmation of absent pupillary and pain response, respiratory effort and circulation after a 5-minute period
- time and date of death
- identity of the practitioner/s performing the evaluation, with their HPCSA number/s.

7. Strengths and limitations

This guidance document offers clear, pragmatic evidence-based medical guidance in the determination of death in the SA context. The provided checklists (Figs 1 - 4) offer a summary of the recommendations for clinical application. A limitation of this document is that a lack of high-quality data from large randomised clinical trials prevented the use of formal analytical techniques (GRADE, AGREE) in the literature appraisal. It is also acknowledged that these recommendations were developed without direct patient, cultural and religious input; however, the panel of experts did represent a broad range of cultural and religious viewpoints from across SA.

7.1 Applicability, barriers, facilitators and cost implications of these guidelines

Barriers to effective implementation of death determination guidelines include a lack of uptake and acceptance by both clinicians and the public as well as the challenge of ensuring cultural and religious engagement and support for the medical determination of death. The endorsement and application of these guidelines is to be formally encouraged across all healthcare disciplines. Undergraduate and postgraduate teaching should consider these guidelines as the standard of care for the determination of death in SA and incorporate them into their training programmes. Hospital policies and standard operating procedures should similarly align themselves with the guidelines to prevent differing practices across institutions, which can cause confusion among clinicians, families and the general public.^[9,69] Use of the provided checklists is expressly recommended in the guidelines to facilitate clinicians to apply them in their daily practice. Continuing professional development programmes should incorporate these guidelines in the education of practising clinicians.

Pursuing a diagnosis of brain death may entail additional resources, and the clinician must judge the appropriateness in each clinical context. The beneficial effects of organ and tissue donation to the SA public and healthcare system are large. Exploring the option of organ and tissue donation at the end of life is a standard of care that should be fully assessed and appropriately explored with all patients and their families. Folder reviews and audits of documentation completed at end of life to assess compliance with these guidelines will assist in improving standards in death determination and end-of-life care.^[70]

8. Conclusion

Death can be determined with complete certainty by medical professionals adhering to these guidelines which offer the clinician the latest evidence in best practice for determining death by either neurological or circulatory criteria. Use of the attached checklists is recommended.

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key questions and supervised the Delphi process, including the drafting and grading of recommendations; drafted the manuscript, supervised revisions to the drafts, and approved the final manuscript.

IJ, DG, FP, KDV chose the participants, developed and supervised the methodology, participated in the Delphi process and face-to-face round-table meeting, performed literature searches, participated in manuscript drafting and revisions to the drafts, and approved the final manuscript.

SM, RM, MM, BM, DB, BR, NM, GAR, NW, BL, MS, NA participated in the Delphi process and face-to-face round-table meeting, drafted and graded associated recommendations; participated in revisions to the drafts and approved the final manuscript.

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BRAIN DEATH CERTIFICATION Summary Recommendations - South African Death Determination Guidelines Checklist				
Name: _____ Hospital Number: _____ Date of Birth: _____				
Doctor 1	Prerequisites	Doctor 2		
	Coma, irreversible and cause known			
	Normothermia (>36°C)			
	Systolic blood pressure ≥100 mm Hg*			
	Significant metabolic, electrolyte, endocrine abnormality excluded			
	Sedative/analgesic/neuromuscular drug effect excluded			
	Examination**			
	Pupils fixed and non-reactive			
	Absent corneal reflex			
	No gag reflex			
	Absent cough reflex on deep suctioning down tracheal tube			
	No facial grimace to supra-orbital pain			
	Absence of motor response in all limbs (spinal reflexes may persist)			
	Oculovestibular (cold caloric) reflex absent			
	Apnoea testing – no spontaneous respiration			
<table style="width: 100%; border: none;"> <tr> <td style="width: 50%; border: 1px solid black; padding: 10px; vertical-align: top;"> Doctor 1 Name: _____ HPCSA Number: _____ Signature: _____ Date and time: _____ </td> <td style="width: 50%; border: 1px solid black; padding: 10px; vertical-align: top;"> Doctor 2 Name: _____ HPCSA Number: _____ Signature: _____ Date and time: _____ </td> </tr> </table>			Doctor 1 Name: _____ HPCSA Number: _____ Signature: _____ Date and time: _____	Doctor 2 Name: _____ HPCSA Number: _____ Signature: _____ Date and time: _____
Doctor 1 Name: _____ HPCSA Number: _____ Signature: _____ Date and time: _____	Doctor 2 Name: _____ HPCSA Number: _____ Signature: _____ Date and time: _____			
* or mean arterial pressure ≥ 60 mmHg / in paediatrics age-appropriate haemodynamic targets apply ** One doctor with more than 5 years' experience, tests may be done by both doctors simultaneously, no doctor may be involved with the transplant team				

Fig. 1. Summary recommendations – Brain death certification checklist (<https://criticalcare.org.za/resource/death-determination-checklists/>).

BRAIN DEATH CLINICAL TESTING Summary Recommendations - South African Death Determination Guidelines Checklist	
Coma	Apply deep pressure to: the temporomandibular joints, the supra-orbital notch bilaterally, the sternal notch and all four extremities via deep nail bed pressure. There should be no true motor response.
Pupillary light reflex	Shine a bright light into the eye and look for a pupillary constrictor response. There should be no pupillary response.
Corneal reflex	Touch the corneas with soft cotton wool or gauze and examine the eyes for blinking or a withdrawal response. There should be no blinking or withdrawal response.
Response to pain in the trigeminal distribution	Apply pain over the trigeminal distribution, e.g. pressure over the supra-orbital nerve and to the condyles at the level of the temporomandibular joints. There should be no facial or limb movement.
Gag reflex	Stimulate the posterior pharyngeal wall, on both sides, with a tongue depressor or Yankauer suction. There should be no gag response.
Cough reflex	Stimulate the tracheobronchial wall with a soft suction catheter. There should be no cough response.
Vestibulo-ocular reflex (cold caloric)	Inspect the external auditory canal with an otoscope to confirm that the eardrum is visible. If the eardrum is not visible, the canal must be cleared before testing. Elevate the head to 30° to align the semi-circular canal and generate a maximal response. Flush 50 mL of ice-cold water into the ear canal using a syringe. Hold eyelids open and observe for eye movement for a minimum of 60 seconds. There should be no eye movement in response to the cold water; the eyes remain in the midline within the socket.
Apnoea test	See separate advice sheet.
Observations that are incompatible with a diagnosis of brain death:	
Decerebrate or decorticate posturing. True extensor or flexor motor responses. Witnessed seizures.	
Observations that are compatible with a diagnosis of brain death:	
Spinal reflexes, sweating, blushing, tachycardia, and a normal blood pressure.	

Fig. 2. Summary recommendations – Brain death clinical testing (<https://criticalcare.org.za/resource/death-determination-checklists/>).

APNOEA TESTING

Summary Recommendations - South African Death Determination Checklist

Background:

Only proceed with the apnoea test to confirm brain death if all other brainstem reflexes are absent and patient is haemodynamically stable (may be on inotropes).

Two doctors should perform a single apnoea test together.

Testing procedure:

Pre-oxygenate the patient with 100 percent oxygen for 10 minutes to prevent hypoxaemia during the test.

Perform a baseline ABG.
Assess baseline CO₂ level (p_aCO₂ 35 - 45 mmHg, 4.6 - 6.0 kPa).

Disconnect the patient from the mechanical ventilator and supply continuous oxygen via a T-piece (preferred) or through a catheter inserted through the endotracheal tube and placed above the carina.*

Observe continuously for any spontaneous breathing.

Halt testing if:

- Spontaneous respirations are witnessed;
- Systolic blood pressure <100 mm Hg or mean arterial pressure <60 mm Hg despite titration of inotropes/vasopressors;
- Sustained oxygen desaturation <85%;
- An unstable arrhythmia occurs.

It is suggested that prior to aborting the apnoea test due to cardiorespiratory instability, an arterial blood gas (ABG) be sent for testing. If the p_aCO₂ target is met, the apnoea test can be considered positive (consistent with brain death).

Interpretation:

In cases of brain death, apnoea must be demonstrated in the presence of an adequate stimulus to spontaneous ventilation, i.e. an arterial p_aCO₂ >60 mmHg (8 kPa) and an arterial pH <7.30.

In patients with pre-existing hypercapnia, it is recommended to wait for a p_aCO₂ rise of >20 mmHg (2.7 kPa) above the chronic level, with a pH <7.30.

Attempt at breathing is defined as any respiratory muscle activity that results in abdominal or chest excursions or activity of accessory respiratory muscles.

Duration (Target p_aCO₂)

The period of observation to achieve an adequate threshold of stimulus of the respiratory centre is variable. Usually p_aCO₂ rises by ~3 mmHg (0.4 kPa) for every minute of apnoea.

If starting from normocapnia, the p_aCO₂ is likely to be >60 mmHg (8 kPa) after 10 minutes.

If this is not the case at 10 minutes and the patient is stable, wait a further 5 minutes and repeat the arterial blood gas.

An option to minimise the time required for the p_aCO₂ to rise to the desired level, is to adjust the minute ventilation to mild hypercapnia (p_aCO₂ ~45 mmHg [6 kPa]) beforehand.

If unable to complete testing, consider:

If hypoxia occurs 1-2 positive pressure breaths can be given, and apnoea testing continued.

Adequate pre-oxygenation and recruitment usually avoids this problem.

*If using a catheter through the endotracheal tube supply oxygen at 4 - 6 L/min.

Fig. 3. Summary recommendations – Apnoea testing (<https://criticalcare.org.za/resource/death-determination-checklists/>).

CIRCULATORY DEATH CERTIFICATION Summary Recommendations - South African Death Determination Guidelines Checklist	
Name: _____ Hospital Number: _____ Date of Birth: _____	
Prerequisites	
Inappropriate to attempt cardiopulmonary resuscitation or attempts at cardiopulmonary resuscitation have failed	
Intensive support (ventilation, inotropes) withdrawn at _____ (time) on _____ / _____ / _____ (date)	
Examination	
Absence of mechanical cardiac function confirmed by one of the following: - Absence of central pulse / heart sounds on auscultation - Absence of pulsatile flow on intra-arterial BP monitoring - Absence of contractile activity on echocardiography	
Patient observed for 5 minutes with no respiratory or circulatory activity seen*	
At end of 5 minutes observation period, lack of pupillary response to light and motor response to supraorbital pain confirmed	
Death certified at _____ (time) on _____ / _____ / _____ (date) by _____	
Doctor 1 Name: _____ HPCSA Number: _____ Signature: _____	Doctor 2 (in case of organ donation**) Name: _____ HPCSA Number: _____ Signature: _____
*Any spontaneous return of circulatory or respiratory activity during the 5-minute observation period requires a reset of the observation period from this point ** In cases of possible organ donation after circulatory death one doctor with more than 5 years experience, neither doctor may be involved with the transplant team.	

Fig. 4. Summary recommendation – Circulatory death certification checklist (<https://criticalcare.org.za/resource/death-determination-checklists/>).

Organ Donation after Circulatory Death (DCD)

Donation after circulatory death

INTRODUCTION

What is donation after circulatory death and what are the different types?

Donation after circulatory death (DCD) was previously referred to as organ donation after cardiac death or non-heart beating organ donation. It entails the retrieval of organs for the purpose of transplantation from patients whose death is diagnosed and confirmed using cardio-respiratory criteria.

There are two main types of DCD, controlled and uncontrolled. Uncontrolled DCD refers to organ retrieval after a cardiac arrest that is unexpected and from which the patient cannot or should not be resuscitated.

Controlled DCD takes place after death which follows the planned withdrawal of life-sustaining treatments that have been considered to be of no overall benefit to a critically ill patient on ICU or in the Emergency Department. As with all organ donation, informed consent is needed from the next of kin in South Africa.

46% of all organ donors in the UK are donors after circulatory death.

These organs are considered extended criteria organs because the organs suffer a degree of damage during the dying process from the warm ischaemic time. This time period cannot be controlled as euthanasia is illegal and the palliative process is to withdraw all non-beneficial treatments and allow a natural death to take place. The only medications allowed are titrated medications for symptom control. Organ recovery teams may have to stand down and not recover organs if the warm ischaemic time is too long. This time period varies between organs. Kidneys and lungs are the most resistant to warm ischaemic damage.

Challenges in the practice of DCD include: Identification and referral of patients as suitable potential DCD donors, supporting and maintaining the trust of bereaved families, integrating the specialist transplant coordinator nurse into the clinical teams managing the end of life process and how to manage the consequences of warm ischaemia in a fashion that is professionally, ethically, and legally acceptable. Given the time pressures of organ recovery in this situation it is important that the concerns about the practice of DCD are addressed by institutional protocols for the processes that support DCD.

The clinical circumstances in which DCD can occur are described by the Maastricht classification. In South Africa we have to date only performed renal transplantation from donors after circulatory death.

Figure 1: The modified Maastricht classification of donors after circulatory death

Category	Circumstances	Controlled/Uncontrolled	Location of care
Category 1	Dead on arrival	Uncontrolled	ED in a transplant centre
Category 2	Unsuccessful resuscitation	Uncontrolled	ED in a transplant centre
Category 3	Anticipated cardiac arrest	Controlled	ICU and ED in transplant and non-transplant centres
Category 4	Cardiac arrest in a brain-dead donor	Controlled	ICU and ED in transplant and non-transplant centres
Category 5	Unexpected arrest in ICU patient	Uncontrolled	ICU in a transplant centre

Procedure

How is DCD different from a brain dead donor?

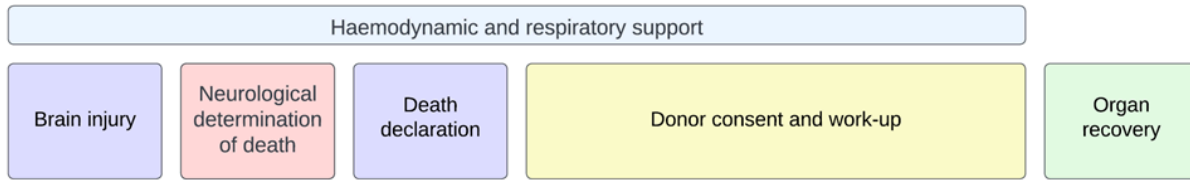
For a brain dead donor because the legal criteria for death certification have already been met while the heart is beating and the patient's body is being maintained on a ventilator the donor can be managed by the transplant team from the point of death certification through to organ recovery in theatre. The heart remains beating right up until cross clamping and cold perfusion of the organs.

For DCD donor support is done by the treating clinical team up until certification of death. Organ recovery surgery can only begin on the cadaver after all circulatory output is halted. The organs are rapidly flushed with preservation solution and removed en bloc to be placed in cold storage with limited dissection.

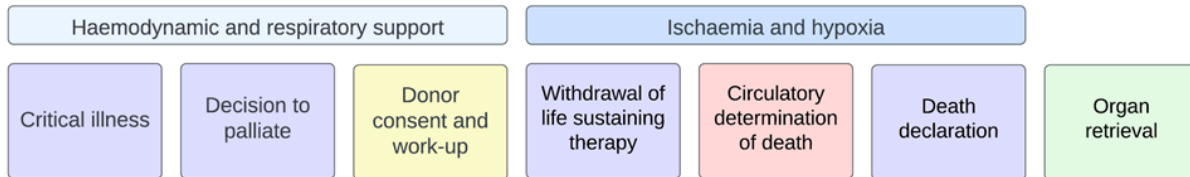
Ideally the surgery should start immediately upon certification. Family wishes and hospital environment may influence the location of the withdrawal. This all forms part of the consent and planning process with the family and treating team and this is the part of the DCD pathway that needs to be addressed through education, training and protocols.

Figure 3: Donation pathways: Donation after brain death, controlled donation after circulatory death, uncontrolled donation after circulatory death

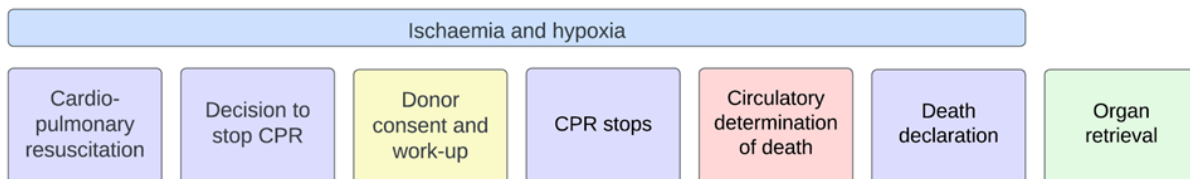
A: Donation after brain death



B: Controlled donation after circulatory death



C: Uncontrolled donation after circulatory death



How is donation after circulatory death integrated into a palliative care pathway without compromising the dead donor rule?

Figure 4: Sequence of steps that comprise a comprehensive palliative withdrawal process for an invasively mechanically ventilated patient

- Identification of a palliative care situation (treating team)
- Clarification of donation potential (donation specialist)
- Initial discussion with the family - Breaking bad news (treating team, family)
- Shared decision making process - DNR form (treating team, family)
- Shared decision making process - Palliative withdrawal agreed to be appropriate (treating team, family)
- Request for consent to organ and tissue donation (donation specialist, family)
- Shared decision making process - Logistics of the withdrawal - timing, location, who will be present, the process (treating team, family, donation specialist in cases where consent for organ donation)
- Preparation for the withdrawal (treating team)
- Implementation of withdrawal (treating team) - Individualised medication for pain, anxiety, agitation, nausea, secretions - Extubation and stopping of all non-comfort treatments - Continual reassessment of clinical situation and appropriate palliative supports added / titrated
- Certification of death or transfer out of ED/ICU for ongoing palliative care (treating team)
- Organ and/or tissue recovery in cases where consent for organ donation given (donation specialist, organ recovery team)
- Release of the body to forensics or undertaker (treating team or in cases of organ donation the donation specialist, theatre team)
- Family bereavement support (treating team, donation specialist)

Figure 5: Differences between DCD and non-DCD palliative withdrawals

Location of the extubation - Typically in the ICU as normal - Can be in the theatre (typically in cases where the family does not want to be present and the treating team is comfortable to palliate in the theatre environment)
Timing of the extubation - This is normally scheduled with the family for a specific time and place - This needs to be coordinated with the surgical recovery team with a theatre open
Pre-mortem testing and medications - Blood tests for infectious screening and tissue typing are sent off - 500 u/kg unfractionated heparin dose is given intravenously at the start of the palliation process in cases where it is not felt to hasten death - As these are not felt to cause the patient pain and is felt to be in accordance with their wishes to be a potential donor
Monitoring - The ECG tracing can be disabled as normal - The saturation monitor is left on until the point of functional warm ischaemia* after which it can be taken off. - The a-line if present is typically left in and the monitor left on to accurately identify the start of functional warm ischaemia* and when there is loss of cardiac output. - Alternatively transthoracic echocardiography or manual palpation of a central pulse can be used.
Certification of circulatory death - The normal standard determination process is followed: - The patient is observed for a minimum period of five minutes to establish that irreversible circulatory arrest has occurred - With confirmation at this point that there is an absence of pupillary responses to light and of any motor response to supra-orbital pressure. - In a DCD scenario a second doctor is needed to certify the circulatory death, both may not be involved with the transplant team, and one of the doctors must have more than 5 years experience.
Time points to be documented - Time of extubation as normal - Time of death as normal - Onset of functional warm ischaemic time* during the palliative process

*Functional warm ischaemia is the time when either the systolic blood pressure is < 50 mmHg or the peripheral saturation is less than 70%

Certification of death in donation after circulatory death?

The standards used in the determination of death are in accordance with published South African Death Determination Guidelines.

Figure 2: Checklist for determination of circulatory death

CIRCULATORY DEATH CERTIFICATION Summary Recommendations - South African Death Determination Guidelines Checklist	
Name: _____ Hospital Number: _____ Date of Birth: _____	
Prerequisites	
Inappropriate to attempt cardiopulmonary resuscitation or attempts at cardiopulmonary resuscitation have failed	☐
Intensive support (ventilation, inotropes) withdrawn at _____ (time) on _____ / _____ / _____ (date)	
Examination	
Absence of mechanical cardiac function confirmed by one of the following: - Absence of central pulse / heart sounds on auscultation - Absence of pulsatile flow on intra-arterial BP monitoring - Absence of contractile activity on echocardiography	☐
Patient observed for 5 minutes with no respiratory or circulatory activity seen*	☐
At end of 5 minutes observation period lack of pupillary response to light and motor response to supraorbital pain confirmed	☐
Death certified at _____ (time) on _____ / _____ / _____ (date) by	
Doctor 1 Name: _____ HPCSA Number: _____ Signature: _____	Doctor 2 (in case of organ donation**) Name: _____ HPCSA Number: _____ Signature: _____

*Any spontaneous return of circulatory or respiratory activity during the 5-minute observation period requires a reset of the observation period from this point

** In cases of possible organ donation after circulatory death one doctor with more than 5 years experience, neither doctor may be involved with the transplant team.



How is donation after circulatory death legal in South Africa?

The exact same permissions are sought as per donors after brain death. Informed consent from the patient's next of kin is required. In cases of an unnatural death forensic pathologist approval for the removal of the specific organs is required. Finally medical authorization from the head of the institution where the organ recovery is taking place is requested. Donation after circulatory death, in alignment with the National Health Act, requires death to be determined by two doctors, one with more than 5 years of experience and both not involved in the transplant team.

How could and should you be involved in DCD?

Donor identification in ICU

You may be working in an ICU and be faced with a palliative care situation and aiming to perform a compassionate extubation and de-escalation of all non-beneficial supports. Best practice for referral of DCD donors is when a defined clinical trigger is met. All patients with a GCS of 4 or less not explained by sedation, or where a planned withdrawal of intensive life support is expected to result in the imminent demise of the patient should be discussed with the transplant team to clarify the potential for donation after circulatory death before end of life discussions take place with the patient or their family. This allows the treating team to be fully informed as to the potential for donation in that particular end of life care situation and keep donation as an option if possible.

It is essential that the decision to palliate is independently made prior to that discussion to clarify the potential for donation in order to avoid a perceived conflict of interests. High performing systems around the world mandate this discussion in end of life care in order to make it easier for clinicians to support it at an emotional and difficult time.

Palliation in the ICU - Integrating the transplant coordinator into end of life discussions

For DCD to take place end of life care and counselling has to be very well done. If you are leading on these discussions, plan the approach for consent away from the bedside. Ensure it is completely decoupled from discussions around palliation so that a perceived conflict of interests. Fully introduce everyone involved. Actively engage with family emotions and assess understanding. The treating team needs to ensure there is family understanding of the clinical situation and that goals of care have changed from cure to comfort and allowing a natural dignified death. Introduce the coordinator as a specialist nurse who helps in these situations and give them time with the family.

Palliation in theatre environment

The location of palliation may be in the theatre environment. In this case the anaesthetist is needed to assist the treating team implementing the withdrawal in an unfamiliar space. Assistance for additional medications and setting up monitors would be needed. Being the second doctor to certify death if only one doctor from the treating team is able to be present may be required. In cases of lung recovery the donor would need to be intubated and the lungs re-expanded (after a period of 10 minutes from loss of circulatory output.)

F

Future advances

Future advances in South Africa may relate to normothermic regional perfusion which after surgical isolation of the circulation from the brain entails the reestablishment of blood flow with an extracorporeal pump and oxygenator with the aim of allowing a period of organ resuscitation and assessment and controlled normothermic organ recovery. This practice is in place in certain countries already to increase the utilisation of organs from donors after circulatory death. It is conceptually and logistically challenging and would need theatre team support with anaesthetist contributions in such cases.

DISCUSSION

There are multiple benefits of a donation after circulatory death program beyond the immediate organs and tissues transplanted. First and foremost it allows more patients and grieving families the ability to have their tragedy become a life-saving gift for someone else. Secondly it helps to ensure consistent practice of end-of-life care focused around good and clear communication and sound ethical practice.

By necessity such a program requires better integration of the consideration of organ donation into end of life care planning. When healthcare teams (and typically this is at the direction of a single lead clinician or purported "institutional practice") at the end of life do not advance discussions to include an assessment of donation potential and support a best practice approach for consent to donation whenever it is possible they have unfortunately, unilaterally and paternalistically denied that family the option to support donation.

We firmly believe it should be the healthcare system that supports the family in this situation to understand all the options and not up to the grieving family to drive the donation process. Bookholane et al. showed that in the private sector the family brought up organ donation in 20% of cases and not the treating team while in the public sector 42.3% of families were not aware of the poor prognosis at the time of the transplant coordinator being asked to make a consent request for donation.

Through a donation after circulatory death program with recognized triggers for referral there are more opportunities to support clinicians and families with expert support in the end of life care situation, regardless of the ultimate decision to donate or not.

Unfortunately there is a common misperception that de-escalation of non-beneficial treatments is euthanasia. This is incorrect. To allow a patient a natural dignified death is the appropriate action when intensive organ support is not felt to offer a meaningful chance of recovery. This includes compassionate extubation and symptom control with medications to prevent suffering. Ensuring that such medication is titrated to the symptoms and that quality of life is prioritised for the patient, family and caregivers is the essence of good palliative care as defined by the World Health Organization and not euthanasia.

The timeous assessment of when it is appropriate to transition to comfort care is difficult to establish and is at the discretion of the treating clinical team. Most high performing countries have established triggers to ensure end of life referral pathways do engage with the right expertise at the right time with quality assurance processes continually auditing practice and multiple education interventions to empower and support all staff working in these areas.

CONCLUSION

The anaesthetist can play multiple roles in supporting donation after circulatory death. Donors after circulatory death offer an opportunity to afford more families and patients the ability to support organ donation. This requires good end of life care and communication with an understanding of the time considerations related to the process of organ recovery. Informed consent, having all needed authorizations in place and rigorous and independent certification of circulatory death are required by law before organ recovery can take place.

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Care and Management of the Brain Dead patient

INTRODUCTION

Organ and tissue donation has saved or enhanced the lives of hundreds of thousands of recipients worldwide. Optimal donor management is essential throughout the emergency department, intensive care unit and the operating room journey to maximise the donation potential and honour the wishes of the patient and their family. Ultimately donor management translates into the function of transplanted organs and the quality of life and survival benefits conveyed to the recipients. Here we will discuss the management of brain dead patients and potential organ donors.

GENERAL COMMENT

Management of the potential organ donor primarily involves the use of conventional critical care therapeutic and supportive measures to reverse or mitigate the physiologic changes that occur after brain death. This process requires balancing the interventions needed for the successful preservation of multiple organs. Interventions that improve the function of one organ may be detrimental to the function of other organs.

As an example, although the renal transplant teams prefer an extremely well-hydrated donor with excellent diuresis, aggressive hydration may cause edema that may jeopardise the transplantability of the heart, lungs and liver. **Current data supports a restrictive fluid balance**, which is intended to improve lung procurement and viability, and has no substantial detrimental effect on rates of kidney procurement or function after transplant.

STEPS TO OPTIMISE ORGAN FUNCTION

Care of the potential organ donor should continue the support that started before the declaration of brain death and must actively manage the associated physiologic deterioration. Unfortunately, it is often needed to make up for a lack of care in the time it takes to determine brain death and to compassionately engage with the family around end of life and a decoupled consent request for donation.

Active management improves the retrieval rate of organs and transplant outcomes. There are key interventions and quality indicators for the management of the donor after brain death which we will review here.

General care components include:

- Appropriately gentle yet effective ventilatory support and good pulmonary.
- Head up at 30 to 45 degrees.
- Temperature control as with the loss of hypothalamic temperature regulation, the brain-dead donor tends to be hypothermic.
- Mild hypothermia (34 to 35°C) has been shown to reduce delayed graft function of transplanted kidneys in some settings with no increase in adverse events.
- Thromboprophylaxis and infection prevention and appropriate treatment.



MONITORING OF PATIENTS

The primary goal for monitoring of the potential organ donor is to better direct intervention and support of the donor to optimise the number and long-term function of transplanted organs.

This monitoring encompasses the usual hemodynamic and respiratory monitoring for critically ill patients: such as serial or continuous monitoring of temperature, blood pressure, heart rate and rhythm, pulse oxygen saturation, and urine output.

An **arterial line** is typically placed for continuous blood pressure monitoring, assessment of pulse pressure variation (PPV) and/or systolic pressure variation (SPV), and access for drawing blood. A **central venous catheter** should be placed to facilitate administration of vasopressor and other medications such as insulin. Up to 81% of brain-dead donors will manifest hypotension.

Hemodynamic parameters and resuscitation can also be measured with pulse contour analysis. Cardiac function can also be assessed by **trans-oesophageal (TOE) or transthoracic (TTE) echocardiography**. Repeated **blood gases** are needed during resuscitation to guide acid-base status and track serum lactate. Laboratory monitoring includes **renal function, electrolytes, and liver function tests**. **Bacterial cultures** of blood, urine, and sputum are routinely obtained. This may guide pre-emptive therapy in recipients of the particular organs.

Upper airway testing for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) is performed, and lower airway testing with tracheal aspirate or broncho-alveolar lavage is required for all lung donors.

HEAMODYNAMIC SUPPORT

Optimum haemodynamic support is an essential component for optimising future organ function in the recipients and increasing the number of organs able to be recovered for transplantation. This needs the referring treating team to be active in managing the brain dead patient. Control of the extremes of blood pressure is a key tenet of donor management. The target MAP is at least 60 to 65 mmHg.

Brain death often results in an initial hypertensive crisis followed by hypotension. The hypertensive crisis (autonomic storm) is attributed to a massive sympathetic discharge following brain death. For patients who manifest the cardiovascular effects of autonomic storm (eg, tachycardia, increased myocardial oxygen consumption, hypertension), any intervention should be easily reversible as this phase tends to be followed by vasodilatation and hypotension. A short acting agent should be used to ameliorate the myocardial damage that can occur during this hypertensive phase.

Hypotension is caused by hypovolemia, cardiac dysfunction, and vasodilation. The initial management requires resuscitation fluid administration to ensure adequate circulating volume. Administration of vasopressors may be needed to support vascular tone (counteract vasoplegia) in brain death. It is important to note that inadequate fluid resuscitation may require excessive vasopressors, which can be harmful to some organs intended to be transplanted. Excessive fluid administration and subsequent oedema can, on the other hand, damage the lungs, liver, and heart which can also be detrimental to their function in recipient.



Repeated assessment of fluid responsiveness through passive leg raising should guide resuscitation fluid (eg. Ringer's lactate or Plasmalyte L). If a cardiac output monitor is in use, then stroke volume variation greater than 15% or a pulse pressure variation greater than 15mmHg would be an indication for a rapid (over 15 minutes) fluid bolus of 500 ml resuscitation fluid. Aim to bring the MAP to greater than 60 to 65 mmHg and urine output to 0.5 to 1 mL/kg per hour.

Hypovolaemia may be secondary to a high urine output from brain-death-associated diabetes insipidus (DI), which needs specific treatment to avoid development of hypernatremia and ongoing fluid losses. A CVP reading of 4 to 10 mmHg is not recommended to guide resuscitation. When the PPV or SPV indicates that the patient is no longer volume responsive, it is appropriate to start a vasoconstrictor or a positive inotrope.

A prospective study utilising an early stroke-volume-based fluid resuscitation protocol for four hours in brain-dead organ donors was associated with donors in the protocol group more likely to donate four or more organs than donors in the control group

Isotonic crystalloids, such as Ringer's lactate solution, and Plasmalyte are the preferred intravenous fluids. Normal saline is not preferred because of its link to hyperchloremic acidosis and hypernatremia. Hydroxyethyl starch has been associated with acute kidney injury and is not recommended.

The target haemoglobin level is based on the success of restrictive transfusion policies for critically ill patients and a haemoglobin level >7 g/dL is suggested, but a higher level may be requested by one of the procuring surgical teams if difficult surgery is anticipated or the lactate is not responding to fluid resuscitation.

Careful attention should be paid to maintaining normal electrolyte levels. According to current guidelines, serum sodium should be maintained below 155 mEq/dL and potassium between 4 and 5 mEq/dL. High sodium levels impact liver transplant outcomes in some studies and a normal potassium will hopefully prevent arrhythmias in the donor causing haemodynamic instability.

PERSISTENT HYPOTENSION

For patients who remain hypotensive (MAP <60 mmHg) despite appropriate fluid replacement (hypotension despite fluid repletion), a low-dose vasopressin is preferred if available. This is effective in treating vasodilatory shock and reducing the need for catecholamines for blood pressure support which can damage the myocardium and cause contraction band necrosis. In addition it can assist in preventing and treating diabetes insipidus. Noradrenalin can be used as an alternative vasopressor. If not available adrenalin or phenylephrine may need to be used.

Myocardial dysfunction can also be due to underlying cardiac disease, cardiac injury from chest trauma, or stress cardiomyopathy. Transthoracic or transesophageal echocardiography can be used to assess left ventricular ejection fraction and guide organ suitability and determine the need for additional monitoring or support. Heart transplant centres prefer the donor to be on minimal or no vasopressors or inotropes during the echocardiogram performed prior to a heart offer.



For **vasopressin** an initial bolus infusion of 1 unit, followed by a continuous infusion of 0.01 to 0.04 units/minute, titrating to a mean arterial pressure (MAP) >60 mm Hg and systemic vascular resistance of 800 to 1200 dynes-sec/cm⁵. Doses >0.04 units/minute have been associated with adverse cardiac effects.

DDAVP (**desmopressin**) is an analog of vasopressin with greater antidiuretic effect and substantially less vasopressor effect. It is used for patients with DI who are not on vasopressin. An initial dose of 1 to 4 micrograms is given intravenously. A typical dose is 1 to 2 micrograms intravenously every six hours. Desmopressin can be used concurrently with vasopressin in patients with severe hyponatremia and hypotension. DI increases urinary losses of magnesium, phosphate, and potassium, so serum levels of these electrolytes should be monitored and replaced as needed.

Steroids are often given – typically methylprednisolone 15 mg/kg as a dose given at the time of brain death certification. Alternative dosing similar to use of steroids in line with septic shock are also practised, with more regular dosing.

Multiple clinical trials have demonstrated that thyroid hormone administration has little to no positive or negative impact on transplant outcomes or haemodynamic stability. Hyperglycemia can cause an osmotic diuresis and hypovolemia and may mislead the care team about the adequacy of the circulation in the donor. For patients with hyperglycemia, we suggest maintaining glycemic control with an insulin infusion following local unit protocols, typically aiming for a blood glucose between 6 to 10 mmol/L.

MECHANICAL VENTILATION

The goals of mechanical ventilation are to maintain tissue oxygenation and protect the lungs for transplantation. Low tidal volume ventilation should be used with safe pressures (driving pressure < 14 cmH₂O, plateau pressure less than 25 – 30cmH₂O) with recruitment manoeuvres performed regularly (after the apnoea test, suctioning or disconnects) and a PEEP level set based on lung compliance and adequate oxygenation. Aim for the lowest FiO₂, preferably 21 to 40 percent, that provides adequate oxygenation (eg, arterial oxygen saturation [SpO₂] >95 percent; arterial oxygen tension [PaO₂] >90 mmHg) to maximise the lung donation potential. Lung recruitment manoeuvres may be used when oxygenation does not meet acceptable parameters for donation. Steps to prevent ventilator-associated pneumonia should be employed, including elevation of the head of the bed to 30 – 45 degrees.

Lung donor assessment is based on a blood gas measured on FiO₂ 1.0 and PEEP 5 cm H₂O with the demonstration of a PaO₂ >300 mmHg generally required for lung transplantation to proceed.

While there have been advancements in ex-vivo lung perfusion to help rehabilitate lungs outside of the body between the donor and recipient this technology is not widely available and comes at great expense and should not replace the fact that the best place to resuscitate organs and make timeous assessment is in the donor prior to recovery. Brain death is associated with endocrine and metabolic dysfunction (hormonal dysfunction) and interventions to address these abnormalities include vasopressin or DDAVP, glucocorticoids, and insulin.

OTHER CLINICAL CONSIDERATIONS

Multiple clinical trials have demonstrated that thyroid hormone administration has little to no positive or negative impact on transplant outcomes or haemodynamic stability. Hyperglycemia can cause an osmotic diuresis and hypovolemia and may mislead the care team about the adequacy of the circulation in the donor. For patients with hyperglycemia, we suggest **maintaining glycemic control** with an insulin infusion following local unit protocols, typically aiming for a blood glucose between 6 to 10 mmol/L

Hypothermia invariably follows brain death (due to loss of hypothalamic control, vasodilatation and decreased muscle activity) without active management. Traditionally, a normothermic body temperature (36.5 to 37.5°C), which may require active warming, has been used for organ donors. **However there are potential therapeutic benefits to donor therapeutic hypothermia (34 to 35°C) shown in renal recipients.**

Pharmacological **thromboprophylaxis** is preferred to mechanical methods (eg, low-molecular weight heparin 40 mg every 12 hours) for patients with normal kidney function. Unfractionated heparin is an alternative for those with renal impairment. **Stress ulcer prophylaxis** can be maintained as per critically ill patients.

SUPPORT FOR FAMILY AND LOVED ONES

The death of a loved one can be intense and emotionally painful, regardless of the manner of death. The process of acceptance of the diagnosis of death by neurological criteria (brain death) or anticipated death by circulatory-respiratory criteria requires patience and empathy on the part of the caregivers. After brain death has been determined and consent has been provided for organ donation, family members and loved ones may or may not wish to stay with the deceased for some period of time, including until transfer to the operating room for organ retrieval. Family members may have questions about the procurement procedure, anaesthesia during procurement, and eventual transfer of the remains of their loved one to a funeral home.

Anaesthesia is given to the deceased to decrease the remaining physiologic response to the procedures involved in the surgical recovery. Specific grief counselling is generally not indicated. Most bereaved individuals are resilient, and acute grief is integrated during a natural adaptive process that typically unfolds with the support and encouragement of close family and friends, as well as spiritual support. Bereaved individuals, especially those who have lost children, often find it very difficult to speak about the loss. However, grief counselling can be helpful for complicated grief.



INTRAOPERATIVE MANAGEMENT

Supportive measures outlined above (eg, monitoring, central venous access, arterial line, mechanical ventilation, vasopressors/inotropic agents) should be continued when the donor is brought to the operating room, or instituted as necessary in the operating room. Theatre considerations may include the amount of cross-matched blood that should be available. The amount of heparin needed at the time of cross clamping and if a repeat dose of hydrocortisone may be required. A typical dose for an adult is 30,000 units of heparin, or 300 units/kg. In cases where lungs are to be procured there may be a need to upsize the existing endotracheal tube, typically to size 8.0 or 8.5 mm.

The brain-dead organ donor usually has a functioning spinal cord, and thus may exhibit unregulated sympathetic and motor spinal reflexes in response to stimulation. Neuromuscular blocking agents (NMBAs) are administered to prevent uninhibited motor responses to surgery as evidenced by zero twitches using a train of four peripheral nerve stimulation monitor).

Reflex sympathetic responses (ie, hypertension and tachycardia) can be controlled with vasoactive drugs, or with inhaled anaesthetics. Temperature can be allowed to drop to 35 degrees intraoperatively.

As multiple organs are retrieved from most deceased donors, more than one procurement team may be present. A midline incision is made from the supraclavicular notch to the symphysis pubis to expose the organs. The remainder of the operation involves clamping and cannulating the major blood vessels associated with the organs to be procured. When the lungs are being procured, the trachea will be dissected and ultimately divided. Donor management and organ procurement occur at the donor hospital where brain death was declared as these patients are physiologically complex and difficult to transfer. Transfer is not impossible in select circumstances.

EX-VIVO ORGAN SUPPORT

Typical preservation and storage entails flushing out blood with a preservation solution and static cold storage. Machine perfusion is being included into the donor-transplant pathway and this can be initiated in the donor hospital or upon arrival at the transplanting institution.

CONCLUSION

Management of the potential organ donor in the intensive care unit (ICU) primarily involves the use of conventional therapeutic and supportive measures to reverse or mitigate the changes in physiology that occur after brain death.

This needs to be an active process from the time of suspected brain death all the way through the communication / grieving / consent process with the family to organ recovery in theatre.

Treating teams supporting donors actively make a marked difference to the number of organs recovered per donor and also impact on their function in recipients.



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GENERAL CARE AND SUPPORT

- ☐ Instrumentation: endotracheal tube, nasogastric tube, urinary catheter, multilumen central line, arterial line, large peripheral intravenous line
- ☐ Monitoring: ECG, pulse oximetry, intra-arterial pressure, core temperature, urine output (hourly)
- ☐ Routine investigations: CXR, ECG, cardiac echo, blood group, coagulation
- ☐ Regular blood testing: FBC, ABG, U&Es, LFTs q6h and as necessary blood gases
- ☐ Review all medications; cease unnecessary orders, continue antibiotics and thromboprophylaxis
- ☐ Maintain core temperature 36–37°C
- ☐ Eye care, mouth care and all other nursing care

RESPIRATORY SUPPORT

- ☐ Aim for S_pO_2 92–97% by FiO_2 (less than 0.6) and PEEP adjustment (minimum PEEP 5cm H_2O)
- ☐ Aim for P_aCO_2 35–45 mmHg by ventilator adjustment (TV 6–8mL/kg ideal body weight, plateau pressure <30 mmHg)
- ☐ Regular turns from side to side; elevate head of bed 30° - 45°
- ☐ Regular tracheal suctioning, chest physio, lung recruitment after apnoea test, bronchoscopy and further recruitment required

CARDIOVASCULAR SUPPORT

- ☐ Ensure patient is neither hypovolemic nor fluid overloaded
- ☐ Aim for MAP 60–80 mmHg, good peripheral perfusion, normal (or decreasing) lactate
- ☐ Aim for urine output ~1 mL/kg/hr (range 0.5–3 mL/kg/hr)
- ☐ Commence vasopressin (infusion of 0.01 to 0.1 units/minute) or noradrenalin if needed
- ☐ Fluid resuscitation with boluses of 500ml of Ringer's lactate or Plasmalyte should be guided by positive response to passive leg raise or strokevolume variation or pulse pressure variation

FLUIDS AND METABOLIC MANAGEMENT

- ☐ Give maintenance fluid as low sodium fluids e.g. glucose 5% 1mL/kg/hr, continue NG feeding or TPN, insulin infusion to keep blood glucose between 6 - 10 mmol/L
- ☐ Give methylprednisolone 15mg/kg stat
- ☐ If polyuric >300 mL/hr assume diabetes insipidus and immediately give desmopressin (DDAVP) as a bolus 1 to 4 micrograms stat followed by 1 – 2 micrograms intravenously 6 hrly
- ☐ If Na >150 mmol/L give additional free water
- ☐ Aim for Hb>70g/L



Australian Government
Organ and Tissue Authority



Best Practice Guideline for Offering Organ and Tissue Donation in Australia

Edition 2 | April 2021



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Foreword

This is the second edition of the *Best Practice Guideline for Offering Organ and Tissue Donation in Australia (the Guideline)*. Evolution of the Guideline is imperative to maintain currency and to continue to deliver a best practice approach in supporting families to make an informed decision about donation. A key component of this update is to incorporate the requirement for referral of all patients with planned end-of-life care in intensive care units and emergency departments to DonateLife so that the possibility of organ and tissue donation can be explored, and the routine involvement of a Donation Specialist Nurse in family donation conversations.

The opportunity for organ donation is an infrequent event. It comes at an intensely emotional time for families and can be challenging for all involved. Specific knowledge and expertise is required to support families during this time and as they consider making a decision about donation. It is critical that information is provided in a clear and sensitive manner and that it is tailored to the needs of individual families. Excellent care and communication is a central part of supporting families during the end-of-life experience of a family member. Evidence suggests that the care families receive and the quality of the communication with staff, including when discussing donation, influences their satisfaction with the process as well as consent rates for donation.

This Guideline recognises the role of many health professionals involved in supporting families during this time. The Intensivist (or Senior Treating Doctor) manages the end-of-life care of a patient and guides their family through this process, with Critical Care Nurses playing an important role in patient care and family support. The guideline is designed to support all health professionals involved in the family donation conversation and to provide information about the role the Donation Specialist can have in these conversations.

The guidance provided in this document has the goal of ensuring families are supported in making an informed decision about donation. This informed decision making should take into consideration their loved one's wishes if known, their own views and preferences, along with accurate information about donation and what it can mean for themselves and for others. The Guideline is part of a key strategy of the Australian Organ and Tissue Authority (OTA) and DonateLife Network to provide better care and support for families who are considering the possibility of donation for their loved one and to provide every family of a potential donor access to a Donation Specialist during the end-of-life care of their family member.

We thank the families that we have had the privilege to work with and who continue to be our most important teachers. We also thank the Australian and New Zealand Intensive Care Society (ANZICS), College of Intensive Care Medicine of Australia and New Zealand (CICM), Australasian College of Emergency Medicine (ACEM) and the Australian College of Critical Care Nurses (ACCCN) for their contributions and support of the Guideline.

Both ANZICS and CICM strongly recommend that intensivists involved in organ donation complete the core Family Donation Conversation (cFDC) workshop, which includes content consistent with this Guideline. This participation recognises the leadership role of the intensivist in end-of-life care and their role in the collaborative approach in offering donation to families.



Ms Lucinda Barry
Chief Executive Officer
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Dr Helen Opdam
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Best Practice Guideline for Offering Organ and Tissue Donation in Australia

Element 1

Routine referral to DonateLife

Patients in the Intensive Care Unit (ICU) and Emergency Department (ED) for whom there is medical consensus for planned end-of-life care are referred to DonateLife (Agency or hospital Donation Specialist staff). Referral enables DonateLife to assess suitability for donation, check the patient's registration status on the Australian Organ Donor Register (AODR), and facilitates the involvement of a Donation Specialist Nurse* to support planning and family communications.

Element 2

Communicating end-of-life

The family are informed and understand that the patient has died, or that death is expected following the withdrawal of treatment.

Element 3

Planning the approach

A planning meeting occurs prior to the family donation conversation attended by the Senior Treating Doctor[^], the Donation Specialist Nurse, relevant Critical Care Nurse/s and other appropriate staff. The discussion will include sharing information about the patient's AODR registration status and planning the roles each staff member will undertake during the family donation discussion.

Element 4

Discussing donation

Donation is discussed with the family in a collaborative approach involving the Senior Treating Doctor and the Donation Specialist Nurse. Information about donation, including the patient's registration status on the AODR, is shared to assist the family in reaching a fully informed decision about donation.

Element 5

Reviewing practice

A team review occurs after each family donation conversation process to provide an opportunity to reflect upon and improve practice.

* The term 'Donation Specialist Nurse' encompasses the roles of Donation Specialist Nurse, Donation Specialist Coordinator and Donation Specialist Nurse Coordinator.

[^] The term 'Senior Treating Doctor' will be the treating Intensivist or Intensive Care Advanced Trainee in most cases or, if the patient is in the ED, the treating ED Physician or ED Advanced Trainee.

1 Introduction

1.1 Purpose and scope

The purpose of the *Best Practice Guideline for Offering Organ and Tissue Donation in Australia* (the Guideline) is to outline the best practice approach for referral of potential organ and tissue donors and discussing donation with their families. This approach is consistent with similar guidance in the United Kingdom, Canada and Spain.¹⁻³

It outlines the elements of the approach and emphasises key principles. Aspects covered include those of routine referral to DonateLife, checking the AODR, planning for family communication, the importance of a collaborative approach, and the requirement for involvement of a Donation Specialist Nurse.

It is appreciated that the individual aspects of each patient's death and their family's circumstances will influence the timing and approach to discussing donation with the family. A flexible approach that is tailored to the specific circumstances is recommended with consistency in applying the principles and key elements in this Guideline where possible.

The Guideline applies to all patients at medical consensus of end-of-life to ensure that organ and tissue donation is considered at the right time. The Guideline does not address the communication approach for eye and tissue only donation, which can occur in a broader range of contexts including in persons dying outside of ICUs and EDs.

The Guideline does not provide clinical advice on assessing medical suitability of potential donors, donor management, or the organ and tissue donation process. These aspects should be managed in consultation with DonateLife and in accordance with the relevant professional and ethical standards including the *Australian and New Zealand Intensive Care Society (ANZICS) Statement on Death and Organ Donation* and the *Transplantation Society of Australia and New Zealand (TSANZ) Clinical Guidelines for Organ Transplantation from Deceased Donors*.^{4,5}

A glossary of key terms used in this document is provided at **Appendix A**.

1.2 Evidence base

The evidence base in this area relies on comparative and observational studies. Randomised controlled trials are very few. While there are areas of uncertainty, the published literature is suggestive that family satisfaction and consent rates are subject to influence by a variety of factors. These include the perceived quality of medical care, provision of adequate information, the process of offering donation and the training of the personnel leading this communication.^{6,7}

Elements of this Guideline are formally monitored through the regular collection and analysis of DonateLife Audit data. The biennial National Donor Family Study⁸ also provides valuable evidence for the ongoing review and enhancement of the care and support provided to families before, during and after donation.

The DonateLife Audit data shows a positive association with higher rates of donor identification and consent when the key elements of the Guideline are implemented. This includes routine referral to DonateLife of ICU and ED patients where end-of-life care is planned, checking of the patient's AODR status before donation is discussed, team planning before the family donation discussion, the involvement of Donation Specialist Nurses in family donation conversations, and team debriefing to review donation discussions.

In particular, the data from the DonateLife Audit highlights the importance of the role of the Donation Specialist Nurse in family donation conversations using a collaborative approach with the Senior Treating Doctor. Data consistently demonstrates that when donation discussions with families include a Donation Specialist Nurse, consent to donation is considerably higher compared to when no Donation Specialist was involved.

These findings are further supported by a study that examined the impact of DonateLife Donation Specialists (nursing or medical) on the consent rate in 'challenging' organ donation conversations. These are circumstances associated with a lower level of patient/family support for or awareness of donation, as characterised by the patient not being registered on the AODR and staff raising the topic of donation rather than family. The study reported the consent rate to be significantly higher

with involvement of DonateLife Donation Specialist staff (54%) as compared with other trained staff (33%) and untrained staff (28%).⁹ Training was defined as having completed the two-day Family Donation Conversation core workshop. The level of experience and professional training of staff raising donation is regularly reported in the literature as being a key factor in the effectiveness of donation discussions.^{1,2,6,10,11} The quality of the donation conversation and information provided to families has also been shown to impact on family decision making.^{6,12}

1.3 Collaborative approach for offering organ and tissue donation

The Guideline recognises that offering organ and tissue donation should be considered a routine part of end-of-life care with the process best served by a collaborative approach involving the Donation Specialist Nurse and Senior Treating Doctor, supported by Critical Care Nurse/s and other members of the healthcare team.

The Senior Treating Doctor has responsibility for managing the end-of-life care of a patient and leading the end-of-life communication with their family. Critical Care Nurses support the process through reinforcing the medical information conveyed, often having developed rapport with the family while providing care to their relative. Consideration should be given to other staff who can assist in these conversations e.g. social workers, Aboriginal or Torres Strait Islander Liaison Officers and interpreters.

The Donation Specialist Nurse role is central to the family donation conversation. They provide detailed and accurate information about donation and have the skills to sensitively introduce the topic of donation and openly explore what donation may mean for a family, thereby assisting them in making a fully informed decision.

Specifically, Donation Specialist Nurses:

- have specialist knowledge related to donation and transplantation and can provide detailed and accurate information about the possibility and process of donation
- have access to real-time information about donation suitability for transplantation
- undergo specific, regular and focused training in communicating with families about donation and are regularly involved in donation conversations
- can provide support and guidance to hospital staff.

Donation Specialist Doctors may also assist in instances where a Donation Specialist Nurse is not available. Donation Specialist Doctors, through their donation role, have specialist knowledge about organ and tissue donation and may be able to assist in instances where a Donation Specialist Nurse is not available. In some regional and remote hospitals, local hospital guidelines will provide advice regarding other hospital-based staff who have been identified to assist in donation conversations. These staff should be called upon when a Donation Specialist Nurse or Donation Specialist Doctor is not available (following discussions with the DonateLife Agency). The DonateLife Agency will be able to provide telephone support to regional and remote hospital staff that can assist when undertaking donation discussions with families.

2 The Best Practice Guideline for Offering Organ and Tissue Donation in Australia

2.1 Guiding principles

The Guideline is underpinned by the following key guiding principles:

- The goal of discussions about donation is for a decision to be aligned with the patient's wishes (if known) and one that the family will be comfortable with for years to come.
- Supporting families during their time of loss and grief is an important part of the provision of care.
- Health professionals have a responsibility in supporting fully informed decision making for patients and their families in all areas of health care including end-of-life care and choices about organ and tissue donation.
- Donation is discussed with families of potential donors by staff who are skilled communicators, knowledgeable about donation and who have received specific training in this area.
- A collaborative approach involving the Senior Treating Doctor, Donation Specialist Nurse and other support staff, and that is tailored to the family circumstances, is critical in the planning, conduct and evaluation of all family donation conversations.

2.2 Key elements of the Guideline

A summary of the five key elements of the Guideline are on page 3 of this guidance document and the checklist at **Appendix B**. These elements are also summarised as:

- 1 Routine referral to DonateLife
- 2 Communicating end-of-life
- 3 Planning the approach
- 4 Discussing donation
- 5 Reviewing practice.

Routine referral to DonateLife

AODR check

Suitability assessment

Planning for Donation Specialist Nurse involvement in end-of-life communication and the family donation conversation

Communicating end-of-life

Senior Treating Doctor informs family of death or expected death following withdrawal of treatment

Family understands death or expected death following withdrawal of treatment

Discussions about death and donation are separated

Planning the approach

Planning meeting between Senior Treating Doctor, Donation Specialist Nurse, Critical Care Nurse and other healthcare staff

Family donation conversation plan agreed

Discussing donation

Donation Specialist Nurse and Senior Treating Doctor collaboratively offer donation to the family

AODR status shared with family

Reviewing practice

Team review

Led by Donation Specialist Nurse in collaboration with Senior Treating Doctor

Element 1

Routine referral to DonateLife

Patients in the Intensive Care Unit (ICU) and Emergency Department (ED) for whom there is medical consensus for planned end-of-life care are referred to DonateLife (Agency or hospital Donation Specialist staff). Referral enables DonateLife to assess suitability for donation, check the patient's registration status on the Australian Organ Donor Register (AODR), and facilitates the involvement of a Donation Specialist Nurse to support planning and family communications.^{6,10,13}

The treating clinical team should contact the DonateLife hospital staff or DonateLife Agency at the earliest opportunity once there is medical consensus that ongoing treatment is not in the patient's interests and the patient will be transitioning to end-of-life care. This ensures all donation opportunities are identified and assessed by DonateLife to determine suitability for organ donation with additional expert donation and transplantation medical advice sought if required. Advice regarding eye and tissue donation suitability can also be provided, in consultation with eye and tissue banks as appropriate.

The referral facilitates the timely involvement of a Donation Specialist Nurse to check the AODR and, if required, advise on the process of donation and plan for family communication. DonateLife can also provide advice, if needed, on death determination using either circulatory or neurological criteria and on donor physiological management.

The Donation Specialist Nurse will ensure that the AODR has been checked prior to the end-of-life conversation and this information will be shared with the treating clinical team. Having this information during the end-of-life conversation is helpful if the family unexpectedly raises organ donation.^{14–16} The Donation Specialist Nurse can assist the treating medical team in planning family communication about end-of-life care and how to best respond if the family raises donation. It may also be helpful to include the Donation Specialist Nurse in the end-of-life conversation to assist and support the family if they do raise donation at this time.

A registered 'no' on the AODR does not preclude donation being raised with the family. Similar to if the patient had registered 'yes', the family donation conversation will seek to confirm whether the family were aware of their relative's donation intentions and whether they had changed their decision since registering on the AODR.

If a patient is deemed medically unsuitable for donation following referral to DonateLife, this outcome can be communicated to the family in order to avoid any future doubt or regret that donation was not explored at the time. This may be particularly important if the person had registered to donate on the AODR and/or previously expressed preferences about donation.

End-of-life care and the family donation conversation are usually best managed in the ICU. For potential donors in the ED, referral to the ICU should occur according to hospital protocols and practices. If this is not possible, it may be necessary to undertake all of the end-of-life care communication with the family within the ED including offering donation in a collaborative approach with a Donation Specialist Nurse.

Roles and responsibilities

 Senior Treating Doctor	Refers patients in the ICU or ED with planned end-of-life care to Donation Specialist staff in the hospital or DonateLife Agency Plans for Donation Specialist Nurse involvement
 DonateLife	Obtains status of patient's registration on the AODR prior to end-of-life conversations Provides advice on donor suitability and the donation process Plans for Donation Specialist Nurse involvement in family conversations and the donation process

Element 2

Communicating end-of-life

The family are informed and understand that the patient has died, or that death is expected following the withdrawal of treatment.

The Senior Treating Doctor discusses the suspected neurological status consistent with death, the conduct and outcome of examinations to determine neurological death or the planned withdrawal of cardio-respiratory support and expected death. It is the Senior Treating Doctor's responsibility to ensure the family understands that death has or is expected to occur before organ and tissue donation is discussed.

It is important the family understands and accepts that the patient has died or that death is expected before they are asked to consider organ and tissue donation.^{4,12} It is generally advisable, if possible, to separate the end-of-life conversation from the donation conversation. A break assists in ensuring that all family members have reached the same point of understanding regarding the end-of-life discussions and enables families to contact other family members and friends, attend to their personal needs, or spend time at the bedside with the patient. A break also provides the treating clinical team with the opportunity to meet with the Donation Specialist Nurse (if this has not already occurred) and to plan next steps including the family donation conversation. The Donation Specialist Nurse can provide advice on how to sensitively separate these topics or defer discussion about donation if the family raise donation early in the process.

The Senior Treating Doctor may choose to invite the Donation Specialist Nurse to attend the family meeting regarding neurological determination of death or treatment withdrawal in order to support the clinical team and the family. This involvement provides the Donation Specialist Nurse with the opportunity to develop rapport with the family, observe family dynamics and be present to offer information about donation if the family initiate and are ready to discuss donation at this meeting.

If the family raise donation early or during the end-of-life conversation, and a Donation Specialist Nurse is not present, it is recommended the Senior Treating Doctor acknowledge the family's request and advise that it would be preferable to defer the discussion. This enables the involvement of a Donation Specialist Nurse who will be able to provide accurate and more detailed information about donation. Once in attendance, and at a time that meets the family's needs, the Donation Specialist Nurse and Senior Treating Doctor can together continue the discussion about donation. If a Donation Specialist Nurse or Donation Specialist Doctor are not available to attend, then the local DonateLife Agency and hospital guidelines should be followed regarding who is the most suitable person to be involved in donation conversations.

Roles and responsibilities

Senior Treating Doctor	Considers inviting Donation Specialist Nurse into end-of-life meeting Discusses patient care and prognosis with family and ensures there is an understanding of death Separates conversations about death and donation to create time and space for family
Donation Specialist Nurse	Provides advice to treating clinical team in separating death and donation conversations Attends family meeting about death when invited

Element 3

Planning the approach

A planning meeting occurs prior to the family donation conversation attended by the Senior Treating Doctor, the Donation Specialist Nurse, relevant Critical Care Nurse/s and other appropriate staff. The discussion will include sharing information about the patient's AODR registration status and planning the roles each staff member will undertake during the family donation discussion.

A multidisciplinary team planning meeting should occur including all professionals who will be involved in the first family donation conversation and others who may have important information to share about the patient and/or family. This includes the Senior Treating Doctor, the Donation Specialist Nurse, Critical Care Nurse/s (e.g. bedside nurse, and/or nurse in charge) and other appropriate staff or personnel as required (e.g. interpreter, social worker, pastoral care, Aboriginal or Torres Strait Islander Liaison Officer, or religious leader).

Planning will include determining which professionals will be involved in the family donation conversation and agreeing the roles and responsibilities of all involved. The Donation Specialist Nurse and Senior Treating Doctor will agree the way the Donation Specialist Nurse will be introduced to the family (if not already introduced), how the opportunity will be created so they can develop rapport with the family, how the conversation will transition to raise donation, and how information about the patient's AODR registration status will be shared with the family.

The collaborative nature of the planning meeting should also include identifying which components of the conversation the Senior Treating Doctor or the Donation Specialist Nurse are best placed to undertake or respond to, based on the expertise, recency of training and experience regarding donation discussions held by each person. Both people should also view the collaborative discussion as a chance to support each other and as a possible training opportunity.

In this planning meeting the Donation Specialist Nurse, if unfamiliar with the patient and family, will seek to understand previous events and the family's experience since admission. This includes gathering relevant clinical and social patient and family information, and confirming that the family understands death has occurred or is expected to occur.

Planning also involves determining the best time for the family donation conversation and ensuring a suitable private and comfortable location for the meeting to be held. A family spokesperson should be informed that the discussion will involve the consideration of some important decisions and then consulted to identify which family members will be present.

The Donation Specialist Nurse will also discuss the plan for a team review after the family donation conversation and seek agreement on the way this review should occur (guidance on team review is in element 5).

Roles and responsibilities

 Treating clinical team and support staff	Actively participates in the planning meeting
	Shares relevant clinical and social patient and family information
	Agrees roles, responsibilities and approach for the family donation conversation
 Donation Specialist Nurse	Actively participates in the planning meeting
	Shares patient AODR registration status information
	Agrees roles, responsibilities and approach for the family donation conversation
	Confirms plan for team review after the family donation conversation

Element 4
Discussing donation

Donation is discussed with the family in a collaborative approach involving the Senior Treating Doctor and the Donation Specialist Nurse. Information about donation, including the patient's registration status on the AODR, is shared to assist the family in reaching a fully informed decision about donation.

A collaborative approach requires involvement of a Donation Specialist Nurse in every conversation when donation is discussed with the family of a potential organ donor. With the expertise of both the Senior Treating Doctor and Donation Specialist Nurse, a collaborative approach ensures that families are provided with support and donation expertise to deliver accurate information to the family in a sensitive and respectful manner that is tailored to their individual circumstances.

It is important that the transition to the topic of donation occurs in a manner that facilitates the open sharing of information and further explores the patient's and family members' values, beliefs and preferences regarding donation.

The Donation Specialist Nurse is best placed to share factual information about organ and tissue donation, what it may mean in terms of enabling transplantation, their relative's AODR registration status, the donation process and to answer any questions that the family may have in relation to these aspects.

The collaborative team will sensitively support the family and assist them in making a fully informed decision about donation. They will explore a family's decision-making process to ensure that it is based on accurate information and that any decision is free of misconceptions. The family will be assured that, whatever they decide, their decision will be supported. The decision-making process can sometimes require a series of discussions over time.

The Senior Treating Doctor may choose to leave the meeting when comfortable to do so and the Donation Specialist Nurse will continue to support the family. This ongoing support may involve the Donation Specialist Nurse answering any questions the family may have about donation and providing further information about the donation process. Once the family have reached a decision, the next steps will be explained and the family will continue to be supported throughout the process, irrespective of their decision.

Roles and responsibilities

Treating clinical team	Senior Treating Doctor commences the meeting, introduces new staff and confirms family understanding of death
	Works in collaboration with the Donation Specialist Nurse in the family donation conversation, as agreed in the planning meeting
	Provides further information as required to the family
Donation Specialist Nurse	Provides ongoing support to family
	Offers donation to family in collaboration with the Senior Treating Doctor
	Provides the family with factual information about donation and transplantation, including patient AODR status, and answers any questions
	Supports the family in their decision-making
	If the family consents, undertakes next steps in the donation process

Element 5

Reviewing practice

A team review occurs after each family donation conversation process to provide an opportunity to reflect upon and improve practice.

A brief review of the family donation conversation provides an opportunity for participants to reflect upon and evaluate how the process went. The Donation Specialist Nurse is responsible for leading a review meeting and at the earlier planning meeting should gain agreement from all staff participating in the family donation conversation to attend this review (element 3).

The review should take place soon after the family donation conversation has concluded where possible. It is an opportunity for attendees to constructively provide feedback on what went well and what may have been done better or differently. The review should discuss whether the principles and elements of the Guideline were followed.

The most important outcome of the family donation conversation is that the family reach a decision that is right for them. An ongoing study⁸ commissioned by the Organ and Tissue Authority examines family experiences of organ and tissue donation for transplantation. The insights gained from this study provide valuable evidence for the ongoing review and enhancement of the care and support provided to families before, during and after donation.

Roles and responsibilities

Treating clinical team and support staff	Provides feedback on the family donation conversation process
Donation Specialist Nurse	Reflects upon the family donation conversation process Complete the family donation conversation review template

3 Training and skills development

Donation Specialist Nurses are required to undertake extensive and regular communication skills training that are specifically focused on the complexities associated with donation conversations and how to provide information about donation to families experiencing acute grief.

All DonateLife staff are required to complete the core and practical Family Donation Conversation (FDC) workshop at employment, and to repeat the practical FDC workshop every two years, as outlined in role descriptions and the Clinical Practice Improvement Program (CPIP).

The practical FDC workshop provides the opportunity for small group skills practice and exploration of responses to family concerns with the support of skilled facilitators. In addition, focused skills training is provided on a regular basis to all Donation Specialist Nurses. This includes simulation, mentoring, coaching and focussed small group training on key elements of the family donation conversation relevant to Donation Specialist Nurses.

Intensive care medical trainees are required by the College of Intensive Care Medicine to complete the core FDC workshop as part of their mandatory training requirements. It is also recommended that ICU and ED medical and nursing staff who have a role in end-of-life care and family donation conversations attend the DonateLife core FDC workshop. The workshop provides core information about the FDC process, including the expectations of treating clinical staff in the collaborative approach to family communication. The workshop provides information about grief, family reactions to catastrophic news and skills for communicating with families to explain death and donation so as to support informed decision making.

Details of the FDC workshops, including the workshop schedule, are available at www.donatelife.gov.au.

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Appendix A

Glossary

Australian and New Zealand Intensive Care Society (ANZICS)	The leading advocate on intensive care related matters in Australia and New Zealand, including research, ICU data collection, resources and professional education.
Australian College of Critical Care Nurses (ACCCN)	The peak professional association representing critical care nurses.
Australian Organ Donor Register (AODR)	The Australian national register for people to record whether they wish to be an organ and tissue donor for transplantation after death.
Australasian College of Emergency Medicine (ACEM)	The body responsible for training emergency physicians and advancement of professional standards in emergency medicine in Australia and New Zealand.
Clinical Practical Improvement Program (CPIP)	A program that comprises seven elements of clinical strategic focus with associated reportable key performance indicators (KPIs). These KPIs are key to achieving best-practice organ and tissue donation in hospitals with a focus on the intensive care and emergency department environments. The CPIP elements are implemented in all DonateLife Network hospitals and are fundamental to the work of hospital Donation Specialists and the DonateLife Agencies.
College of Emergency Nursing Australasia (CENA)	The peak professional association representing emergency nurses.
College of Intensive Care Medicine of Australia and New Zealand	The body responsible for intensive care medicine specialist training and education in Australia and New Zealand.
Critical Care Nurse	Nursing staff working in intensive care units who are involved in caring for critically ill patients and their families and may be involved in the donation process. Includes bedside nurses, nurse unit managers and senior nurse-in-charge.
DonateLife	The Australian Government brand for all initiatives undertaken as part of the national program to increase organ and tissue donation for transplantation.
DonateLife Agencies	Agencies that provide organ donation coordination services and through which the national DonateLife program is implemented in their respective state or territory. They employ staff to provide organ and tissue donation coordination services, professional education, donor family support, community awareness raising, and audit and data collection.
DonateLife Network Hospitals	Hospitals in Australia with DonateLife Donation Specialist staff that are embedded in the hospital or provide outreach services including undertaking the DonateLife Audit.
DonateLife Network	The national network of organ and tissue donation agencies, hospital-based staff and the OTA, focused on optimising organ and tissue donation practices and increasing rates of donation for transplantation.

Donation Specialist	Includes nurses in the roles of Donation Specialist Nurse, Donation Specialist Nurse Coordinator, Donation Specialist Coordinator and doctors in the role of Medical Donation Specialists. These staff are employed by hospitals or DonateLife Agencies and have specialist knowledge and experience in the process of organ donation from deceased donors for transplantation.
Donation Specialist Doctor	Includes doctors in the roles of Donation Specialist Medical and Medical Donation Specialist.
Donation Specialist Nurse	Includes nurses in the roles of Donation Specialist Nurse, Donation Specialist Nurse Coordinator and Donation Specialist Coordinator.
Emergency Department Physician	An Emergency Medicine specialist, or other specialist with rostered responsibility for patients in the ED.
Family	In this document, 'family' means those closest to the person in knowledge, care and affection, including the immediate biological family; the family of acquisition (related by marriage or contract); and the family of choice and friends (not related biologically or by marriage or contract). ^{4,17}
Family Donation Conversation workshops	OTA/DonateLife workshop-based training that provides health professionals with knowledge and skills in relation to communication with families about death and donation and supporting families to make an informed donation decision.
Intensivist	An Intensive Care specialist, or other specialist with rostered responsibility for patients in the ICU.
Organ and Tissue Authority (OTA)	Australian Government statutory body established under the <i>Australian Organ and Tissue Donation and Transplantation Act 2008</i> to implement the national program.
Senior Treating Doctor	The treating Intensivist or Intensive Care Advanced Trainee in most cases or, if the patient is in the ED, the treating ED Physician or ED Advanced Trainee.
Transplantation Society of Australia and New Zealand (TSANZ)	The peak representative body for transplantation professionals in Australia and New Zealand.

Appendix B

Checklist – Best Practice Guideline for Offering Organ and Tissue Donation in Australia

Elements of the Guideline	Element summary	Resources
Routine referral to DonateLife	1 Routinely refer all patients in the ICU and ED with planned end-of-life care to DonateLife (Agency or hospital Donation Specialist staff) AODR check Planning for Donation Specialist Nurse involvement in end-of-life conversation and FDC	DonateLife AODR
Communicating end-of-life	2 Family understands death or expected death following the withdrawal of treatment Discussions about death and donation are separated	ANZICS Statement DonateLife FDC training
Planning the approach	3 Participate in a planning meeting to plan the FDC and share relevant information including the AODR status Agree roles, responsibilities and introductions for the FDC Confirm intentions for team review after the FDC	DonateLife FDC training
Discussing donation	4 Donation is offered to the family in a collaborative approach involving the Senior Treating Doctor and the Donation Specialist Nurse, supported by other healthcare personnel AODR status shared with family	DonateLife FDC training
Reviewing practice	5 Team review led by Donation Specialist Nurse	Donation Specialist Nurse

Appendix C

FDC planning template

Each FDC process should involve a team planning meeting before the donation conversation. This template may be used to guide the team planning meeting and to record information before, during and after the meeting.

Patient details	
Name of patient	
DOB, age, sex	
Cause of death	
Details and status of the determination of neurological death or the planned withdrawal of treatment, and related family conversations <i>(include anticipated timing of family conversations about end-of-life if these have not already occurred)</i>	
Coroners case / details	
Transfer from another facility	
Clinical picture	
Course of admission	
Length of stay in unit	
Any complications / treatments / misadventure	
Current clinical considerations	
Any previous conditions / co-morbidities / high risk behaviours	
Family details	
Name and relationship of important family members, carers and friends	

Decision makers in the family / senior available next of kin	
How did the family react to the news of neurological death or the decision to withdraw treatment?	
What family members were present?	
How did they show their understanding that their relative had died or would die?	
Has donation been mentioned or volunteered by family or staff? If so, when was it raised, by whom and what was the outcome?	
Is there any indication of the family's current attitudes towards donation? If so, what is it?	
Are there specific issues that should be considered, such as: — Family dynamics? — Relationships with patient? — Grief risk factors? — Cultural, language or religious considerations for patient and for the family?	

Family Donation Conversation

Planned time & location of FDC	Time: Location:
FDC attendees	Family: Staff:
Staff roles	
Introductions (agree exact wording)	
Who will offer donation and how? (agree exact wording)	
Outcome of AODR check and the approach for sharing this with the family	Patient AODR decision (please tick): ☐ Yes ☐ No ☐ Not registered
Plan for team review after FDC	Time: Location:

Appendix D

FDC review template

Each FDC process should be reviewed and discussed by the team as soon as practical after the FDC. This template may be used to guide the team review meeting and to record team discussions.

Date of team review

Case identifiers:

Referral to DonateLife and organisation of the Donation Specialist Nurse

What was done well?

What are the lessons from this case?

Communication with family about end-of-life

What was done well?

Were there alternative methods that could have been used to describe death or expected death to this family?

Were family conversations about death or withdrawal of treatment separated from the donation conversation? (please tick)

☐

Yes

☐

No

Was this appropriate and why?

Team planning

What was done well?

What are the lessons from this case?

Was the following information shared and discussed in the team planning meeting? (please tick)

a Clinical picture of the potential donor	☐ Yes	☐ No
b Outcomes of previous family meetings	☐ Yes	☐ No
c Family dynamics and background	☐ Yes	☐ No
d Australian Organ Donor Register status of potential donor	☐ Yes	☐ No
e Decisions about roles for the Family Donation Conversation	☐ Yes	☐ No
f Agreement on how participants will be introduced	☐ Yes	☐ No
g Discussion of who will offer donation and how	☐ Yes	☐ No
h Planning about location for the Family Donation Conversation	☐ Yes	☐ No

If above information not discussed, why not?

What other information could have been discussed in the team planning meeting to enable the team to better support the family?

Did it feel like a team approach? (please tick)

☐ Yes ☐ No

Any suggestions on alternatives that could have been used?

Family Donation Conversation

What was done well?

What are the lessons from this case?

Was the donation conversation conducted as planned in the team planning meeting?
(please tick)

☐

Yes

☐

No

Why? Why not?

Was any change to the planned approach appropriate?

Did it feel like a team approach? (please tick)

☐

Yes

☐

No

Any suggestions on alternatives that could have been used?

Were there any contributing factors or circumstances specific to this case
(e.g. language barriers, religious or cultural issues, complex family dynamics)? (please tick)

☐

Yes

☐

No

Were there any additional ways that could the team have supported this particular family?

Do you believe the family received sufficient information to make an informed decision
about donation? (please tick)

☐

Yes

☐

No

What additional information could have been provided to this family?

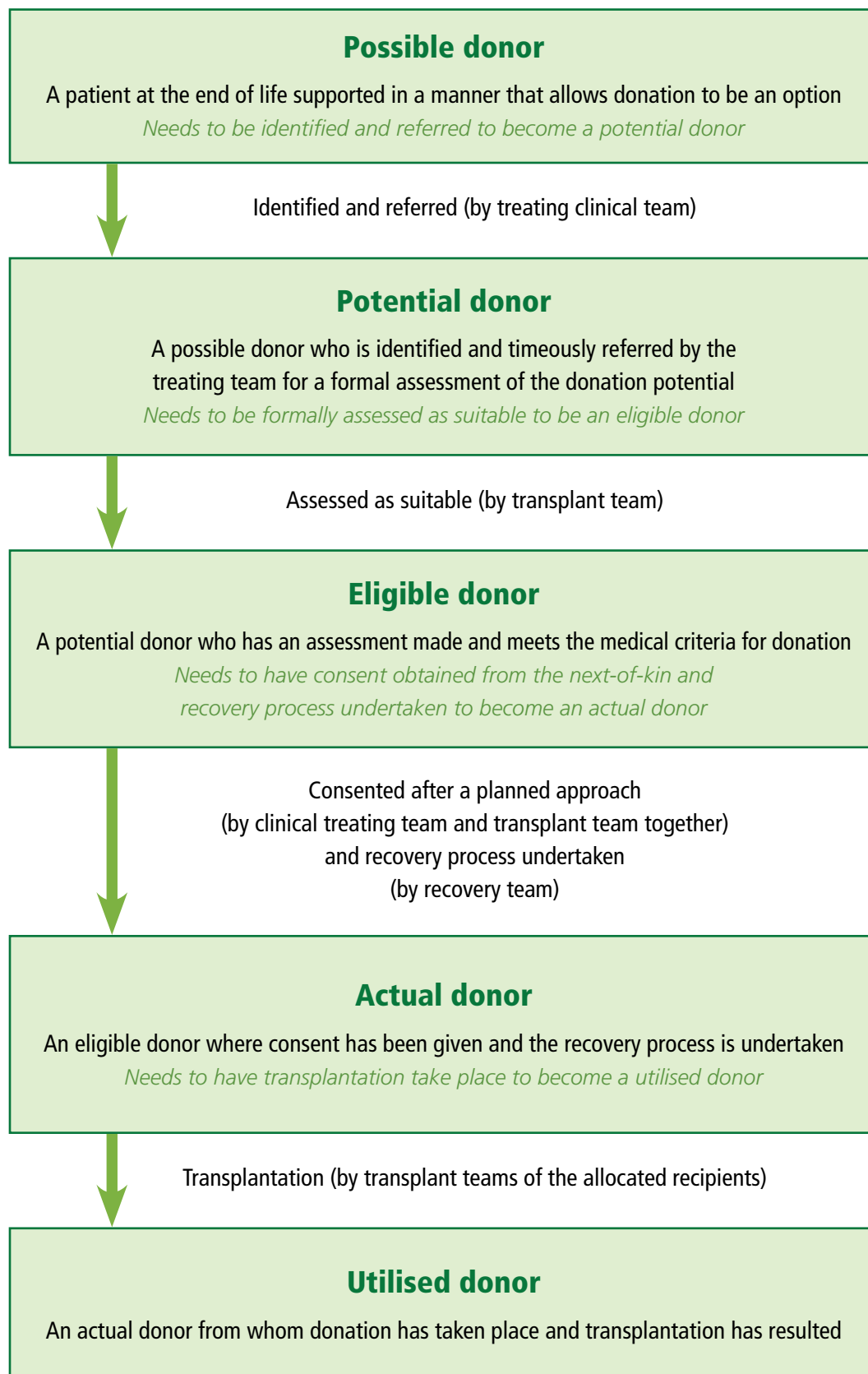
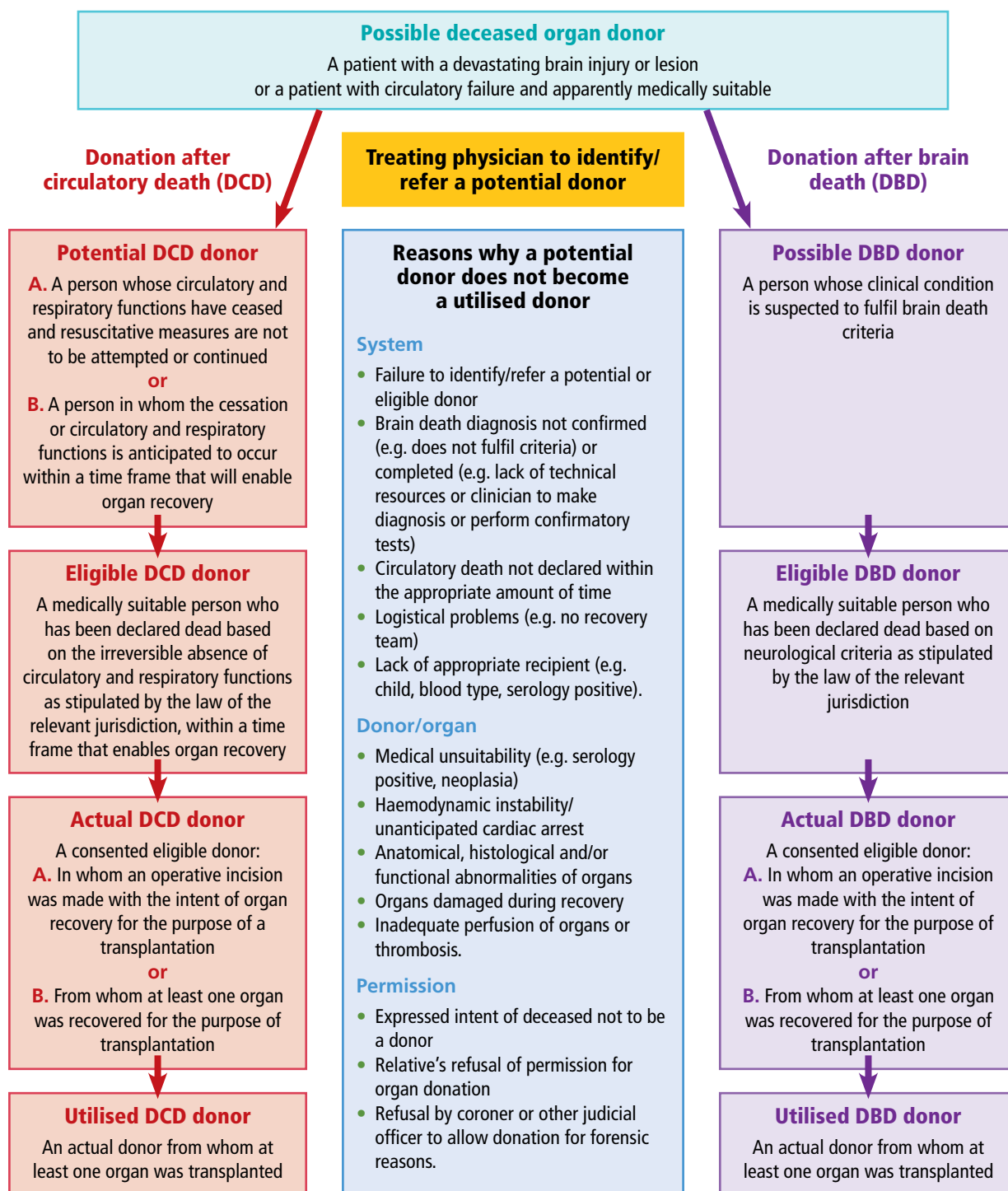


Figure 1 | The critical pathway of organ and tissue donation

Adapted from: Domínguez-Gil B, Delmonico FL, Shaheen FA, *et al.* The critical pathway for deceased donation: reportable uniformity in the approach to deceased donation. *Transpl Int.* 2011 Apr; **24**(4):373-8. doi: 10.1111/j.1432-2277.2011.01243.x. PMID: 21392129. <https://onlinelibrary.wiley.com/doi/full/10.1111/j.1432-2277.2011.01243.x>

Appendix 1

Critical pathways for deceased organ donation



* The "dead donor rule" must be respected. That is, patients may only become donors after death, and the recovery of organs must not cause a donor's death

Domínguez-Gil B, Delmonico FL, Shaheen FA, *et al.* (2011). The critical pathway for deceased donation: reportable uniformity in the approach to deceased donation. *Transplant International*; 24(4): 373-378.

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MEDICINE AND THE LAW

Human tissue and organ transplant provisions: Chapter 8 of the National Health Act and its Regulations, in effect from March 2012 – what doctors must know

David McQuoid-Mason

Where provisions that were in the Human Tissue Act have been left out of the National Health Act they have been included in its regulations. However, new provisions in the latter Act provide strict controls for the transplantation of organs into non-South African citizens or non-permanent residents, and outlaw the charging of fees for human organs. The provisions also expand the list of persons who can give consent to donations from deceased persons to include 'partners', who now take precedence over all other relatives except spouses. Some of the matters in the Human Tissue Act that were not covered by the National Health Act have

now been included in the regulations, such as: (i) the parties responsible for determining death of a person whose organs are to be removed for transplantation purposes; (ii) the requirement that tissue must be harvested within 24 hours from donated bodies; (iii) the removal of eye tissue; (iv) the exclusivity of rights in respect of tissue donations; and (v) confidentiality and publicity regarding tissue and organ transplants.

S Afr Med J 2012;102(9):733-735. DOI:10.7196/SAMJ.6047

On 1 March 2012 the Human Tissue Act¹ was repealed and the provisions in chapter 8 of the National Health Act² (the Act) dealing with the control and use of blood, blood products, tissue and gametes in humans came into effect.³ On 2 March 2012 regulations⁴ were promulgated to incorporate certain provisions in the Human Tissue Act that were not included in the National Health Act. Some of the provisions of the latter and its regulations that doctors need to know about include: (i) tissue donation by living persons; (ii) organ transplantations into non-South African citizens and non-permanent residents; (iii) donations of human bodies and tissue of deceased persons; (iv) consent to donations of human tissue on behalf of deceased persons; (v) allocation and use of human organs of deceased persons; (vi) revocations of donations; (vii) exclusive rights in donations; and (viii) confidentiality and publicity regarding donations.

Tissue donations by living persons

The Act does not state who may consent to donations of human tissue in the case of living persons. However, the regulations provide that tissue, blood or gametes may not be removed from the body of another living person unless written consent has been granted by donors themselves, if they are older than 18 years, or by the parents or guardians of donors aged under 18 years.⁵ In terms of the Act, persons younger than 18 years may not donate tissue that is not replaceable by natural processes, nor may they donate gametes.⁶

The Act does not list the institutions or persons to whom tissue, blood, blood products or gametes from living persons may be donated, but these are provided for in the regulations: (i) hospitals; (ii)

universities or universities of technology; (iii) authorised institutions; (iv) medical practitioners or dentists; or (v) a tissue bank or any person who requires therapy in which the tissue concerned can be used.⁷

The provisions in the Act restricting payment for the importation, acquisition or supply of tissue, blood, blood products or gametes are specific: it is an offence for a donor of tissue, a gamete, blood or a blood product to receive financial or other reward for such donation except for reimbursing their reasonable costs incurred (e.g. travel costs, medical bills). It is also an offence to sell or trade in tissue, gametes, blood or blood products except as provided for in the Act.⁸ A person convicted of either of these offences may be fined or imprisoned for up to 5 years or both.⁹ However, the Act allows a health care provider registered with a statutory health professional council to receive remuneration for any professional service rendered.¹⁰

Organ transplants into non-South Africans or non-permanent residents

A human organ may not be transplanted into a person who is not a South African citizen or a permanent resident without the Minister's written authorisation.¹¹ This provision outlaws the type of situation that arose when Israeli patients came to South Africa to receive kidney transplants from their Brazilian 'relatives'.¹² It makes no difference that a non-South African or non-permanent resident potential donor is a genuine relative of the donee; approval must still be sought from the Minister. However, if the parties are genuinely related, it is highly likely that the Minister will consent.

Doctors who transplant organs into persons who are not South African citizens or permanent residents without authorisation from the Minister, or who charge a fee for a human organ, as distinguished from a fee for professional services, are guilty of an offence and on conviction may be fined or imprisoned (not exceeding 5 years) or both.¹³

Donations of human bodies and tissue of deceased persons

Under the Act,¹⁴ persons competent to make a will may donate their bodies or tissues in a will, or document signed by them and two

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competent witnesses, or in an oral statement in the presence of two competent witnesses. Persons older than 16 years are competent to make a will;¹⁵ competent witnesses are older than 14 years.¹⁶ The will, document or statement must nominate the intended donee institution or person, and should specify how the donor's body or tissue should be used after his or her death. Failure to specify a donee renders the donation null and void. However, this does not apply to organs that have been donated for transplantation or treatment, or for medical or dental training or research, provided they are used in the prescribed manner and the donee determined in accordance with the prescribed procedure.¹⁷

The regulations determine how the donation should be dealt with where no specific institution or donee is named. In such instances, the institution in the appropriate category that is nearest the place where the donor's body or tissue is kept shall be deemed to be the donee. Similarly, if a donation is made to a specific donee who is not within easy reach at the time and place of the death of the donor, the institution in the appropriate category that is nearest to the place of the donor's death shall be deemed to be the donee. Regarding conflicting donations, the regulation states that effect should be given to the donation made last.¹⁸

While the Act defines death as 'brain death', it does not state by whom this must be determined. The regulations provide that in the case of organ transplantation, the death of the deceased has to be determined by two doctors, one of whom must have been qualified for at least 5 years and neither of whom may be members of the transplant team.¹⁹

A person who fails to comply with the prescribed conditions regarding the use of human organs obtained from a deceased person for the purpose of transplantation or treatment or medical or dental training or research, or does not allocate human organs in accordance with the prescribed procedures, is guilty of an offence and on conviction is subject to a fine or imprisonment (up to 5 years) or both.¹³

Consent to donations of human tissue on behalf of deceased persons

Where a donation of tissue has not been made or where the deceased has not prohibited a donation while alive, certain relatives may consent to a donation. The Act provides an order of preference for whom may make a donation in these circumstances, namely the spouse, partner, major child, parent, guardian, major brother or major sister of that person.²⁰ Partners take precedence over all other family members except spouses when it comes to donating tissue or the organs of a deceased person, and other family members may not overrule their decision. While other family members may approach the courts if they can show good cause for why the partner's decision should be overruled, the courts are likely to be reluctant to interfere with a partner's decision made according to the provisions of the Act.

The Act provides that where persons legally empowered to consent to tissue or organ donation on behalf of a deceased person cannot be located after



**THE HEALTHCARE PROFESSIONS COUNCIL OF SOUTH
AFRICA**

**GUIDELINES FOR GOOD PRACTICE
IN THE HEALTHCARE PROFESSIONS**

**SEEKING PATIENTS' INFORMED CONSENT:
THE ETHICAL CONSIDERATIONS**

BOOKLET 4

REVISED: DECEMBER 2021

THE SPIRIT OF PROFESSIONAL GUIDELINES

Good clinical practice is based on a trust relationship between patients and healthcare professionals. Being a good healthcare practitioner requires a life-long commitment to sound professional and ethical practice and an overriding dedication to the interests and wellbeing of one's fellow human beings and society. This makes the practice in the healthcare profession a moral enterprise. It is in this spirit that the HPCSA presents the following ethical guidelines to guide and direct practice by healthcare practitioners. These guidelines are an integral part of the standards of professional conduct against which professional conduct is evaluated.

[Note: The terms "healthcare practitioner" and "healthcare professional" in these guidelines refer to persons registered with the HPCSA].

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1. GUIDANCE TO HEALTHCARE PRACTITIONERS

Being registered under the Health Professions Act No. 56 of 1974 (as amended) gives healthcare practitioners certain authority and privileges. In return, practitioners must meet the standards of competence, care and conduct set by the Health Professions Council of South Africa (HPCSA).

This booklet sets out the principles of good practice which all healthcare practitioners are expected to follow when seeking patients' informed consent to investigations, treatment, screening or research.

2. INTRODUCTION

- 2.1 Successful professional relationships between healthcare practitioner and patients depend upon mutual trust. To establish that trust, practitioner must respect patient's autonomy - their right to decide whether or not to undergo any medical intervention - even where a refusal may result in harm to themselves or in their own death. Patient must be given sufficient information in a manner that they can understand, to enable them to exercise their right to make informed decisions about their care. This is what is meant by an informed consent
- 2.2 The right to an informed consent flows from the South African Constitution, the National Health Act, various other statutes, the common law and the HPCSA's ethical rules and guidelines. Healthcare practitioner is expected to be aware of the laws in this regard. The laws prescribe the minimum requirements when seeking informed consent from patient.
- 2.3 Effective communication is key to enabling patients to make informed decisions. Healthcare practitioner must take appropriate steps to ensure that patient is provided with information about their condition and its treatment. Such dialogue with patient leads to clarity of objectives and understanding and strengthens the quality of the relationship between healthcare practitioner and patient. It provides an agreed framework within which healthcare practitioner can respond effectively to the individual needs of patient. Patient who makes properly informed decisions about their healthcare is more likely to co-operate fully with the agreed management of their conditions.

3. CONSENT TO INVESTIGATION AND TREATMENT

3.1 PROVIDING SUFFICIENT INFORMATION

- 3.1.1 Patient is entitled to information about his/her condition and the treatment options available to them. The amount of information that must be given to each patient will vary according to factors such as the nature of the condition, the complexity of the treatment, the risks associated with the treatment or procedure, and the patient's own wishes. For example, patient may need more information to make an informed decision about a procedure which carries a high risk of failure or adverse side effects, or about an investigation for a condition which, if present, could have serious implications for the patient's employment, social or personal life.
- 3.1.2 The National Health Act requires patients to be given information about:
 - 3.1.2.1 Their patient's health status except in circumstances where there is substantial evidence that the disclosure of the patient's health status would be contrary to the best interests of the patient;

- 3.1.2.2 The range of diagnostic procedures and treatment options generally available to the patient;
 - 3.1.2.3 The benefits, risks costs and consequences generally associated with each option; and
 - 3.1.2.4 The patient's right to refuse health services and explain the implications, risks and obligations of such refusal.
- 3.1.3 Patient has a right to information about any condition or disease from which they are suffering. This information should be presented in a language that the patient understands. The information which patient want or ought to know, before deciding whether to consent to treatment or an investigation, include:
- 3.1.3.1 Details of the diagnosis and prognosis (and the likely prognosis if the condition is left untreated);
 - 3.1.3.2 Uncertainties about the diagnosis, including options for further investigation prior to treatment;
 - 3.1.3.3 Options for treatment or management of the condition, including the option not to treat;
 - 3.1.3.4 The purpose of a proposed investigation or treatment; details of the procedures or therapies involved, including subsidiary treatment such as methods of pain relief; how the patient should prepare for the procedure; and details of what the patient might experience during or after the procedure including common and serious side effects;
 - 3.1.3.5 For each option, explanations of the likely benefits and the probabilities of success; and discussion of any serious or frequently occurring risks, and of any lifestyle changes which may be caused or necessitated by the treatment;
 - 3.1.3.6 Advice about whether a proposed treatment is experimental;
 - 3.1.3.7 How and when the patient's condition and any side effects will be monitored or re-assessed;
 - 3.1.3.8 The name of the practitioner who will have overall responsibility for the treatment and, where appropriate, names of the senior members of his or her team;
 - 3.1.3.9. Whether students will be involved, and the extent to which students may be involved in an investigation or treatment (patients should be informed of their right to refuse the involvement of students);
 - 3.1.3.10 A reminder that patient can change his/her mind about a decision at any time;
 - 3.1.3.11 A reminder that patient have a right to seek a second opinion;
 - 3.1.3.12 Details of costs or charges which the patient may have to meet.
- 3.1.4 When providing information, healthcare practitioner must do their best to find out about patients' individual needs and priorities. For example, patients' beliefs, culture, occupation or other factors may have a bearing on the information they need in order to reach a decision. Healthcare practitioner should not make assumptions about patients' views but should discuss these matters with them and ask them whether they have any concerns

about the treatment or the risks it may involve. Healthcare practitioner should provide patient with appropriate information, which should include an explanation of any risks to which they may attach particular significance. Patient should be asked whether they have understood the information and whether they would like more before making a decision.

- 3.1.5 Healthcare practitioner must not exceed the scope of the authority given by a patient, except in an emergency. Therefore, healthcare practitioner providing treatment or undertaking investigations must give the patient a clear explanation of the scope of consent being sought. This will apply particularly where:
 - 3.1.5.1 Treatment will be provided in stages with the possibility of later adjustments;
 - 3.1.5.2 Different healthcare practitioners provide particular elements of an investigation or treatment (for example anaesthesia during surgery) and where such situation arise, it is incumbent for each practitioner to engage the patient separately and fully for the consent to be completely informed;
 - 3.1.5.3 A number of different investigations or treatments are involved;
 - 3.1.5.4 Uncertainty about the diagnosis or about the appropriate range of options for treatment may be resolved only in the light of findings once an investigation or treatment is underway, and when the patient may be unable to participate in decision making.
- 3.1.6 In the cases referred to in para 3.1.5 above, healthcare practitioner should explain how decisions will be made about whether or when to move from one stage or one form of treatment to another. There should be a clear agreement about whether the patient consent to all or only parts of the proposed plan of investigation or treatment, and whether further consent will have to be sought at a later stage.
- 3.1.7 Healthcare practitioner should raise with patient the possibility of additional problems emerging during a procedure when the patient is unconscious or otherwise unable to make a decision. They should seek consent to treat any problems which they think may arise and ascertain whether there are any procedures to which the patient would object, or prefer to give further thought to before they proceed. Healthcare practitioner must abide by patients' decisions on these issues. If in exceptional circumstances healthcare practitioners decide, while the patient is unconscious, to treat a condition which falls outside the scope of the patient's consent, their decision may be challenged in the courts, or be the subject of a complaint to their employers or the HPCSA. Healthcare practitioner should therefore seek the views of an experienced colleague, wherever possible, before providing the treatment. They must be prepared to explain and justify their decisions based on such consideration as preservation of life. Healthcare practitioner must tell the patient what they have done and why, as soon as the patient is sufficiently recovered to understand.

3.2 RESPONDING TO QUESTIONS

Healthcare practitioner must respond honestly to any questions the patient raises and, as far as possible, answer as fully as the patient wishes. In some cases, a patient may ask about other treatments that are unproven or ineffective. Some patients may want to know whether any of the risks or benefits of treatment are affected by the choice of institution or practitioner providing the care. In any given situation where there's limitation in their competence/experience to provide the fullest possible answer, they should commit to seek the additional information from credible sources and provide same to the patient. Healthcare practitioner must answer such questions as fully, accurately and objectively as possible.

3.3 WITHHOLDING INFORMATION

- 3.3.1 Healthcare practitioner should not withhold information necessary for decision making, unless they judge that disclosure of some information could cause the patient serious harm. In this context, serious harm does not mean the patient would become upset or decide to refuse treatment.
- 3.3.2 The South African courts have held that patients must be informed of all “material risks” in order to give a proper informed consent. A risk is “material” if:
 - 3.3.2.1 A reasonable person in the position of the patient, if warned of the risk, would attach significance to it; and
 - 3.3.2.2 The healthcare practitioner should reasonably be aware that the patient, if warned of the risk, would attach significance to it.
- 3.3.3 No-one may make decisions on behalf of a mentally competent adult. If patients ask healthcare practitioner to withhold information and make decisions on their behalf or nominate a relative or third party to make decisions for them, the healthcare practitioner should explain the importance of patient knowing the options open to them, and what the treatment they may receive will involve. If patient insist that they do not want to know in detail about their condition and its treatment, the healthcare practitioner should still provide basic information about the treatment. If a relative asks a healthcare practitioner to withhold information, the latter must seek the views of the patient. Again, healthcare practitioner should not withhold relevant information unless they consider that it would cause the patient serious harm.
- 3.3.4 The National Health Act provides that healthcare practitioner must provide patient with information about their health status, unless “there is substantial evidence that the disclosure of the patient’s health status would be contrary to the best interests of the patients”.
- 3.3.5 In any case where healthcare practitioner withholds relevant information from the patient, they must record this, and the reason for doing so, in the patient's health records and they must be prepared to explain and justify their decision.

3.4 PRESENTING INFORMATION TO PATIENTS

- 3.4.1 Obtaining informed consent cannot be an isolated event. It involves a continuing dialogue between healthcare practitioner and their patient or their next of kin which keeps them abreast of changes in the condition of patients and the treatment or investigation the practitioners propose. Whenever possible, healthcare practitioner should discuss treatment options at a time when the patient is best able to understand and retain the information.
- 3.4.2 To be sure that their patient understand, healthcare practitioner should give clear explanations and give the patient time to ask questions. In particular, healthcare practitioner should:
 - 3.4.2.1 Use up-to-date written material, visual and other aids to explain complex aspects of the investigation, diagnosis or treatment where appropriate and practicable;
 - 3.4.2.2 Make arrangements, wherever possible, to meet particular language and communication needs, for example through translations, independent interpreters, people who sign on behalf of patients, or the patient's representative;

- 3.4.2.3 Where appropriate, discuss with patients the possibility of being accompanied by a relative or friend, as appointed by patient;
- 3.4.2.4 Explain the probabilities of success, prognosis, the risk of failure of, or harm associated with options for treatment, using accurate data;
- 3.4.2.5 Ensure that information which patients may find distressing is given to them in a considerate way. Provide patients with information about counseling services and support groups, where appropriate;
- 3.4.2.6 Allow patient sufficient time to reflect, before and after making a decision, especially where the information is complex or the severity of the risks is great. Where patients have difficulty understanding information, or there is a lot of information to absorb, it may be appropriate to provide it in manageable amounts, with appropriate written or other back-up material, over a period of time, or to repeat it;
- 3.4.2.7 Involve nursing or other members of the healthcare team in discussions with the patient, where appropriate. They may have valuable knowledge of the patient's background or particular concerns, for example in identifying what risks the patient should be told about);
- 3.4.2.8 Obtaining of informed consent belongs exclusively with the primary practitioner and should not be delegated.
- 3.4.2.9 Ensure that, where treatment is not to start until sometime after informed consent has been obtained, patients are given clear instructions on how to review their decision with the healthcare practitioner providing the treatment.
- 3.4.2.10 Consent will not be informed if it was given as a result of duress, coercion, manipulation, misrepresentation or mental impairment (e.g. under the influence of alcohol, drugs, including premedication in the theatre).

4. WHO OBTAINS CONSENT?

- 4.1 A healthcare practitioner providing treatment or undertaking an investigation has the responsibility to discuss it with the patient and obtain informed consent, as the practitioner will have a comprehensive understanding of the procedure or treatment, how it is to be carried out, and the risks attached to it. Where this is not practicable, healthcare practitioner may delegate these tasks provided they ensure that the person to whom they delegate:
 - 4.1.1 Is suitably educated, trained and qualified;
 - 4.1.2 Has sufficient knowledge of the proposed investigation or treatment and understands the risks involved; and
 - 4.1.3 Acts in accordance with the guidance in this Booklet.
- 4.2 A healthcare practitioner will remain responsible for ensuring that before he or she starts any treatment, the patient has been given sufficient time and information to make an informed decision and has given consent to the investigation or procedure.

5. THE RIGHT OF PATIENTS TO INFORMATION

- 5.1 Patients have a right to information about the healthcare services available to them, presented in a way that is easy to follow and use.
- 5.2 The National Health Act provides that healthcare providers (this includes healthcare practitioners) must inform users (patients) of the following:
 - 5.2.1 The user's health status except in circumstances where there is substantial evidence that the disclosure of the user's health status would be contrary to the best interests of the user;
 - 5.2.2 The range of diagnostic procedures and treatment options generally available to the user;
 - 5.2.3 The benefits, risks costs and consequences generally associated with each option; and
 - 5.2.4 The user's right to refuse health services and explain the implications, risks and obligations of such refusal.

6. ENSURING VOLUNTARY DECISION MAKING

- 6.1 It is for the patient, not the healthcare practitioner, to determine what is in the patient's own best interests. Nonetheless, practitioners may wish to recommend a treatment or a course of action to patient, but they must not put pressure on patient to accept their advice. In discussions with patients, healthcare practitioners should:
 - 6.1.1 Give a balanced view of the options;
 - 6.1.2 Explain the need for informed consent.
- 6.2 Healthcare practitioner must declare any potential conflicts of interest, for example where they or their organisation benefit financially from the use of a particular drug or treatment, or treatment at a particular institution if permitted by the HPCSA.
- 6.3 Pressure may be put on patients by employers, insurance companies or others to undergo particular tests or accept treatment. Healthcare practitioner should do their best to ensure that patient has considered the options and reached their own decision. Healthcare practitioner should take appropriate action if they believe patient is being offered inappropriate or unlawful financial or other rewards.
- 6.4 Patients who are detained by the police or immigration authorities, or are in prison, and those detained under the provisions of any mental health legislation may be particularly vulnerable. Where such patients have a right to decline treatment, healthcare practitioners should do their best to ensure that they know this and are able to exercise this right.

7 EMERGENCIES

- 7.1 In an emergency, where consent cannot be obtained, healthcare practitioner may provide medical treatment to anyone who needs it, provided the treatment is limited to what is immediately necessary to save life or avoid significant deterioration in the patient's health.
- 7.2 However, healthcare practitioner must respect the terms of any valid advance refusal by the patient which they know about, or which is drawn to their attention.

- 7.3 After the emergency healthcare practitioner should tell the patient what has been done and why, as soon as the patient is sufficiently recovered to understand.

8 ESTABLISHING CAPACITY TO MAKE DECISIONS

8.1 ASSESSING MENTAL CAPACITY

- 8.1.1 Healthcare practitioner must work on the presumption that every adult has the capacity to decide whether to consent to, or refuse, proposed medical intervention, unless it is shown that they cannot understand information presented in a clear way.
- 8.1.2 If a patient's choice appears irrational, or does not accord with the healthcare practitioner's view of what is in the patient's best interests, this is not evidence in itself that the patient lacks competence. In such circumstances it may be appropriate to review with the patient whether all reasonable steps have been taken to identify and meet their information needs.
- 8.1.3 Where healthcare practitioner need to assess a patient's capacity to make a decision, they should consult the guidance issued by the relevant professional bodies.
- 8.1.4 In the case of children who have legal capacity to give consent in terms of the Child Care Act (Act No. 35 of 2005) or the Choice on Termination of Pregnancy Act (Act No.92 of 1996), health care practitioners should make sure that the children have sufficiently mental maturity to understand the nature and effect of the treatment or procedure to which they are consenting (see below para 8.5).

8.2 FLUCTUATING CAPACITY

- 8.2.1 Where patient has difficulty retaining information or are only intermittently competent to make a decision, healthcare practitioner should provide any assistance they might need to reach an informed decision.
- 8.2.2 Healthcare practitioner should record any decision made while the patient was competent, including the key elements of the consultation.
- 8.2.3 Healthcare practitioner should review any decision made whilst the patient was competent, at appropriate intervals before treatment starts, to establish that their views are consistently held and can be relied on.

8.3 MENTALLY INCAPACITATED PATIENTS

- 8.3.1 The National Health Act makes provision for certain persons to consent on behalf of mentally incompetent patients to an operation or medical treatment where such patients are unable to give the necessary consent and have not mandated - while still mentally competent- somebody else in writing to give consent on their behalf.
- 8.3.2 The Act sets out a priority list of persons who may consent in such circumstances:
- 8.3.2.1 A person authorized by the court (e.g. a curator); or
 - 8.3.2.2 In order of priority, the patient's spouse, partner, parent, grandparent, major child or brother or sister.
- 8.3.3 Healthcare practitioners should also consult the provisions of the Mental Health Care Act (Act No.17 of 2002) when dealing with mentally ill patients.

8.4 THIRD PARTY NOMINATIONS IN REGARD TO CONSENT

- 8.4.1 The National Health Act allows patient – while still mentally competent - to mandate another person in writing to give consent on their behalf
- 8.4.2 If healthcare practitioner is treating a patient who has lost the capacity to consent to or refuse treatment, for example through the onset or progress of a mental disorder or other disabilities, they should try to find out whether:
 - 8.4.2.1 The patient has previously mandated someone else in writing to make decisions on their behalf; or
 - 8.4.2.2 Have indicated preferences in an advance statement (e.g. an “advance directive” or “living will”).
- 8.4.3 Healthcare practitioner must respect any refusal of treatment given when the patient was competent, provided the decision in the advance statement is clearly applicable to the present circumstances, and there is no reason to believe that the patient has changed his or her mind. Where an advance statement of this kind is not available, the patient's known wishes should be taken into account

8.5 CHILDREN

The ages as stipulated in this document are a reflection of the Children's Act, 2005 (Act No. 38 of 2005).

- 8.5.1 Healthcare practitioner must assess a child's capacity to decide whether to consent to or refuse a proposed investigation or treatment before they provide it.
- 8.5.2 In general, a competent child will be able to understand the nature, purpose and possible consequences of the proposed investigation or treatment, as well as the risks and consequences of non-treatment.
- 8.5.3 A healthcare practitioner's assessment must take account of the following:
 - 8.5.3.1 A person over the age of 18 years is an adult and is legally competent to decide on all forms of treatment and medical procedures (Children's Act, 2005).
 - 8.5.3.2 A child who is 12 years of age is legally competent to consent to medical treatment if the child is of sufficient maturity and has the mental capacity to understand the benefits, risks, social and other implications of the treatment (Children's Act, 2005 (Act No. 38 of 2005) section 129(2)).
 - 8.5.3.3 Where a child is under 12 years of age or not of sufficient maturity and does not have the necessary mental capacity his or her parent, guardian or care-giver may give consent to medical treatment (Children's Act, 2005 section 129(4)).
 - 8.5.3.4 A child who is 12 years of age is legally competent to consent to a surgical operation if the child is of sufficient maturity, has the mental capacity to understand the benefits, risks, social and other implications of the surgical operation, and is duly assisted by his or her parent or guardian (Children's Act, 2005 section 129(3)).

- 8.5.3.5 Where a child is under 12 years of age or not of sufficient maturity and does not have the necessary mental capacity his or her parent or guardian may give consent to a surgical procedure (Children's Act, 2005 section 129(5)).
- 8.5.3.6 The superintendent of a hospital or the person in charge of a hospital in the absence of the superintendent may consent to the medical treatment of or a surgical operation on a child if the treatment or operation is necessary to preserve the life of the child or to save the child from serious or lasting physical injury or disability; and the need for the treatment or operation is **so** urgent that it cannot be deferred for the purpose of obtaining consent that would otherwise have been required (Children's Act, 2005 section 129(6)).
- 8.5.3.7 The Minister of Health may consent to the medical treatment of, or surgical operation on a child if the parent or guardian of the child unreasonably refuses to give consent or to assist the child in giving consent, is incapable of giving consent or of assisting the child in giving consent, cannot readily be traced, or is deceased (Children's Act, 2005 section 129(7)).
- 8.5.3.8 The Minister may consent to the medical treatment of or surgical operation on a child if the child unreasonably refuses to give consent (Children's Act, 2005 section 129(8)).
- 8.5.3.9 A High Court or children's court may consent to the medical treatment of or a surgical operation on a child in all instances where another person that may give consent refuses or is unable to give such consent (Children's Act, 2005 section 129(9)).
- 8.5.3.10 No parent, guardian or care-giver of a child may refuse to assist a child or withhold consent by reason only of religious or other beliefs, unless that parent or guardian can show that there is a medically accepted alternative choice to the medical treatment or surgical operation concerned (Children's Act, 2005 section 129(10)).
- 8.5.3.11 A female of any age is legally competent to consent to a termination of pregnancy (Choice on Termination of Pregnancy Act, 1996 (Act No. 92 of 1996)) provided she has the necessary mental capacity to give an informed consent by understanding and appreciating the benefits, risks, social and other implications of the termination of pregnancy.

9. THE "BEST INTERESTS" PRINCIPLE
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- 9.1 In deciding what options may be reasonably considered as being in the best interests of a patient who lacks capacity to decide, healthcare practitioner should take into account:
 - 9.1.1 The options for investigation or treatment which are clinically indicated;
 - 9.1.2 Any evidence of the patient's previously expressed preferences, including an advance statement;
 - 9.1.3 Their own and the healthcare team's knowledge of the patient's background, such as cultural, religious or employment considerations;

- 9.1.4 Views about the patient's preferences given by a third party who may have other knowledge of the patient, for example, the patient's partner, family, carer, or a person with parental responsibility;
- 9.1.5 Which option least restricts the patient's future choices, where more than one option (including non-treatment) seems reasonable in the patient's best interests.
- 9.2 The South African Constitution provides that “a child’s best interests are paramount in every matter concerning a child”.

10. APPLYING TO THE COURT

- 10.1 Where a patient's capacity to consent is in doubt, or where differences of opinion about his or her best interests cannot be resolved satisfactorily, or third party is not nominated, healthcare practitioner should consult more experienced colleagues and, where appropriate, seek legal advice on whether it is necessary to apply to the court for a ruling.
- 10.2 Healthcare practitioner should seek the court where a patient lacks capacity to consent to a medical intervention and the situation is contentious for example, parent withholding consent to life-saving treatment for children under the age of 12 years, in other words, in any situation where the interest of the patient is protected by a constitutional right.
- 10.3 Where healthcare practitioner decides to apply to a court they should, as soon as possible, inform the patient, or his or her representative of their decision and of his or her right to be represented at the hearing.

11. FORMS OF CONSENT

To determine whether patients have given informed consent to any proposed investigation or treatment, healthcare practitioners must check how well the patients have understood the details and implications of what is proposed, and not simply rely on the form in which their consent has been expressed or recorded – especially where the initial consent was obtained by a third party.

12 EXPRESS CONSENT

- 12.1 Patients can indicate their informed consent either recorded or in writing.
- 12.2 In some cases, the nature of the risks to which the patient might be exposed makes it important that a written record is available of the patient's consent and other wishes in relation to the proposed investigation and treatment. This helps to ensure later understanding between healthcare practitioner, the patient and anyone else involved in carrying out the procedure or providing care.
- 12.3 Except in an emergency, where the patient has the capacity to give consent, healthcare practitioners should obtain written consent in the following cases, although this list is not exhaustive:
 - 12.3.1 The treatment or procedure is complex or involves significant risks and/or side effects;
 - 12.3.2 Providing clinical care is not the primary purpose of the investigation or examination;

12.3.3 There may be significant consequences for the patient's employment, social or personal life;

12.3.4 The treatment is part of a research programme.

12.4 Healthcare practitioner must use the patient's formal health records or the consent form to detail the key elements of the discussion with the patient, including the nature of information provided, specific requests by the patient, and details of the scope of the consent given.

13. STATUTORY REQUIREMENTS

Some statutes require particular forms of consent to be obtained for specific procedures (for example, sterilizations (Sterilization Act (Act No. 44 of 1998), terminations of pregnancy (Choice on Termination of Pregnancy Act (No.92 of 1996) and removal of organs from deceased people (National Health Act, 2003). Healthcare practitioners need to consult the law in this regard when carrying out such procedures.

14. IMPLIED CONSENT

Healthcare practitioner should be careful about relying on a patient's apparent compliance with a procedure as a form of consent. Submission in itself may not necessarily indicate consent. For example, the fact that a patient lies down on an examination couch does not indicate that the patient has understood what the health care practitioner proposes to do and why. Consent must at all times be expressed and not implied.

15. REVIEWING CONSENT

15.1 A signed consent form is not sufficient evidence that a patient has given, or still gives, informed consent to the proposed treatment in all its aspects. Healthcare practitioners must review the patient's decision close to the time of treatment, and especially where:

16.1.1 Significant time has elapsed between when the consent was obtained and the start of treatment;

16.1.2 There have been material changes in the patient's condition, or in any aspects of the proposed treatment plan, which might invalidate the patient's existing consent;

16.1.3 New, potentially relevant information has become available, for example about the risks of the treatment or about other treatment options.

16. CONSENT TO SCREENING AND TESTING

16.1 Screening or testing of healthy or asymptomatic people to detect genetic predispositions or early signs of debilitating or life threatening conditions can be an important tool in providing effective care. However, the uncertainties involved in screening or testing may be great, for example the risk of false positive or false negative results. Some findings may potentially have serious medical, social or financial consequences not only for the individuals, but for their relatives. In some cases the fact of having been screened or tested may itself have serious implications.

16.2 Healthcare practitioner must ensure that anyone considering whether to consent to screening or testing can make a properly informed decision. As far as possible, practitioners should ensure that screening or testing is not contrary to the individual's interests. Healthcare practitioner must pay particular attention to ensuring that the information the person wants or ought to have is identified and provided. Practitioner should be careful to explain clearly:

- 16.2.1 The purpose of the screening or test;
- 16.2.2 The likelihood of positive or negative findings and the possibility of false positive or negative results;
- 16.2.3 The uncertainties and risks attached to the screening or testing process;
- 16.2.4 Any significant medical, social or financial implications of screening or testing for the particular condition or predisposition;
- 16.2.5 Follow up plans, including the availability of counseling and support services.

17. ICD10 CODING AND INFORMED CONSENT

Informed consent is an exercise of an **informed choice** by a patient who **has the capacity** to give consent:-

- a) in instances where there are multiple options or alternatives to treatment; or
- b) in making a decision whether to withhold or disclose information or allow someone else to disclose information on their medical condition to a defined third party; or
- c) in making a decision for purposes of reimbursement by a Medical Scheme,

based on adequate information and a detailed analysis or unpacking of each of the options or alternatives as well as the legislative requirements for disclosure of such information.

This means there must be a full and frank disclosure of all the material facts to enable the patient to decide from an informed basis. With regards to ICD 10, for instance, the patient should be given information as to who will access their information, for what purpose and what would be the implications of the utilization of such information etc.

17.1 PROVIDER RESPONSIBILITIES: ICD-10

Healthcare provider have the following obligations, the list not being exhaustive as any other ethical obligation in handling and dealing with patient information and respecting their confidentiality will be required:-

- a) to provide information to the patients about the legislative requirement of supplying ICD-10 codes to the medical schemes for purposes of reimbursements and the inevitable consequences of the medical scheme becoming aware of the diagnosis of the patient/member;
- b) to procure patient's consent to release ICD-10 coding to the medical scheme and/or (where required) to the other health professional (within the healthcare team);
- c) to advise the patient of their choice not to have their ICD-10 coding divulged to the medical scheme which would mean the patient has to settle the provider's account directly; and

- d) to indicate that the practitioner does not have control over the management and utilization of this information once divulged over to the medical scheme and that the medical scheme takes responsibility for any further disclosure or utilization of such information for whatever purpose.

It is strongly suggested that written consent be procured from the patient by the provider in order to safeguard the interests of both parties. Consent by a patient may be once-off in relation to the treatment of a similar condition provided there is a verbal reminder to the patient about their initial commitment to confirm if they are still comfortable with the disclosure. It would be advisable for a provider to note the verbal reminder on that patient's file. Where a patient presents with a new or different condition, a fresh consent should be obtained from the patient and appropriately documented.

Provider without a direct patient contact like Pathologists and Radiologists act on referrals from other providers. Their responsibility would be to ensure that the referring provider has procured consent for that other provider (in this case a pathologist or radiologist) to access and also disclose the information to the medical scheme for reimbursement purposes.

17.2 INFORMED CONSENT FOR INTER-COUNCIL INTERACTION OF HEALTH PROFESSIONALS AS MEMBERS OF A HEALTHCARE TEAM
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Sharing of information with members of a healthcare team providing a health service to a patient would be permissible to the extent that it is necessary to enhance the quality of care to be provided to that patient and the patient has given consent to treatment and disclosure of such information to another healthcare practitioner. This would include members beyond the HPCSA.

Ethical guidelines for good practice in the health care professions

The following Booklets are separately available:

- Booklet 1: General ethical guidelines for healthcare professions***
- Booklet 2: Ethical and professional rules of the health professions council of South Africa as promulgated in government gazette R717/2006***
- Booklet 3: National Patients' Rights Charter***
- Booklet 4: Seeking patients' informed consent: The ethical considerations***
- Booklet 5: Confidentiality: Protecting and providing information***
- Booklet 6: Guidelines for the management of patients with HIV infection or AIDS***
- Booklet 7: Guidelines withholding and withdrawing treatment***
- Booklet 8: Guidelines on Reproductive Health management***
- Booklet 9: Guidelines on Patient Records***
- Booklet 10: Guidelines for the practice of Telehealth***
- Booklet 11: Guidelines on over servicing, perverse incentives and related matters***
- Booklet 12: Guidelines for the management of healthcare waste***
- Booklet 13: General ethical guidelines for health researchers***
- Booklet 14: Ethical Guidelines for Biotechnology Research in South Africa***
- Booklet 15: Research, development and the use of the chemical, biological and nuclear weapons***
- Booklet 16: Ethical Guidelines on Social Media***
- Booklet 17: Ethical Guidelines of Palliative Care***

their death, the Director-General of Health may donate any specific tissue of that person to an institution or a person, provided all the 'prescribed steps' have been taken to locate the person authorised to give consent.²¹ While the General Regulations do not mention what the 'prescribed steps' are, these steps have been published for donations of human biological material for genetic testing, genetic training and genetic health research for therapeutic purposes, in the Regulations for the Use of Human Biological Material.²² The latter Regulations require the Director-General to obtain the name, address and telephone number of the spouse, partner, major child, parent, guardian, major brother or major sister of the deceased person from: (i) any person working in the relevant hospital, institution or facility where the deceased died; or (ii) any person who visited the deceased before he or she died.²³ It is submitted that these steps should also be followed for other donations of human tissue by the Director-General under the National Health Act.

Allocation and use of human tissue and organs from deceased persons

The Act states that, except in the case of a whole-body donation, where a body has been donated to an institution for removal of tissue, the tissue must be harvested within a period of 24 hours, after which the body may be claimed by the following in the order specified: a spouse, partner, major child, parent, guardian, major brother or major sister.²⁴ Relaxed provisions regarding the removal of eye tissue are also included in the regulations.²⁵

Except as provided for in any other law, any person who acquires, uses or supplies a body of a deceased person or any tissue, blood or gametes of a living or deceased person in any manner or for any purpose, other than as provided for by the Act and its regulations, is guilty of an offence and liable on conviction to a fine or imprisonment for a period of 10 years or to both.²⁶

Revocation of donations

A donor may, before transplantation of the relevant organ into the donee, revoke the donation in the same way in which it was made. In the case of donation by a will or other document, revocation may be by the intentional destruction of such will or document by the person making it.²⁷ Donors may therefore change their mind at any time before transplantation of the organ into the donee. Persons consenting to the donation of tissue or organs on behalf of a deceased person may also change their minds and revoke the donation at any stage. However, it is submitted that such revocations must be done within the limits of good medical practice, for instance not at the last minute when the transplant operation on the donee is already underway.

Exclusive rights in donations

The Act does not provide that the persons who acquire the body of a deceased person or any tissue, blood or gamete in terms of the Act or its regulations have exclusive rights over them, but the regulations provide for such rights by the donee.²⁸

Confidentiality and publicity regarding donations

The regulations prohibit the publication, without the consent of the appropriate person concerned, of any information that would identify: (i) a deceased person whose body or any specific tissue thereof was donated; (ii) the donor of the body of a deceased person or any specific tissue thereof; (iii) a living person from whose body any tissue, blood or gamete has been removed or withdrawn for any purpose; or (iv) a person who consented to the removal of any tissue, blood or gametes from a living person for such purposes. The regulations also prohibit the publication, without the consent of the appropriate person, of information that would identify the recipient of any tissue removed from the body of a deceased or living person.²⁹

1. Human Tissue Act No. 65 of 1983.
2. National Health Act No. 61 of 2003.
3. Proclamation No. 11 in Government Gazette No. 35081 of 27 February 2012.
4. Government Notices R175-R183 in Government Gazette No. 35099 of 2 March 2012.
5. Regulation 2 of the Regulations Regarding the General Control of Human Bodies, Tissue, Blood, Blood Products and Gametes in Government Notice R180 of Government Gazette No. 35099 of 2 March 2012.
6. Section 56(2) of the National Health Act No. 61 of 2003.
7. Regulation 4 of the Regulations Regarding the General Control of Human Bodies, Tissue, Blood, Blood Products and Gametes in Government Notice R180 of Government Gazette No. 35099 of 2 March 2012.
8. Section 60(4) of the National Health Act No. 61 of 2003.
9. Section 60(2) of the National Health Act No. 61 of 2003.
10. Section 60(3) of the National Health Act No. 61 of 2003.
11. Section 61(3) of the National Health Act No. 61 of 2003.
12. Khoza SS. The human organ trade: The South African tragedy. *South African Journal of Bioethics and Law* 2009; 2(2):46-47.
13. Section 61(5) of the National Health Act No. 61 of 2003.
14. Section 62(1) of the National Health Act No. 61 of 2003.
15. Section 1 of the Wills Act No. 7 of 1953.
16. Section 4 of the Wills Act No. 7 of 1953.
17. Section 62(1) read with section 61(1) and (2) of the National Health Act No. 61 of 2003.
18. Regulation 6 of the Regulations Regarding the General Control of Human Bodies, Tissue, Blood, Blood Products and Gametes in Government Notice R180 of Government Gazette No. 35099 of 2 March 2012.
19. Regulation 9 of the Regulations Regarding the General Control of Human Bodies, Tissue, Blood, Blood Products and Gametes in Government Notice R180 of Government Gazette No. 35099 of 2 March 2012.
20. Section 62(2) of the National Health Act No. 61 of 2003.
21. Section 62(3) of the National Health Act No. 61 of 2003.
22. Regulations relating to the Use of Human Biological Material in Government Notice R177 of Government Gazette No. 35099 of 2 March 2012.
23. Regulation 4(2) of the Regulations relating to the Use of Human Biological Material in Government Notice R177 of Government Gazette No. 35099 of 2 March 2012.
24. Regulation 8(1) and (2) of the Regulations Regarding the General Control of Human Bodies, Tissue, Blood, Blood Products and Gametes in Government Notice R180 of Government Gazette No. 35099 of 2 March 2012.
25. Regulation 8(3) of the Regulations Regarding the General Control of Human Bodies, Tissue, Blood, Blood Products and Gametes in Government Notice R180 of Government Gazette No. 35099 of 2 March 2012.
26. Section 25(a) of the Regulations Regarding the General Control of Human Bodies, Tissue, Blood, Blood Products and Gametes in Government Notice R180 of Government Gazette No. 35099 of 2 March 2012.
27. Section 65 of the National Health Act No. 61 of 2003.
28. Regulation 26 of the Regulations Regarding the General Control of Human Bodies, Tissue, Blood, Blood Products and Gametes in Government Notice R180 of Government Gazette No. 35099 of 2 March 2012.
29. Regulation 24 of the Regulations Regarding the General Control of Human Bodies, Tissue, Blood, Blood Products and Gametes in Government Notice R180 of Government Gazette No. 35099 of 2 March 2012.



COUNCIL OF SOUTH AFRICA

GUIDELINES FOR GOOD PRACTICE IN THE HEALTH CARE PROFESSIONS

ETHICAL GUIDELINES ON PALLIATIVE CARE

EDITED BY THE HUMAN RIGHTS, ETHICS AND PROFESSIONAL PRACTICE

BOOKLET 17

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THE INTENT OF PROFESSIONAL GUIDELINES

Practicing as a health care professional is based upon a relationship of mutual trust between patients and health care practitioners. The term "profession" means "a dedication, promise or commitment publicly made".¹ To be a good health care practitioner, requires a life-long commitment to sound professional and ethical practices and an overriding dedication to the interests of one's fellow human beings and society. In essence, the practice of health care professions is a moral enterprise. The HPCSA presents the following ethical guidelines to guide and direct the practice of health care practitioners. These guidelines form an integral part of the standards of professional conduct against which a complaint of professional misconduct will be evaluated.

[Note: The term "health practitioner" in these guidelines refers to persons registered as such with the HPCSA].

¹ Pellegrino, ED. Medical professionalism: Can it, should it survive? *J Am Board Fam Pract* 2000; 13(2):147-149 (quotation on p. 148).

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1 PREAMBLE

- 1.1 The World Health Organisation (WHO) defines palliative care as ‘an approach that improves the quality of life of patients and their families facing the problem associated with life-threatening illness, through the prevention and relief of suffering by means of early identification and impeccable assessment and treatment of pain and other problems, physical, psychosocial and spiritual’.
- 1.2 Palliative care:
 - 1.2.1 provides relief from pain and other distressing symptoms;
 - 1.2.2 affirms life and regards dying as a normal process;
 - 1.2.3 intends neither to hasten or postpone death;
 - 1.2.4 integrates the psychological and spiritual aspects of patient care;
 - 1.2.5 offers a support system to help patients live as actively as possible until death;
 - 1.2.6 offers a support system to help the family cope during the patient’s illness and in their own bereavement;
 - 1.2.7 uses a team approach to address the needs of patients and their families, including bereavement counselling, if indicated;
 - 1.2.8 will enhance quality of life, and may also positively influence the course of illness;
 - 1.2.9 is applicable early in the course of illness, in conjunction with other therapies that are intended to prolong life, such as chemotherapy or radiation therapy, and includes those investigations needed to better understand and manage distressing clinical complications.’
- 1.3 With the spread of HIV, South Africa shoulders a significant disease burden, and the Department of Health’s *National Policy Framework and Strategy on Palliative Care 2017–2022* acknowledges the importance of integrating palliative care as an essential component of health service delivery.

2 WHEN IS PALLIATIVE CARE PROVIDED

- 2.1 Palliative care is provided to those living with and dying from any advanced, progressive and incurable condition regardless of age or setting.
- 2.2 Palliative care is provided throughout the journey, from diagnosis to end of life.

- 2.3 As the patient's disease progresses, the need for palliative care will increase and the level of curative treatment will decrease.
- 2.4 The WHO recognises that palliative care is required for a wide range of diseases, including but not limited to:
 - 2.4.1 Cardiovascular disease;
 - 2.4.2 Cancer;
 - 2.4.3 Chronic respiratory diseases;
 - 2.4.4 AIDS;
 - 2.4.5 Diabetes;
 - 2.4.6 Kidney failure;
 - 2.4.7 Chronic liver disease;
 - 2.4.8 Multiple sclerosis;
 - 2.4.9 Parkinson's;
 - 2.4.10 Rheumatoid arthritis;
 - 2.4.11 Neurological disease;
 - 2.4.12 Dementia;
 - 2.4.13 Congenital anomalies;
 - 2.4.14 Drug-resistant tuberculosis

3 WHY IS PALLIATIVE CARE PROVIDED

- 3.1 Palliative care is recognised by the United Nations as part of the human right to health.
- 3.2 The goal of palliative care is the achievement of the best possible quality of life for patients and their families, even if life expectancy is short.
- 3.3 Palliative care is an extension of the South African Constitutional rights of human dignity and access to healthcare.
- 3.4 The Constitution also guarantees the right to basic healthcare for children which includes the provision of paediatric palliative care, and not merely access to such care, as is the case with adults.
- 3.5 Evidence suggests that palliative care can reduce unnecessary hospitalisation and use of health care services, which lessens the costs of the health system.

4 WHO PROVIDES PALLIATIVE CARE?

- 4.1 Palliative care is provided by all health practitioners and other health care professionals working with people with life-threatening conditions.

- 4.2 Health practitioners providing palliative care should possess the requisite knowledge, skills and attitudes to meet the physical, psychological, practical, social and spiritual needs of their patients.
- 4.3 Health practitioners providing palliative care should keep their skills up to date.
- 4.4 As a result of the current inadequate provision of palliative care services in the country, the National Department of Health has developed a strategy to increase the scope of palliative care services in its *National Policy Framework and Strategy on Palliative Care 2017-2022*.
- 4.5 It is imperative that all healthcare professionals play an advocacy role on the importance of palliative care services in the public health sector and for palliative care to be included in the curriculum of health practitioners.

5 WHERE IS PALLIATIVE PROVIDED

- 5.1 Palliative care can be provided in public, private facilities and patient's homes.
- 5.2 According to the National Department of Health National Policy Framework and Strategy on Palliative Care the different palliative care service delivery settings, are:
 - 5.2.1 Home-based palliative care;
 - 5.2.2 Mobile outreach services;
 - 5.2.3 Outpatient care;
 - 5.2.4 Inpatient palliative care facilities;
 - 5.2.5 Hospital based palliative care teams;
 - 5.2.6 Day care palliative services;
 - 5.2.7 Frail care and other care homes;
 - 5.2.8 Workplace programs; and
 - 5.2.9 Correctional services.

6 HOW PALLIATIVE CARE IS USED

- 6.1 Health practitioners must deliver care in a manner that upholds these principles, and the rights and values of the patient and their family, even in the midst of a dehumanising environment or in noisy facilities/areas”
- 6.2 Health practitioners providing palliative care should apply the bioethical principles of autonomy, beneficence, non-maleficence and justice that require respect for the worth, dignity and human rights of their patients when managing them.
- 6.3 Patients and their families faced with life-threatening illnesses, are likely to be vulnerable and anxious during this time, health practitioners providing palliative care will find the bioethical principles helpful in guiding decision-making.
- 6.4 Health practitioners must establish trust in the patient and their family through availability, listening, providing honest answers, and having a non-judgmental attitude.
- 6.5 In administering palliative care, health practitioners should also adhere to the HPCSA’s general ethical guidelines and the National Department of Health’s *National Policy Framework and Strategy on Palliative Care 2017–2022*.

7 PATIENT AUTONOMY AND PALLIATIVE CARE

7.1 MEANING OF PATIENT AUTONOMY

- 7.1.1 Patient autonomy recognises the ability of patients to make decisions for themselves regarding palliative care, and is in accordance with the constitutional rights to human dignity and bodily and psychological integrity.
- 7.1.2 Patient autonomy presumes that patients have the required information and capacity to make rational decisions about palliative care and that the circumstances allow them to give an informed consent.
- 7.1.3 When obtaining informed consent from patient’s health practitioners providing palliative care must comply with the National Health Act 61 of 2003 and the HPCSA’s guidelines on *Seeking Patients’ Informed Consent: The Ethical Considerations* (Booklet 9).

7.2 CAPACITY TO CONSENT TO PALLIATIVE CARE

- 7.2.1 Health practitioners providing palliative care should assume that every mentally competent adult patient and every sufficiently mature child patient over the age of 12 years have legal capacity to consent to, or oppose, medical interventions - unless it is shown that they clearly cannot understand the information given to them.
- 7.2.2 It is for the patient, and not the health practitioner providing palliative care, to determine what is in the patient's best interests.
- 7.2.3 Although health practitioners providing palliative care may recommend a course of treatment, they may not pressurize or manipulate the patient into accepting their advice.
- 7.2.4 Health practitioners providing palliative care must understand and respect that a patient's decision-making may be guided by values, customs and beliefs different from the practitioner's.
- 7.2.5 If a mentally and legally competent patient, who is fully informed of the benefits and risks of treatment, consents to or refuses a particular course of treatment, their decision must be respected, even if the health practitioner providing palliative care believes it will result in serious harm or even death.
- 7.2.6 Respecting patient autonomy, however, does not mean that patients are entitled to illegal, unethical or medically inappropriate treatment simply because they have requested it.
- 7.2.7 Health practitioners providing palliative care should only recommend and provide treatment options that are both scientifically grounded, and are, in their best medical judgement, reasonably expected to yield the intended benefits.
- 7.2.8 Should a patient request illegal, unethical or medically inappropriate treatment, the health practitioner providing palliative care should reassure them and their family of the sound treatment options available to the patient - subject to consent by the latter or his or her surrogate.

7.3 ADVANCED PALLIATIVE CARE PLANNING

- 7.3.1 Patient autonomy includes the need to protect patients with diminished autonomy.
- 7.3.2 Health practitioners providing palliative care should encourage competent patients to indicate their wishes regarding treatment options in an advance directive, such as a living will, and in such a document or a written nomination of a surrogate in terms of the National Health Act, to mandate a person to make decisions on their behalf if they become incapacitated.
- 7.3.3 An advance directive helps ensure consideration of the patient's values and preferences, even when they lack the capacity to express them.
- 7.3.4 Health practitioners providing palliative care should record any conversations with a competent patient regarding their goals for care in their medical records, and should review such goals regularly and as circumstances change.
- 7.3.5 The advance directive or other records of the patient's wishes should be kept inside the patient's file, with stickers indicating that the advance directive is available for ease of reference.
- 7.3.6 The *Patient's Rights Charter* introduced by the National Department of Health requires patients to advise their healthcare provider of their wishes regarding death, and to take care of any medical records in their possession – see *HPSCA National Patient's Rights Charter* (Booklet 3).
- 7.3.7 An advance directive should include the following:
 - 7.3.7.1 the patient's wishes and preferences with regard to future treatment;
 - 7.3.7.2 any beliefs or values that may influence the patient's decisions and preferences;
 - 7.3.7.3 the family members or surrogates that should be involved in decision-making;
 - 7.3.7.4 any interventions that should be considered and implemented in case of emergency, such as CPR;
 - 7.3.7.5 the patient's preferred place of care; and
 - 7.3.7.6 the patient's need for religious, spiritual, and other personal support.
- 7.3.8 As is the case with life-threatening illness, towards the end-of-life, it is not uncommon for a patient's mental capacity to be impaired.
- 7.3.9 When a patient becomes mentally incompetent, and an advance directive is available, health practitioners providing palliative care should give effect to the patient's wishes – provided that the directive is applicable to the present circumstances and represents the patient's current wishes.

- 7.3.10 If no advance directive is available, and no surrogate has been appointed, the health practitioner providing palliative care must, in consultation with the patient's family, determine the treatment option that is in the patient's best interest.
- 7.3.11 In determining an incapacitated patient's best interests during the decision-making process, the health practitioner providing palliative care must consider:
- 7.3.11.1 the medically appropriate treatment options available;
 - 7.3.11.2 any previous requests of the patient when he or she was mentally competent;
 - 7.3.11.3 what the practitioner knows about the background and preferences of the patient, in consultation with the patient's healthcare team;
 - 7.3.11.4 information regarding the patient's values, beliefs and preferences from third parties who may have other knowledge of the patient due to their relationship (e.g. a parent, sibling, child or spouse);
 - 7.3.11.5 where more than one treatment option is reasonably in the patient's best interests, the practitioner must, in consultation with the patient or the patient's surrogate, choose the option that least restricts the patient's future choices.
- 7.3.12 The health practitioner providing palliative care must take appropriate action in the case of illegal, unethical or inappropriate behaviour that jeopardises the best interests of the patient.

7.4 COMMUNICATION

- 7.4.1 Health practitioners providing palliative care must communicate effectively and timeously with the patient and their family, and the rest of the palliative care team.
- 7.4.2 Health practitioners providing palliative care must take into account the emotional toll that the circumstances regarding the use of palliative care may have on the patient and their family, and should communicate with them in an empathic and understandable manner.
- 7.4.3 If a patient's communication skills are compromised by their illness, health practitioners providing palliative care must communicate with the patient through verbal and non-verbal means.
- 7.4.4 Irrespective of the patient's level of consciousness, health practitioners providing palliative care should communicate any proposed interventions to the patient, based on the understanding that hearing is the last sense to die.

8 BENEFICENCE AND PALLIATIVE CARE

8.1 MEANING OF BENEFICENCE

- 8.1.1 The principle of beneficence requires the health practitioner providing palliative care to do good for their patients.
- 8.1.2 The health practitioner providing palliative care must weigh up the costs, risks and benefits of particular forms of palliative care and recommend those that are most beneficial to the patient, while leaving the final decision to the patient or their surrogate.
- 8.1.3 From the moment of diagnosis of a life-threatening condition, the health practitioner providing palliative care must monitor the patient and keep abreast of relevant medical advancements, and ensure that their knowledge and understanding of palliative care is up to date.
- 8.1.3 If the health practitioner providing palliative care is not able to manage severe or refractory symptoms, it is important for such practitioner to refer the patient to a health practitioner who is qualified to do so.

8.2 WITHHOLDING OR WITHDRAWAL OF TREATMENT

- 8.2.1 The health practitioner providing palliative care is required to balance the intended benefits of palliative care against the risks and burdens of treatment.
- 8.2.2 In some instances, the quality of life which follows palliative treatment may raise questions as to whether such treatment is in the best interests of the patient.
- 8.2.3 Treatment can legally and ethically be withheld or withdrawn if the patient refuses further treatment, further treatment is futile, or if it is no longer in the patient's best interests (e.g. when treatment merely prolongs the dying process).

- 8.2.4 The decision to withhold or withdraw treatment must not be taken lightly, and must be considered by the health practitioner providing palliative care in consultation with the patient and their family.
- 8.2.5 When the possibility of withholding or withdrawal of treatment arises, the health practitioner providing palliative care should discuss with the patient and their family, arrangements for basic care and other appropriate treatments.
- 8.2.6 At end-of-life, discussions about withholding or withdrawal of treatment should include plans to manage the final stages of the lives of patients, before they are incapacitated, and include personal matters such as wills, as well as any other concerns that patients believe are important to ensure that they die with dignity.
- 8.2.7 When dealing with end-of-life decisions health practitioners providing palliative care must approach the withholding or withdrawal of treatment in a manner consistent with the HPSCA's *Guidelines for the Withholding and Withdrawal of Treatment* (Booklet 7).

9	NON-MALEFICENCE AND PALLIATIVE CARE
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- 9.1 Non-maleficence requires health practitioners providing palliative care not to harm their patients and complements beneficence and the balancing of risks and benefits.
- 9.2 Euthanasia and doctor-assisted suicide are often perceived as inconsistent with the principle of non-maleficence.
- 9.3 Euthanasia and doctor-assisted suicide are presently prohibited under South African law, and the courts frequently do not distinguish between the two when it comes to culpability.
- 9.4 Euthanasia is the employment of any medical intervention primarily aimed at ending a patient's life (e.g. giving a patient lethal drug or injection).
- 9.5 Doctor-assisted suicide occurs when the health practitioner provides the means necessary to enable the patient to end their own life (e.g. handing a patient a lethal drug or prescribing a lethal drug for a patient).
- 9.5 At present, South African courts have acknowledged that both euthanasia and doctor-assisted suicide are fundamentally incompatible with a practitioner's role as a healer, and a practitioner guilty of either is regarded as having acted unethically and unlawfully.

- 9.6 Due to the stress of extreme pain and the prospect of facing a life-threatening illness, a patient may request a health practitioner providing palliative care to end their life.
- 9.7 The role of the health practitioner providing palliative care in this instance is to discuss the concerns and fears that have led to the patient's request, and to provide alternate approaches to address these issues.
- 9.8 When medical treatment relieves suffering - but has the effect of accelerating the dying process - the health practitioner providing palliative care must consider whether the palliative benefits justify the shortened life expectancy before pursuing that course of treatment.
- 9.9 Palliative care treatment where life is shortened as a side effect, is not regarded as unlawful or unethical – provided the patient or their surrogate has given informed consent.

10 JUSTICE OR FAIRNESS AND PALLIATIVE CARE

- 10.1 The World Health Organisation (WHO) states that 'palliative care needs to be provided in accordance with the principles of universal health coverage'.
- 10.2 This means that everyone, 'irrespective of income, disease type or age, should have access to a nationally determined set of basic health services, including palliative care' and is consistent with the South African Constitution.
- 10.3 The WHO also states that 'financial and social protection systems need to take into account the human right to palliative care for poor and marginalized population groups'.
- 10.4 Palliative care should be available to all patients from birth until death, and should be accessible at all levels of the health care service and this is what the National Department of Health's *National Policy Framework and Strategy on Palliative Care 2017–2022* aspires to do.
- 10.5 Health practitioners providing palliative care must treat patients equally and without discrimination.
- 10.6 According to the National Department of Health's *National Policy Framework and Strategy on Palliative Care 2017–2022* health practitioners providing palliative care must give special consideration to the following groups of people in their provision of palliative care services:

- 10.6.1 Persons with disabilities;
 - 10.6.2 Children (including neonates and adolescents);
 - 10.6.3 Older persons including those living in residential care settings and frail care facilities;
 - 10.6.4 Asylum seekers and refugees;
 - 10.6.5 Inmates of correctional services; and
 - 10.6.6 Persons in long term care facilities such as TB and psychiatric hospitals or residential care facilities.
- 10.6 Palliative care practitioners should use the country's limited health care resources responsibly, fairly and effectively to ensure all patients receive appropriate palliative care.

11 REFERENCES

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SPIKES

Embrace a Patient-first Approach to
Advance Care Planning Conversations



S

Setting

Choose a private, comfortable,
non-threatening setting



P

Perception

Uncover what patient & family think
is happening



I

Invitation

Ask patient what they would like to know



K

Knowledge

Explain disease and care options in
plain language



E

Emotion

Respect feelings, respond with empathy



S

Summarize

Recap and decide what's next

Source: Baile, W. F., Buckman, R., Lenzi, R., Glober, G., Beale, E. A., & Kudelka, A. P. (2000). SPIKES—a six-step protocol for delivering bad news: application to the patient with cancer. *The oncologist*, 5(4), 302-311.

S	<u>S</u>etting up	-Arrange for privacy. - Involve significant others, sit down, make connection and establish rapport with the patient, manage time constraints and interruptions.	-What time would suit you and your family members for a chat about your diagnosis?
P	<u>P</u>erception of condition/seriousness	-Determine what the patient knows about the medical condition. - Listen to the patient's level of comprehension, accept denial but do not confront at this stage	-Explain to me what you understand of your recent diagnosis?
I	<u>I</u>nvitation from the patient to give information	-Ask patient if s/he wishes to know the details of the medical condition, accept patient's right not to know. -Offer to answer questions later if s/he wishes	-Would you like me to explain exactly what your diagnosis means?
K	<u>K</u>nowledge: giving medical facts	-Use language intelligible to patient -Consider educational level, socio-economic background, current emotional state, give information in small chunks -Check if patient understand what you have said, respond to patient's reactions as they occur, give any positive aspect first	-When we examined your chest x-ray, we saw a small visible mass, this is usually an indication of cancer, is this all making sense to you?
E	<u>E</u>xplore emotions and sympathize	-Prepare to give an empathetic response -Identify emotion expressed by the patient -Identify cause/source of emotion, give the patient time to express their feelings	-Has your diagnosis come as a shock to you? Explain to me how you are feeling?
S	<u>S</u>trategy and summary	-Close the interview. -Ask whether they want to clarify something else. -Offer agenda for the next meeting	-Has this all made sense to you? Have you anymore questions? When do you wish to arrange our next meeting?

EXCELLENCE IN DECEASED DONATION AND END OF LIFE CARE COURSE



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Apply South African laws, and ethical considerations and guides for the donation process



Communication skills

Develop your skills in breaking bad news to families, and supporting fully informed decision making



Roles and role players

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